

Instructions in preparing an application

Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent. No submission will be accepted after the deadline, March 15th at 13:00 CET. Applications should be sent in PDF format to **Flagships@erasmusmc.nl**.

Title: Development of the microbial safe hospital – the ‘MI-Safe-Hospital flagship’

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name (abbreviation)</i>	<i>Institution</i>	<i>Dept.</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position / Tenure-track/ Other (please specify)</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship [Main Lead/Lead/Applicant]</i>
Prof. dr. Margreet Vos (MV)	EMC	MMIZ	m.vos@erasmusmc.nl	Tenured position	mix	Main Applicant-Project Lead
Dr. Juliëtte Severin (JS)	EMC	MMIZ	j.severin@erasmusmc.nl	Tenured position	mix	Coordination Team
Dr. Anne Voor in 't holt (AV)	EMC	MMIZ	a.voorintholt@erasmusmc.nl	Tenured position	researcher	Coordination Team
Dr. John Hays (JH)	EMC	MMIZ	j.hays@erasmusmc.nl	Tenured position	researcher	Management Team
Dr. L. Georgievska (LG)	EMC	MMIZ	l.georgievska@erasmusmc.nl	Tenured position	researcher	Management Team
Dr. Robert Kraaij (RK)	EMC	Inwendige Geneeskunde	r.kraaij@erasmusmc.nl	Tenured position	researcher	Applicant
Prof. dr. André Uitterlinden (AU)	EMC	Inwendige Geneeskunde	a.g.uitterlinden@erasmusmc.nl	Tenured position	researcher	Applicant
Prof. dr. Monique van Dijk (MvD)	EMC	Inwendige Geneeskunde/	m.vandijk.3@erasmusmc.nl	Tenured position	researcher	Applicant

		Kinderchirurgie				
Dr. Erwin Ista (EI)	EMC	Inwendige Geneeskunde / Kinderchirurgie	w.ista@erasmusmc.nl	Tenured position	researcher	Applicant
Prof. dr. Marco Bruno (MBr)	EMC	MDL	m.bruno@erasmusmc.nl	Tenured position	mix	Applicant
Prof. Dr. Ed van Beeck (EvB)	EMC	iMGZ	e.vanbeeck@erasmusmc.nl	Tenured position	researcher	Applicant
Dr. Vicki Erasmus (VE)	EMC	iMGZ	v.erasmus@erasmusmc.nl	Tenured position	researcher	Applicant
Liesbeth van Heel (LvH)	EMC TUD	Public Health & Fac. ABE	m.vanheel@erasmusmc.nl	Tenured position (till June 2023 & PhD)	Researcher	Applicant
Dr. Jeroen van Kampen (JvK)	EMC	Viroscience	j.vankampen@erasmusmc.nl	Tenured position	mix	Applicant
Dr. ir. Nick van de Berg* (NvdB)	EMC TUD	Gynaecological Oncology & 3mE, dept. BMechE	n.vandeberg@erasmusmc.nl	Postdoc		Applicant
Dr. ir. Arjo J. Loeve (AL)	TUD	Fac. 3mE, dept. BMechE	A.J.Loeve@tudelft.nl	Tenured position		Lead
Prof. dr. ir. Atze Boerstra (AB)	TUD	Architecture & the Built Environment	A.C.Boerstra@tudelft.nl	Tenured position (currently until March 2026)		Applicant
Prof. ir. Peter Luscuere (PL)	Inspired Ambitions		pluscuere@gmail.com	External		Applicant
Prof. dr. Jenny Dankelman (JD)	TUD	Fac. 3mE, dept. BMechE	J.Dankelman@tudelft.nl	Tenured position		Applicant
Ir. Daniel Robertson (DR)	TUD	Fac. 3mE, dept. BMechE	P.D.Robertson@tudelft.nl	PhD		Applicant

Dr. Ir. Sonja Paus-Buzing (SPB)	TUD	Fac. Industrial Design Engineering	S.N.Paus-Buzink@tudelft.nl	Lecturer		Applicant
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Dr. Jeroen van Wijngaarden (JvW)	EUR	ESHPM	vanwijngaarden@eshpm.eur.nl	Tenured position		Applicant
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Corresponding Lead: Dr. J. Severin (j.severin@erasmusmc.nl)

Planned start date of seed funding: 31st December 2022

Primary Convergence Theme: Improving Health Journeys

Secondary convergence theme (if applicable): Fundamentals of Health and Disease

Tertiary convergence theme (if applicable): Technology Supported Transitions in Health(care)

Quaternary convergence theme (if applicable): Human Centered Technology and AI for Health

Keywords: 7 keywords needed. Healthcare-Associated Infections, Prevention of Infection, Design of Hospital Structures, Design of Hospital Appliances, Microbial Detection, Behavior and Education of Healthcare Workers, Sterilization and Disinfection

1.b Summary (max 300 words) 299 words

Healthcare-associated infections (HAI) are either endogenous or exogenous and (in)directly acquired from the body, environment or medical devices^{1,2,3,4} and approximately **5% of patients acquire a HAI while hospitalized.**

Economic and Societal Impact

In European hospitals alone, a total of **4.5 million HAI occur yearly** with an estimated annual **financial loss of 7 billion euros**. Bacterial infections account for 3.4 million extra days of Intensive Care Unit (ICU) stay, **and 8,700 attributable deaths yearly**^{5,6,7}. Further, many microorganisms causing HAI can evolve to become highly resistant to antibiotics (HRMO) and thereby enhance their transmission. **Globally, 1,27 million deaths are attributable to HRMO** (~73,000 per year in Europe)^{8,9} with 1 out of 3 HAIs in Europe being caused by HRMO. On top of this burden are the dynamics and risks of viral HAI. For example, the recent **SARS-CoV-2 pandemic was responsible for up to 65% of HAIs**.^{10,11,12,13,14} Such infections complicate treatment options, patient morbidity/mortality, length of stay in hospital and increase the burden on healthcare economies.

The Flagship

This Flagship will initially focus on exogenous HAI, which are not well understood, involve many complex and diverse factors, and require an integrated, comprehensive, multi-faceted and multi-disciplinary approach (involving collaboration between medical, microbiological, technological and behavioral sciences) in order to create the **Microbial-SAFE hospital** (Figure 1)!

Objectives

The prevention of approximately 25% of EU HAI in the next 10 – 20 years via the following core aspects: **1) adaptation of the environment** by design to prevent contamination by microorganisms^{15,16,17}, **2) development of novel detection methods** for microorganisms in the environment¹⁸ and **3) identification of HAI-related facilitators and barriers to best practices / healthcare worker (HCW), patient and hospital visitor behaviors**^{19,20}. These aspects will **utilize state-of-the-art knowledge and technologies and generate new research, prototypes and educational materials.**



Figure 1. Overview of the 'MI-SAFE hospital' Flagship

1.c Public Summary (max 300 words) 299 words

Ongeveer een op de twintig patiënten loopt een infectie op tijdens de ziekenhuisopname. Deze zogenoemde zorginfecties kunnen het ontslag naar huis vertragen, extra ziekte veroorzaken, en in sommige gevallen zelfs de dood tot gevolg hebben. Alleen al in Europese ziekenhuizen krijgen jaarlijks 4,5 miljoen patiënten een zorginfectie, wat neerkomt op 7 miljard euro aan extra kosten. Een toenemend probleem is dat bacteriën minder of niet meer gevoelig zijn voor veel gebruikte antibiotica. Wereldwijd overlijden er jaarlijks zo'n 1,27 miljoen patiënten aan een infectie met dit soort bacteriën, waarvan zo'n 73.000 patiënten per jaar in Europa. Ook virussen (SARS-CoV-2 bijvoorbeeld) kunnen zorginfecties veroorzaken. Zorginfecties kunnen ontstaan doordat het micro-organisme al aanwezig is in het lichaam van de patiënt ('endogene' infectie), of door contact met een gecontamineerde ziekenhuisomgeving ('exogene' infectie). Endogene infecties kunnen soms worden voorkomen door bijvoorbeeld preventief antibiotica te geven voor een operatie, maar dit ligt lastiger bij exogene infecties. Dit MI-SAFE hospital Flagship (oftewel 'mijn veilige ziekenhuis') zal zich richten op deze exogene infecties, welke vaak niet goed worden begrepen en waarbij complexe en diverse factoren betrokken zijn. Hiervoor is een geïntegreerde, veelzijdige en multidisciplinaire aanpak vereist, waarbij samenwerking tussen medische, microbiologische, technologische en gedragswetenschappen essentieel is om een microbieel-veilig-ziekenhuis te creëren!

Onze aanname is dat exogene zorginfecties te voorkomen zijn door besmetting en overdracht van micro-organismen binnen ziekenhuizen beter te detecteren, beter te begrijpen en daarmee beter aan te pakken. De hoofdaspecten van MI-SAFE hospital zijn onder meer: 1) het aanpassen van de ziekenhuisomgeving om besmetting door micro-organismen te voorkomen, 2) op een nieuwe, betere manier micro-organismen in de omgeving detecteren, en 3) de identificatie van factoren van zorginfecties en contaminatie-veilig gedrag van zorgpersoneel, patiënten en bezoekers van ziekenhuizen. De resultaten van MI-SAFE hospital zullen nieuwe kennis en technologieën opleveren, inclusief nieuw onderzoek, prototypes en educatief materiaal.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)**218 words (Total = 1,989 words)**

Contribution: The MI-SAFE hospital flagship is an integral/essential part of the H&T mission *prevention and early diagnostics in order to proactively maintain health (mission 3)*, and overlaps with missions 1) *Life-long health and well-being, from pre-birth to end of life* and 2) *Socio-economic equality in health(care)*. Healthcare-associated infections (HAI) lead to a considerable patient morbidity and mortality at all ages (including at Erasmus MC). Further, HAI may be largely preventable by implementing effective prevention measures and applying early diagnostic and disinfection technologies. MI-SAFE hospital's final contribution to healthcare will be: 1) fewer HAI, 2) fewer deaths of patients and a 3) reduced burden on healthcare systems.

Fit: The central themes of the MI-SAFE hospital flagship are to *Improve Health Journeys* and to develop *Technology Supported Transitions in Health(care)* to prevent HAI. In this respect, the flagship will identify, measure and proactively prevent the diverse parameters associated with the acquisition of HAI during a patient's journey through the whole process of hospitalization. This journey not only includes hospital admission, hospital stay, medical procedures and hospital discharge of the patient, but also the interactions of the patient with his/her environment, visitors and various healthcare workers (HCW). Technological innovations (including creating inherent microbial safety by design and placement of hospital structures and appliances) will support the transition towards truly HAI-safe hospitals.

2.b Synergy between disciplines 423 words

MI-SAFE hospital will address the problem of HAI using a wide variety of consortium expertise to generate synergistic disciplinary technological, microbiological and behavioral science solutions. **Technological sciences** will develop more microbiologically safe hospital structures and appliances (e.g., ventilation, sinks, disinfection). **Clinicians, technologists and research microbiologists** will assess the presence, quantity and species of microorganisms contaminating these structures before and after technological or procedural interventions (using for example hyperspectral imaging, forensic alternative light source imaging, optical coherence tomography etc.). Synergy between **behavioral sciences** and **human factors engineering**, will indicate how patients, visitors and HCW interact with each other, generating improved hospital structures and appliances that prevent contamination by the transfer of (antibiotic resistant) microbial pathogens. Finally, understanding the behaviors of, patients, visitors and HCW will facilitate the development of educational resources that will inform and help change current habits, behaviors and processes. The insights of behavioral science on the target groups will serve as input for the design process.

Consortium members involved in the MI-SAFE hospital flagship already have track records of interdisciplinary collaboration and of obtaining relevant funding from (inter)national funding bodies and businesses. New collaborations are being formed via the financing and building of a 'Lely-living-lab' (a modular designed patient room already planned to be established at Erasmus MC), in which adaptations to existing hospital structures will be explored. These collaborations also include cross-cutting research into the duration of infectivity and survival time of infective viruses e.g. SARS-CoV-2 and the presence of antibiotic resistant biofilms.

Table 1: Synergy between MI-SAFE hospital funded positions, institute and work packages.

Position (name of employee if planned)	Fte	Years	Institute	WP	Deliverables - topics focused on	Deliverables linked to	Budget source
Research assistant	0.5	5	EMC	1	1.1-1.4 Optimized designs	3.3-3.4	Flagship
PostDoc	1	2	TUD	1	1.2-1.3 Optimized designs	3.2	Flagship
PostDoc (M.Loth)	0.5	5	TUD-IO	1	1.1-1.4 / Sink Design	3.1-3.3	Flagship

PhD	0.5	4	TUD-IO	1	1.1-1.4 / Optimized designs	3.1-3.3	Flagship
	0.5		EMC				
PostDoc	1	5	EMC	2	2.1-2.4 / Virus detection	1.2-1.3, 3.2	Flagship
PhD	0.5	4	TUD-3mE	2	2.1-2.4 / Localizing biofilm	1.2-1.3, 3.2	Flagship
	0.5		EMC	2			
PhD (D. Hoek)	0.5	8	TUD-3mE	2	2.1-2.4 / UV-C disinfection	1.2-1.3, 3.2-3.3	UV-Smart
PostDoc (D. Robertson)	0.5	5	TUD-3mE	2	2.1-2.4 / Localizing microorganisms	1.2-1.3, 3.2	Flagship
Research assistant	0.5	3	EUR	3	3.2 & 3.3 / Guidance, teaching and education	1.2-1.3	Flagship
PhD	1.0	4	EUR	3	3.1-3.3 / Guidance, teaching and education	1.1 – 1.4	Flagship
PostDoc	0.8	5	EUR	3	3.1-3.4 / Guidance, teaching and education	1.1-1.4	Flagship

2.c Long term sustainability plan 449 words

The feasibility and long-term sustainability of the MI-SAFE hospital flagship is described in the Risk and Mitigation table below

Risk and Mitigation Table

Risk	Mitigation
Lack of short term funding (years 5 – 10)	Funding for the activities described in this proposal have already been secured via the matching process. All project PIs have a proven experience in securing external funding which will be further matched during the 5 years. There already exists strong interest from commercial partners.
Lack of long-term funding (years 11 – 20)	- The many existing connections to private stakeholders within the flagship, means that there is a strong potential for obtaining further funding from private stakeholders. - The multi-disciplinarity of the project will create a number of spin-off projects to be investigated in follow-up research and new grant applications from both public and private sources.

	The project is composed of applicants in different stages of their academic careers, including post-docs and tenure track researchers, who will strengthen their academic profile in the flagship increasing their capacity to secure external funds.
Lack of project sustainability.	- In the Lely-living lab, innovative experiments can be performed and hypothesis tested that were not previously possible and without risks to patients. The results will be novel, evidence-based HAI interventions that can be implemented worldwide, generating global interest and new collaborations. - The varied results of the flagship will form a template for future spin-off projects involving, for example, investigations into the prevention of HAIs in operating rooms, visitor waiting rooms etc.
Lack of international networking capacity and Implementation.	- We will recommend, advise and change current (inter)national Infection Prevention and Control (IPC) regulations, guidelines and standards via our novel flagship results - flagship interdisciplinarity will help convince scientific and clinical communities, as well as relevant policymakers. - We will use the network ENAH (European Network Architecture for Health), Architects for Health, European Health Property Network and S.ARCH (sustainable architecture) to put the results of this Flagship project on the agenda. Beyond Europe we can involve the Center for Health Design. Also to be included is the ESGNI (ESCMID Study Group for Nosocomial Infections) of the European Society for Clinical Microbiology & Infectious Diseases (MV is chair). - Most of the participants in the MI-SAFE Hospital are leaders in their respective field, nationally or internationally, and the current project will confirm their leading position. - The participants have relevant memberships through which results, including tools, products (e.g. the washbasin), and new guidelines and standards of the Flagship can be disseminated.
Lack of synergistic collaboration between partners.	Using existing expertise and by employing new and early stage researchers we will bring novel ideas into the project, further stimulating synergistic collaboration and facilitating new cross-links and impactful intersectoral and interdisciplinary collaborations.

2.d Visibility, attracting talents and new partners 508 words**Visibility**

An important part of the Flagship is the communication of the findings and the promotion of the project to broad audiences to maximize impact. MI-SAFE Hospital will have a central communication platform, a dedicated website, from which flagship progress and results will be regularly communicated and disseminated to stakeholders, including researchers, clinicians, nurses, medical businesses, the general public, policy makers etc.). Via this website, we will also spread newly published insights via twitter, Instagram, YouTube videos, press releases MI-SAFE hospital will utilize a wide range of social media outlets. For fellow technologists and researchers we aim to publish in scientific journals such as 'Lancet Infectious Diseases', 'Clinical Infectious Diseases', 'Health, Environment, Research and Design Journal', 'Ergonomics', 'Antimicrobial Resistance and Infection Control', 'Analytical Chemistry', 'IEEE Transaction on Biomedical Engineering' etc.

One of the key themes behind the social media presence of the flagship will be that the Lely-lab as a 'living lab', not only with respect to the detection and destruction of pathogenic microorganisms (!), but also in the sense of the flagship's adaptability and the breadth of research being performed. The website will ensure that interested stakeholders continue to return for further information and to interact and provide feedback to the MI-Safe hospital Flagship. Further, the website will be extremely helpful in translating HAI-prevention science into more easily accessible and impactful messages for the general audience. The website will be extremely helpful in translating MI-SAFE hospital concepts and achievements science into more easily accessible and impactful messages for all stakeholders, as well as a broader audience interested in HAI in general.

Talents

In order to engage early career researchers and other younger stakeholders (PhD students, postdocs, assistant professors etc.), the visibility of the project will be internationally and interdisciplinary advertised. Further advertising of vacant positions for the flagship will be placed on international vacancy websites such as Linked-In, Academic Transfer, EURAXESS etc.

PIs of the consortium already have strong teaching links with various educational courses taught at TUD, EMC and EUR. For example, AL is coordinator of the final projects of the BSc Clinical Technology and Coordinator of the MSc Biomedical Engineering at TU Delft, while JS, RK, JH, JvK and others teach on the Master Infection & Immunity, BSc medicine and BSc Clinical Technology at EMC. These contacts will further facilitate easy and constant access to a talented pool of BSc and MSc students for MI-SAFE hospital-matching internships, projects and theses, possibly leading to PhD career paths.

New Partners

As a 'living lab' the flagship will be attract not only to currently participating companies (e.g., **Pentax Medical, Boston Scientific, Inspired Ambitions, UV Smart, Medexs, Van Straten Medical** etc.), but also to companies to which the flagship context may be relatively new. These potentially new partners include :

1) Building design e.g.

- **EGM architects,**
- **Royal Haskoning dhv,**
- **Villery and Boch,**
- **Van Dorp,**
- **Virobuster International Gmbh;**

2) Medical device and appliance developers e.g.,

- Olympus
- Applied Biosystems

3) Developers of cleaning and disinfection compounds e.g.,

- Tork,
- Delabie,
- Artimo,
- the Interclean Network and

4) Diagnostic companies e.g.,

- Roche,
- bioMérieux etc.

2.e Long term societal impact 391 words

The possible pathways for a long term significant contribution to healthcare, society and scientific impact are shown in the 'Contribution and Parameter' table below.

Long Term Contribution to Scientific, Societal and Economic Impact

CONTRIBUTION	PARAMETER
Clinical impact	• 25% estimated reduction in HAI in EU per year resulting in lower mortality and morbidity.
Technological Impact	• Novel tech for fast localization of microbial unsafe contamination. • State-of-the-art hospital structures and appliances that are designed to be better optimized for cleaning and disinfection. • New disinfection methods that do not put a burden on HCW but do make the hospital safe.
Scientific Impact	• Improved understanding of the transfer of micro-organisms within hospital settings (human factors). • Groundbreaking publications e.g. better understanding infectivity of patients and duration of infectious period. • Improved understanding of how infections spread - via who, what how and why. • New training methods for achieving microbial safe behavior by stakeholders.

Educational Impact	• More advanced and better targeted stakeholder protocols and guidelines for the prevention of HAI.
Societal Impact	• Less time spent in isolation for patients. • Reduced chance of requirement for Intensive Care Unit admission. • Better psychological health for patients and HCW.
Environmental impact	• Reduction in use of medical PPE and disposables/consumables. • Reduction in energy demand (e.g. alternatives to steam cleaning) • Reduction in use of environmentally unfriendly chemical s
Economic Impact	• Healthcare savings with respect to reduced morbidity, mortality, days spent in hospital and a reduction in the spread of antibiotic resistance.

Concrete research strategies associated with long term societal impact (including expected TRL levels and time scale of impact) are shown in the table below.

Deliverable	Description	Final TRL Level	Year Realized (after project start)
D1.2- Investigate	New Sanitation (e.g. sink) Designs	6	4/5
D1.4- Educate	Guidelines and educational material available for improved system design.	4	4/5
D1.4- Educate	Guidance document that describes how to ideally ventilate and apply air cleaning devices in patient rooms.	4	4/5
D2.2- Develop	Prototype microorganism and biofilm detection device.	6/7	5
D2.2- Develop	Develop a prototype molecular infectivity assay for viruses.	6	5

D2.2-Develop	New microbial methods for microbiota detection.	6/7	5
D2.3-Educate	Guidance document for UV-C disinfection of medical devices and hospital appliances.	6	5
D2.3-Educate	Design guidance for making medical devices and hospital appliances suitable for disinfection.	6	5
D3.3-Develop	New behavioral nudging strategies	4	4
D3.4-Educate	Teaching and educational material available for stakeholders (HCW, patients, visitors etc.).	4	4/5

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions 498 words (Total = 1,999 words)

The ambitions of the MI-SAFE hospital project involves novel interdisciplinary research that will impact on, hospital structure and appliance design, (public and private) HAI-control product development, new knowledge for use in updated HAI guidelines and educational/training materials relating to the spread of HAI.

Measurable impact description	Target
The Lely-living-lab will be available to other stakeholders to answer study questions on patient rooms, impacting on the efficiency of new HAI product development.	>1 request from external research consortium or business for use of the lab.
New interdisciplinary education programs (Desiderius school) for students/HCW in caring for infectious and non-infected patients e.g., pandemic preparedness etc.	At least 2 programs by year 5.

Companies associated with cleaning, disinfection of materials, antimicrobial coatings, heavy metals etc. will be invited (contract research) to test new products (including robotic methods) that will eventually impact on hospital microbial safety.	> 2 replies or requests. (years 2 – 5). More requests in following years.
Spin off design concepts, knowledge and IP for managing microbial contamination in a variety of sensitive environments e.g. public washrooms, train toilets etc.	Design concepts and IP funding requests based on flagship results submitted to companies and public authorities (beginning year 5). >2 publications by year 5.
Active promotion and highlighting of the need for the inclusion of <i>in silico</i> computational activities in preventing HAI.	Experts in, advanced computational fluid dynamics and crowd behavior modelling will be encouraged to offer advice and make use of Lely lab and flagship facilities.
Attracting funding for extension of HAI-sensitive areas from the Lely Lab model to, for example, operating theatres, patient waiting rooms etc.	Funding requests based on flagship results submitted (beginning year 5).
Design of alternatives for current steam in air humidification protocols.	Funding requests based on flagship results submitted (from year 4).
Offer expertise and facilities to companies to help develop prototypes of microbial detection & localization devices and fast scanning of large surfaces.	>2 replies or request (years 2 – 5). More requests in following years.
Collaboration with expert committees and societies e.g.: 1) Samenwerkingsverband Richtlijnen Infectiepreventie (SRI); 2) ESCMID guideline for Highly Resistant organisms in Europe; 3) SRI guidelines on isolation rooms; 4) World Toilet Organisation; 5) European Health Property Network, 6) Platform Kennisdeling Vastgoed Ziekenhuizen etc.	>6 news reports (from month 1), regular dissemination and communication of progress via dedicated MI-SAFE hospital website and social media. Collaboration in >2 guideline updates (years 4 - 5). More collaborations in following years.
Develop new methods and protocols for the rapid detection of a wide range of HAI microorganisms and biofilms.	At least 1 prototype (TRL6) and 1 protocol published (year 4/5).

Develop a prototype molecular infectivity assay for viruses.	Develop at least one prototype assay (TRL 6) for at least one respiratory virus that is able to discriminate between non-infectious and infectious virus before the end of the MI-SAFE hospital project (year 5).
Design guidelines for microbiologically safe hospital structures and appliances optimized for UV-C disinfection.	At least 1 set of guidelines (year 4/5).

3.b Aims and feasibility 289 words

The flagship is divided into 3 logical WPs, with interactions between the different WPs being clearly defined. The relevant expertise is available for all WPs, with defined partners from all 3 institutions participating in WPs. The Deliverables of the WPs have been determined based on the number of researchers and (co)financing available, as well as on previous/continuing collaborative projects between the participants. **The involvement of 42 participants with 11 new positions is adequately covered by the flagship budget of €2,933,307.** PhD and postdoc positions will also be shared between WPs in order to further facilitate collaboration and maximize the (co-)financing obtained (see also Table 1. Section 2.b). The budget for matching is foreseen as **€975,339 (private sector – industry)**, and approximately **€1,178,000 (public sector)**.

Concrete Plan for attracting funding:

Further funding will be obtained by new synergistic collaborations, IPR and grant applications with stakeholders, including **stakeholders with special interests hospital structures and appliances such as waiting rooms, operating theatres, wash rooms, endoscopes, disinfection, sterilization, infection prevention etc.** Examples of businesses to be approached for potential collaboration in the next 5 years include :

EGM architects, Royal Haskoning dhv, Villery and Boch, Van Dorp, Virobuster International GmbH, Olympus, Applied Biosystems, Tork, Delabie, Artimo, the Interclean Network, Roche, bioMérieux etc. See also Section 2d and 3a.

For public funding general calls appear regularly, including

- **ZonMW (Goed Gebruik Geneesmiddelen, Veni Vici Vidi)** and the
- EU (e.g. **Horizon Europe, Eurostars**, etc.)

Student Involvement:

Participants teach on and lead a range of courses at the 3 flagship institutions including the **BSc Clinical Technology, MSc Technical Medicine, BSc / MSc Medicine BSc and MSc Biomedical Engineering**. This allows active recruitment of a range of ambitious and knowledgeable students for research internships involving flagship WPs.

3.c Work plan 929 words.

MI-SAFE hospital will deliver a growing set of tools, methods and training modules that significantly contribute to a microbial-safe hospital. Each Work Package (WP - see Figure 2) has a fitting ratio of inputs from social sciences (EUR) clinical sciences (EMC) and technical sciences (TUD).

WP1 DESIGN-Safe.

WP leaders: JS

Task leaders: D1.1) ML, AV D1.2) AB, PL, JS, MV; D1.3) LvH, JS, ML; D1.4) MBu, AB, AvH

Participants: TUD: AB, AL, ML; **EMC:** JS, LvH, JvK, MV, AV; **EUR:** MBu; External; PL

Goals: *Ensuring inherent microbial safety by design of the built environment, technical systems, medical devices, appliances. Focus on 1) air, i.e. ventilation system(short-range and long-range airborne transmission of respiratory infection, air temperature, humidity, air exchange rates and air flow direction and pressure difference) 2) water, i.e. sanitary products and 3) surfaces and materials.*

D.1.1-Identify: Review (Journals & case reports) [M01-12]: The relevance of ventilation & air cleaning aspects in relation to the airborne transmission route. Improved sanitary products and materials for surfaces, furniture and fittings to reduce microbial contamination and transmission.

Prepare: Realization of dedicated, adjustable ventilation-air cleaning system for Lely-living-lab.

D.1.2-Investigate [M01-60]: **Report A** : adequately designed ventilation / air cleaning systems that mitigate airborne microbial exposure. Insight in ‘within room’ airborne cross-contamination risks and mitigating effect of air exchange rate manipulation & operation of local air cleaning device.

Report B: microbial transmission with different sink designs. **Report C:** applicability of cleaning and disinfection strategies (e.g. H₂O₂ and UV-C disinfection from WP2), materials used, effect of surface’s and position of elements. **Report D:** position of furniture and fittings enhancing safe behavior by HCW (with WP3).

D1.3-Develop[M12-60]: **Prototype:** verified room design, including ventilation system and sanitation, lay-out and position of furniture

D.1.4-Educate [M48-60]: **Guidance document:** Design guidance for improved system design for patient rooms. Guidance document for design of ventilation/air cleaning systems for patient room incl. pressure difference aspects. Best practices and evidence based design guidance on materials used in and positions of surfaces, furniture and fittings to support microbial safety in use.

WP2 KEEP-Safe.**WP leaders:** AL & JS**Task leaders:** **D2.1)** JS, AL; **D2.2)** NB, RK; **D2.3)** DR, NB, AL; **D2.4)** DH, AL, JS, MBr**Participants:** TUD: AL, NB, DR, DH, SPB; **EMC:** MV, JS, AV, JvK, RK, AU, MBr; **EUR:** MBu

Goals: *To ensure long-term preservation of microbial safety by 1) developing techniques for rapid detection & localization of microorganisms and biofilms and 2) fail-safe, sustainable and robust elimination of microorganisms via novel disinfection methods.*

D2.1-Identify [M01-12]: **Review & analysis (literature, case reports, existing data):** Requirements, methods, key determinants and opportunities for microorganisms (including (RNA) viruses, fungi, bacteria (with their plasmids) both infectious and non-infectious) and biofilm detection and localization in hospital structures and appliances. Retrospective analysis of existing patient data relating to the hospital environment.

D2.2-Investigate [M06-30]: **Reports: A)** Rapid metagenomic detection, including new sample collections and sequencing methods for microorganisms. Assessing the influence of environment, organic material on microbial persistence. **B)** Performance of imaging and sensing modalities for localizing microorganisms and biofilms e.g., hyperspectral imaging, forensic alternative light source imaging and optical coherence tomography) **C)** Limitations and opportunities of disinfection methods, e.g. automated UV-C.

D2.3-Develop [M48-60]: **Prototype:** A. Verified devices, fast metagenomic methods and protocols for microorganisms and biofilm detection and localization on hospital structures and appliances surfaces. B. Molecular assay that discriminates between detection of infectious and non-infectious virus shedding.

D2.4-Educate [M24-60]: Guidance document: Best practices for novel monitoring methods and novel disinfection methods, such as UV-C, of medical devices and appliances, including design guidance for suitability.

WP 3 USE-SAFE.**WP leader:** MBu & MvD**Task leaders:** D3.1 EI & ML / D3.2 MB & ML / D3.3 ML & MvD / D3.4 EvB & SPB**Participants:** TUD: ML, AL, SPB; **EMC:** MvD, EI, EvB, VE; **EUR:** MBu, KA, JvW, CvO, AB, MvW;

Goals: *HCWs, patients, and hospital visitors perform preventive microbial safe behavior when interacting with each other and with the hospital structures and appliances. This safe behavior is established through education, design and nudges.*

D3.1-Identify [M01-12]: **A set of facilitators and barriers** for each user group (i.e. HCWs, patients, and visitors) to perform safe behavior through: 1) reviewing the existing literature; 2) building on the experience and knowledge of the new Erasmus MC building design; 3) creating new knowledge (interviews, focus groups, observations of best practices).

D3.2-Investigate [M6-42]: **Co-created** prototype implementation strategies and behavioral nudges to enhance safe human behavior and compliance to safety (design) guidelines. **Report** insights of how/why experimental designs and prototypes (WP1), influence implementation strategies and nudges contribute to safe behavior. Includes (Group) Interviews.

D3.3-Develop [M30-42]: **A strategy** for implementation and nudges for safe behavior will be developed. **Materials** for innovative education about triggers for (un)safe behavior, which take the diversity between user groups into account. **Report** input for developing human-factors based design guidance documents (D1.3).

D3.4-Educate [M36-60]: **Training modules** tailored to the needs of HCWs, patients, and visitors to deal with the barriers and make optimal use of the facilitators of a microbial safe environment.

WP4 PROJECT MANAGEMENT.

WP leaders: JH, LG

Task leaders: EMC JH, LG

Participants: All

Goals: *Responsible for administrative and financial matters, proper project execution (timely, high-quality results), communication with the Flagship project office, the definition and management of the consortium's IPR aspects and dissemination of project results.*

D4.1-Project coordination (admin and finance) [M01-60]: Collation and submission of yearly Management Reports (Y1-Y5)

D4.2-Dissemination and communication [M01-60]: Scientific publications, Newsletter, website, social media

D4.3-IPR management [M01-12]: IPR-sharing agreement between TUD, EMC and EUR agreed and relevant IPR-sharing document signed.

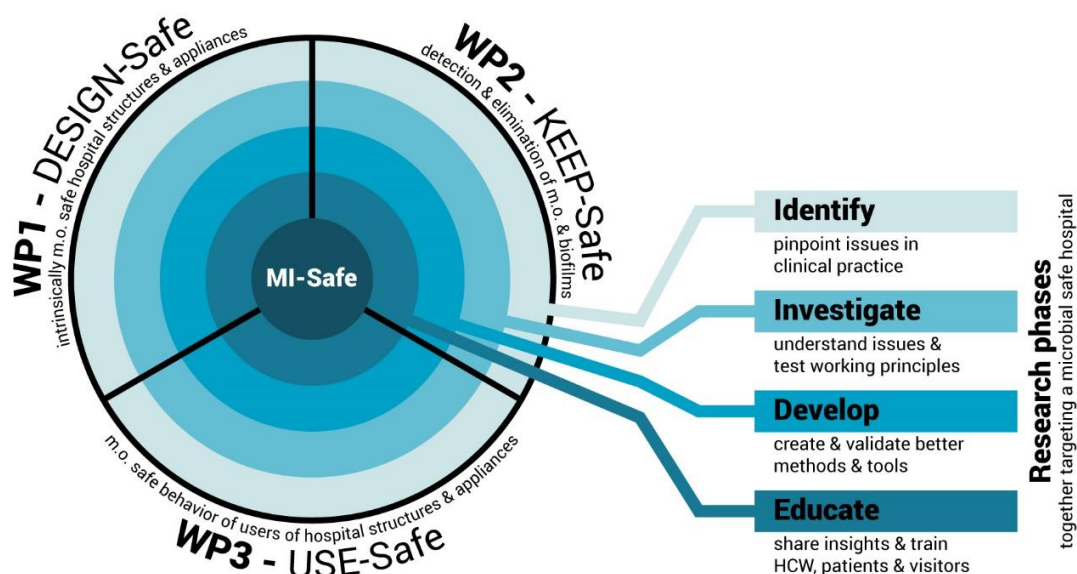


Figure 2. Work Plan for MI-SAFE hospital WPs of WPs

3.d Alignment of the program with relevant national and international research agendas 283 words

MI-SAFE hospital will develop beyond state-of-the-art collaborations according to the requirements and demands of relevant national and international research agendas:

- **EPBD / European Performance of Energy Directive (2022)** - Recently upgraded Net Zero Energy Building requirements
- **WHO (2021)**. "A minimum of two functional, improved sanitation facilities should be available on-site, one for patients and the other for staff; both should be equipped with menstrual hygiene facilities"
- **REHVA and ASHRAE (European and American HVAC research organisations), WHO and CEN/ISO (2020)** – a range of hospital related ventilation and air cleaning guidance documents
- **Dutch Society for Medical Microbiology (2019)**: Research agenda: one of the 10 questions was: What is the best method for disinfection?"
- **EU-JAMRAI - Joint Action Antimicrobial Resistance and Healthcare-Associated Infections (2018)**: "Prioritizing and implementing research innovation".
- **CDC (2015)** : National Action Plan for Combatting Antibiotic-Resistant Infections. "Consider knowledge gaps for prevention of antibiotic-resistant, healthcare-associated infections and identify potential interventions for development, field testing, and eventual widespread implementation".
- **European Union (2010 – 2014)**: Projects on antimicrobial resistance and healthcare-associated infections (**FP7**): PROHIBIT – "Prevention of hospital infections by intervention and training
- **CDC (2009)** National Action Plan To Prevent Health Care-Associated Infections: "There is a need to pay additional attention to projects focusing on environmental issues, including those addressing new technologies and engineering controls, which can potentially affect HAIs."
- **European Study Group of Nosocomial Infections of ESCMID**; defined focus on the impact of the built environment on HAI

High Risk/High Gain

Developing and managing collaborations between EMC, TUD, EUR and multiple private partners necessarily involves a measure of *high risk* (e.g. D2.2). However, the MI-Safe hospital flagship will deliver *high gain* objectives, generating significant reductions in HAI, patient morbidity/mortality and healthcare costs for (inter)national healthcare agencies.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

213 words (Total 1,153 words)

Project members have significantly contributed to academic research having **more than 1,700 publications** among them.

Apart from their contribution to the scientific community, many of these publications have led to **novel mechanisms, inventions of new detection-methods, interviews in newspapers, radio and television, national and international guidelines, and even some exhibitions**, making this way an observable impact on our society.

Engaging with young scientist is also a major part of their work, having taught or supervised over **1,200 BSc and MSc students** so far, both on theses and in other research projects. The topics of these projects include, among others, newly designed hospital buildings, sanitation, infection prevention and contamination-free environment, all of which are closely related to the topic of this proposal.

Additionally, consortium members have a long experience in scientific projects, not only joining as researchers, but also as coordinators, with **more than 80 projects. The total funding acquired for all these projects exceeds 33 million euros.**

Currently, **project members are participating in over 70 currently running projects** and thus, are more than capable of attracting external funding, ensuring the long-term sustainability of the program.

MI-SAFE hospital participants have also been members of **85 boards and commissions**, having received **16 awards**, including **awards for excellence in teaching, for innovation and for best publication.**

4.b Expertise of the consortium members in relation to the aims and objectives of the program 451 words

The consortium is already well experienced in developing and implementing a range of new protocols, technologies and designs in order to improve the health-journey of hospitalized patients. For example, many consortium members are already connected in several (previous) projects relating to the prevention of exogenous HAI. These include:

- HUM (handhygiene)
- MOVE (contamination in the environment)
- REDUCE/DETECT/PREVENT (duodenoscopes)
- SAMPAN (reservoirs of Pseudomonas by JS)
- WELCOME / CHANGE (measuring single room occupancy and infections)
- PE-ONE (measuring the effect of hospital relocation on infection)
- SCIP-COVID-19 (infection prevention and control of SARS-CoV-2)

In more detail, PE-ONE involves LvH (PI), MvD, EI, MV, AV, JS and MBr and is a project instigated by the Executive board of Erasmus MC to study the effect of EMC's recent onsite relocation in maintaining and encouraging healthcare.

From EMC: MV, JS and AV are experienced microbiologists/epidemiologists specializing in HAI, while MvD, EI, MBr, EvB, VE have researchers with interests in HAI. JvK is a clinical virologist and RK and AU are experts in the detection and bioinformatics analysis of microbiota. LvH is currently investigating the influence of design (processes) on HAI outcomes using Erasmus MC as a case study. She is appointed both at TUD/ABE and Erasmus MC, bridging the two faculties in this subject.

From TUD: JD and DR bring in decades of experience in developing medical devices that are, by design, easy to clean while staying affordable, also for low and middle income countries. NvdB works with AL on using hyperspectral imaging for detecting microorganisms and biofilms on medical devices, and on the spread of (potentially microorganism containing) aerosols due to sawing bones and coagulating/cutting soft tissue. AL is an experienced forensic investigator handling medical device related infection outbreaks, regularly hired as independent expert investigator. The TUD faculty of Industrial Design Engineering covers the societal challenges Health, Mobility and Sustainability. ML is a sanitary researcher and designer, pur sang. SPB is lecturer and researcher at the specialization Medesign of IDE faculty. IDE focuses on their Health research on a human-centered approach in relation to technology to understand and positively influence/nudge human behavior through design.

From: EUR (ESHPM) and TUD (3mE) projects are currently investigating how an operating room robot affects individual/team behavior and how advanced designs can improve the implementation of technological innovations. This also relates to the research domain of the Faculty of Industrial Design Engineering (IDE) of TUD.

Finally, although valuable knowledge has been gained by these collaborative projects, it has become obvious that **this experience needs to be translated into a 'laboratory environment' where many different types of appliances, devices and protocols may be tested in a controlled environment, without risk to patients.** This is the main objective and legacy of the MI-SAFE Hospital flagship.

4.c Involvement of partners 263 words

The MI-SAFE hospital flagship is well connected to external partners in private and public sectors, some of which are involved at the start of the planned work programs:

- **Lely-lab** (via Erasmus MC Foundation) is an international family business in the agri-sector, with innovative solutions connected to automation, emissions and business management (AB, PL, AV, MV, JS). [WP1].
- **Pentax Medical** is a company connected to the member MBr of the consortium. It is a globally diversified company with R&D innovation and manufacturing centers in Japan, Europe and the United States. The company focuses on delivering endo-imaging solutions through leading-edge optical technologies. [WP2].

- **Boston Scientific** is dedicated to transforming lives through innovative medical solutions that improve the health of patients around the world, including in the area of infection prevention (MBr). [WP2].
- **UV Smart** is a Delft based company that contributes to a safer healthcare environment with innovative UV-C disinfection machines for the medical sector. UV Smart is a spin-off of DH. [WP1].
- **Inspired Ambitions** is a company active in the area of dynamic particle behavior applied to Operating Rooms in hospitals used to improve the environment and air quality and reduce the risk of infections. The company is an initiative of PL. [WP1].
- **Medexs** is a leading specialist in the design and realization of classified areas such as operating theatres , cleanrooms and laboratories. The company is connected to JL and PL. [WP1].
- **Van Straten Medical** is a Dutch company which is involved in manufacturing, repair and supply of surgical instruments, trays and disposables. This company is connected to DR and AL.

4.d Governance, organization and collaboration strategy of the consortium 226 words

The Project Coordination Team (PCT) will be responsible for the planning, execution and controlling of the project. The following key project positions constitute the PCT:

- **Project Coordinator:** Responsible for a) Administration and scientific coordination activities, b) Implementation of all action plans, c) Maintain contacts with budget and schedule-control representatives, d) Ensure compliance with of quality assurance systems.
- **IPR Coordinator:** Ensure IPR rules establishment and compliance.
- **Dissemination Coordinator:** responsible for social media, public outreach, institutional website, internal newsletter, etc.
- **TUD, EMC, EUR Project Secretariat:** Provide financial control and help desk support in regards to the management of the project consortium and corresponding administrative issues.
- **Ethics Team:** will include key members from TUD, EMC and EUR and be responsible for communication with the Ethics committees of the 3 institutions.
- **Project Management Team:** consists of the representatives of all WP Leaders, who are responsible for convening (physically or remotely) once per month to review and plan project work, identify risks and devise corrective actions.
- **Scientific Advisory Board (SAB):** Advisory will be set up during the first 3 months of the project including stakeholder experts in microbiology, technology and education, as well as user representatives (patients, patient organizations etc.).

Collaborative Strategy:

The MI-SAFE hospital flagship collaborative strategy is outlined in the MI-SAFE and Section 2.b 'Synergy between disciplines' and Section 3.c 'Work Plan'.

Annex I: Budget table (including the external funding table)

Flagship:	MI-SAFE
Flagship acronym:	
Naam Penvoerder:	
Startdatum en looptijd van het project:	
Totaal project budget:	1-7-2022 t/m 30-6-2027

EMC	€ 952,622
TUD	€ 1,241,790
EUR	€ 738,160
Totaal	€ 2,932,572

1.

1.	Loonkosten:							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind		
	Onderzoekorganisaties																					
	EMC (MMI2)	EMC	Research assistant	0.50	01-07-2022	01-07-2027	Schaal 9-1	€ 3,336	€ 152,622	€ 19,077.75	€ 38,156	€ 38,155.50	€ 38,156	€ 19,077.75			€ 152,622	€ 0		€ 152,622.00		
	EMC (Viro)	EMC	Poetdoc	1.00	01-07-2022	01-07-2027	Schaal 10	€ 4,802	€ 439,383	€ 82,000.00	€ 85,000	€ 88,000.00	€ 91,000	€ 93,383.00			€ 439,383	€ 0		€ 439,383.00		
	TUD (Marian)	TUD	PostDoc	0.50	01-07-2022	30-06-2027	PD TUD	€ 4,490	€ 202,050	€ 18,500.00	€ 39,000	€ 39,500.00	€ 41,050	€ 42,500.00	€ 22,500.00		€ 202,050	€ 0		€ 202,050.00		
TUD (new)	TUD	PhD	1.00	01-07-2022	30-06-2026	P TUD	€ 3,122	€ 224,784	€ 25,000.00	€ 52,784	€ 56,000.00	€ 60,000	€ 62,784.00			€ 224,784	€ 0		€ 224,784.00			
TUD (Architecture)	TUD	Postdoc	1.00	01-07-2022	30-06-2024	PD TUD	€ 4,490	€ 161,640		€ 40,410	€ 80,620.00	€ 80,620	€ 40,410			€ 161,640	€ 0		€ 161,640.00			
TUD (Daniel Robel)	TUD	PhD	0.50	01-07-2022	30-06-2026	P TUD	€ 3,122	€ 112,392		€ 20,205.00	€ 40,410	€ 40,410.00	€ 40,410	€ 20,205.00		€ 112,392	€ 0		€ 112,392.00			
TUD (Daan Hooik)	TUD	PhD	0.50	01-07-2022	30-06-2026	P TUD	€ 3,122	€ 224,784		€ 14,049.00	€ 28,098	€ 28,098.00	€ 28,098	€ 14,049.00		€ 224,784	€ 0		€ 224,784.00			
TUDIMMZ	TUD	PhD	1.00	01-07-2022	30-06-2026	P TUD	€ 3,122	€ 224,784		€ 28,098.00	€ 56,196	€ 56,196.00	€ 56,196	€ 28,098.00		€ 224,784	€ 0		€ 224,784.00			
EUR	EUR	Research assistant	0.50	01-07-2023	30-06-2026	Research assistant	€ 4,181	€ 112,887		€ 18,814	€ 37,629.00	€ 37,629	€ 18,814.50			€ 112,887	€ 0		€ 112,886.99			
EUR	EUR	PhD	1.00	01-07-2022	30-06-2026	PhD	€ 3,189	€ 229,617	€ 28,702.11	€ 57,404	€ 57,404.21	€ 57,404	€ 28,702.11			€ 229,617	€ 0		€ 229,616.86			
EUR	EUR	PostDoc	0.80	01-07-2022	30-06-2027	PostDoc	€ 5,342	€ 384,656	€ 38,465.60	€ 76,931	€ 76,931.21	€ 76,931	€ 38,465.60			€ 384,656.04	€ 0		€ 384,656.04			
	Ondernemingen																					
	Onderneming 1						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Onderneming 2						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Onderneming 3						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Onderneming 4						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Overige (private) partners																					
	Naam 1						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Naam 2						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Naam 3						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Naam 4						€ -	€ 0									€ 0	€ 0	€ 0.00			
	Totaal							€ 2,406,455										Totaal loonkosten	€ 2,406,455			

Viro
TUD
TUD
TUD
TUD
EUR
EUR
EUR

2.	Kosten van materialen en hulpmiddelen:			2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
	Partner die de kosten maakt	Materiaal / hulpmiddel	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind		
Vire TUD TUD TUD TUD EUR EUR EUR	Onderzoekorganisaties																		
	EMC	Metagenomics and nanopore sequencing laboratory supplies	€ 200,000	€ 20,000.00		€ 40,000		€ 40,000.00		€ 40,000		€ 40,000.00		€ 20,000		€ 200,000	€ 0	€ 200,000.00	
	EMC		€ 160,817	€ 30,000.00		31000		€ 32,000.00		33000		€ 34,617.00				€ 160,817	€ 0	€ 160,817.00	
	TUD	Development of Sink Prototypes (2x)	€ 7,000					€ 3,500.00		€ 3,500						€ 7,000	€ 0	€ 7,000.00	
	TUD	recruiting and contribution of participants	€ 3,000					€ 1,500.00		€ 1,500						€ 3,000	€ 0	€ 3,000.00	
	TUD	Unforeseen IDE	€ 750		€ 93.75		€ 188		€ 187.50		€ 188		€ 93.75			€ 750	€ 0	€ 750.00	
	TUD	WP 2: Detection materials, prototypes production 22.5k per year	€ 112,500		€ 14,062.50		€ 28,125		€ 28,125.00		€ 28,125		€ 14,062.50			€ 0	€ 112,500	€ 112,500.00	
	EUR	Camara's observation	€ 2,000	€ 500.00		€ 500		€ 500.00		€ 500						€ 2,000	€ 0	€ 2,000.00	
	EUR	Data collection	€ 3,000	€ 1,000.00		€ 1,000		€ 1,000.00								€ 3,000	€ 0	€ 3,000.00	
	EUR	Open access	€ 6,000			€ 2,000				€ 2,000				€ 2,000.00		€ 6,000	€ 0	€ 6,000.00	
	Ondernemingen																		
	Onderneming 1																€ 0	€ 0	€ 0.00
	Onderneming 2																€ 0	€ 0	€ 0.00
	Onderneming 3																€ 0	€ 0	€ 0.00
	Onderneming 4																€ 0	€ 0	€ 0.00
	Overige (private) partners																		
	Naam 1																€ 0	€ 0	€ 0.00
	Naam 2																€ 0	€ 0	€ 0.00
	Naam 3																€ 0	€ 0	€ 0.00
	Naam 4																€ 0	€ 0	€ 0.00
Totaal			€ 494,867													Totaal kosten materialen en hulpmiddelen		€ 494,867	

6.

Ondernemingen																		€ 0	€ 0	€ 0,00
Onderneming 1																		€ 0	€ 0	€ 0,00
Onderneming 2																		€ 0	€ 0	€ 0,00
Onderneming 3																		€ 0	€ 0	€ 0,00
Onderneming 4																		€ 0	€ 0	€ 0,00
Overige (private) partners																		€ 0	€ 0	€ 0,00
Naam 1																		€ 0	€ 0	€ 0,00
Naam 2																		€ 0	€ 0	€ 0,00
Naam 3																		€ 0	€ 0	€ 0,00
Naam 4																		€ 0	€ 0	€ 0,00
Totaal																		Totaal kosten derden	€ 750	€ 750
Totale kosten																			€ 2.932.572	€ 2.932.572

FINANCIERING - Bijdragen per partner per jaar

	In Euro	2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind
Onderzoeksorganisaties																	
1	EMC	€ 52,676	€ 0	€ 194,156	€ 0	€ 198,156	€ 0	€ 202,156	€ 0	€ 187,078	€ 0	€ 20,000	€ 0	€ 854,220	€ 0	€ 854,220	€ 0
2	TUD	€ 0	€ 76,508	€ 0	€ 153,017	€ 0	€ 153,017	€ 0	€ 153,017	€ 0	€ 76,508	€ 0	€ 0	€ 0	€ 612,066	€ 612,066	€ 0
3	Organisatie 3													€ 0	€ 0	€ 0	€ 0
...	Organisatie 4													€ 0	€ 0	€ 0	€ 0
	Totale bijdrage onderzoeksorganisaties	€ 52,676	€ 76,508	€ 194,156	€ 153,017	€ 198,156	€ 153,017	€ 202,156	€ 153,017	€ 187,078	€ 76,508	€ 20,000	€ 0	€ 854,220	€ 612,066	€ 1,466,286	€ 0
Ondernemingen																	
4	Onderneming 1													€ 0	€ 0	€ 0	€ 0
5	Onderneming 2													€ 0	€ 0	€ 0	€ 0
6	Onderneming 3													€ 0	€ 0	€ 0	€ 0
...	Onderneming 4													€ 0	€ 0	€ 0	€ 0
	Totale bijdrage ondernemingen	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Overige (private) partners																	
7	Naam 1													€ 0	€ 0	€ 0	€ 0
8	Naam 2													€ 0	€ 0	€ 0	€ 0
9	Naam 3													€ 0	€ 0	€ 0	€ 0
...	Naam 4													€ 0	€ 0	€ 0	€ 0
	Totale overige bijdrage	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
PPS-toeslag / subsidies / sponsors																	
10	PPS-toeslag	€ 146,629		€ 293,257		€ 293,257		€ 293,257		€ 293,257		€ 146,629		€ 1,466,286	€ 0	€ 1,466,286	€ 0
11	<naam>													€ 0	€ 0	€ 0	€ 0
12	<naam>													€ 0	€ 0	€ 0	€ 0
...														€ 0	€ 0	€ 0	€ 0
	Totaal subsidies	€ 146,629		€ 293,257		€ 293,257		€ 293,257		€ 293,257		€ 146,629		€ 1,466,286	€ 0	€ 1,466,286	€ 0
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	Totaal Incl. PPS-toeslag			
	Totaal budget per jaar	€ 199,304	€ 76,508	€ 487,413	€ 153,017	€ 491,413	€ 153,017	€ 495,413	€ 153,017	€ 480,335	€ 76,508	€ 165,629	€ 0	€ 2.932.572			
														Funding gap			
														€ 0			
	Totaal budget			€ 640,429		€ 644,429		€ 648,429		€ 723,472				€ 1.933,288			

Table 1. Budget obtained from the industry/ external funders
(Letter of Commitment & Letter of Intent)

no	Company	Contact person	Type	Budget
1	Erasmus MC Foundation	L.M.L. Tuijt	LoC	345.390,00 euros (in cash)
2	Inspired Ambitions	Peter Luscuere	LoC	18.309,00 euros (in kind)
3	UV Smart Technologies B.V.	Daan Hoek	Lol	250.000,00 euros (in cash) 75.000,00 euros (in kind)
4	Medexs	Jos Lans	LoC	125.000,00 euros (in kind)
5	Van Straten Medical	Daniel Robertson	Lol	161.640,00 euros (in cash)
	TOTAL			757.030,00 euros (in cash) <u>218.309,00 euros (in kind)</u> 975.339,00 euros

Table 2. Budget obtained from public sector
(externally funded projects)

no	PI	Project		Budget
1	Dr. Martina Buljac	ZonMW: Safe Transfer NICU	EUR	82.564,00 euros
2	Dr. Martina Buljac	ZonMW: Met verve: Verpleegkundigen in the lead voor kwaliteit en veiligheid!	EUR	74.594,00 euros
3	Dr. Martina Buljac	"Treant 2021-2023"	EUR	66.250,00 euros
	Dr. Margreet Vos	HUM3	EMC (MMIZ)	48.093,00 euros
5	Dr. Margreet Vos/ Dr. Anne Voor in t'holt	SCIP COVID19	EMC (MMIZ)	282.375,00 euros
6	Dr. Juliette Severin	SAMPAN	EMC (MMIZ)	150.000,00 euros
7	Dr. Marco Bruno/Dr. M.Vos	Pentax Medical	EMC (GE/MMIZ)	50.000,00 euros
8	Dr. Marco Bruno/Dr. M.Vos	Boston Scientific, Natick, USA	EMC (GE/MMIZ)	100.000,00 euros
9	Dr. Jeroen Kampen	NIH grant	EMC (Virology)	150.000,00 euros
10	Dr. Jeroen Kampen	ROSETTA project	EMC (Virology)	200.000,00 euros
11	Dr. Jeroen Kampen	SCIP COVID19 project	EMC (Virology)	25.000,00 euros
	TOTAL			1.228.876,00 euros

**Annex II: Letters of intent, Letters of commitment & Letters project
awarded**

Erasmus MC
t.a.v. prof. dr. M.C. Vos
Afdeling Medische Microbiologie en Infectieziekten (MMIZ)
Dr. Molewaterplein 40
3015 GD Rotterdam
m.vos@erasmusmc.nl

Doorkiesnummer: 010 – 70 34 802

Intern postadres: Gk-625

E-mail: l.tuijt@erasmusmc.nl

Date: February 17th 2022

Call for proposals: Call for Flagship Programs –
Convergence Alliance

Letter of commitment

Dear professor Vos,

It is with great pleasure and pride that I can confirm that the Erasmus MC Foundation, represented by Leonore Tuijt, is committed to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled *“From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship”*.

In terms of contribution to the costs of the project, the Erasmus MC Foundation will provide a cash contribution of 345.390 euros.

Postadres

Postbus 2040
3000 CA Rotterdam, NL

Bezoekadres

Burg 'sJacobplein 51
3015 CA Rotterdam, NL

Contact

foundation@erasmusmc.nl
010 – 70 34 802
www.erasmusmcfoundation.nl

Signature:



(L.M.L. Tuijt, Legal representative)

Date:

February 17, 2022

**LETTER OF COMMITMENT FROM PARTNERS/ORGANISATIONS
PARTICIPATING IN, AND/OR PROVIDING A (CASH/OTHER)
CONTRIBUTION TO THE PROPOSED PROJECT**

Call for proposals: Call for Flagship Programs – Convergence Alliance

Letter of commitment

This letter is to confirm that *Medexs*, represented by *Jos Lans* is committed to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled “*From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship*”.

In terms of contribution to the costs/other of the project, (name of partner) will provide a cash contribution of 0 euros, and/or in kind contribution of 125,000.00 euros.

Signature: J.L.A. Lans (Legal representative)

Date: 11.03.2022

Inspired Ambitions

Nieuwkoopseweg 26
2641 PB Pijnacker

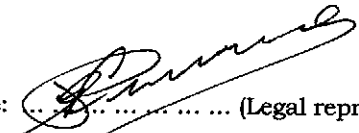
telephone :+31 15 369 68 50
SNS-bank :92.55.11.978
IBAN :NL81 SNSB 0925 5119 78
BIC :SNSBNL2A
Trade Register :27 17 56 46
VAT-number :8076.24.901.B01

Pijnacker: 21 02 2022
Attribute:
Regarding: Letter of commitment

Dear Sir/Madam,

This letter is to confirm that Inspired Ambitions, represented by Prof.ir. Peter Luscuere is committed to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled "From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship".

In terms of contribution to the project, Prof.ir. Peter Luscuere will provide in kind contribution of 18.309 euros, being 50% of 0,1 fte time spent over 2 years @ salary scale 17.

Signature:  (Legal representative)

Date:: 08 03 2022

Sincerely yours,

Inspired Ambitions
Prof.ir. P.G. Luscuere

VAN STRATEN MEDICAL

'Providing Value To Life'

To: Prof. Dr. Margreet Vos

MMIZ Department, Erasmus MC

Doctor Molewaterplein 40, 3015 GD Rotterdam

10.03.2022

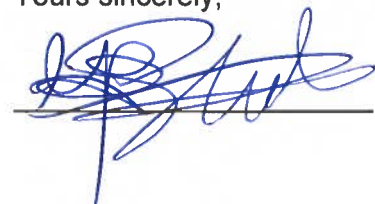
LETTER OF INTENT
for the
MI-SAFE-Hospital PROJECT

Dear Prof. Dr. Vos,

I, Bart van Straten, in my capacity of member of the Board of Directors at Van Straten Medical hereby confirm that Van Straten Medical intends to contribute to the project entitled "From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship", as applied for by the main applicant, Prof. Dr. Margreet Vos, at MMIZ Erasmus MC.

Van Straten Medical intends to contribute € 161.640 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Yours sincerely,



Name: Bart van Straten

Position: Board of Directors

Date: 11-03-2022



UVsmart

Call for proposals

Call for Flagship Programs – Convergence Alliance

Letter of Intent

This letter is to confirm that *UV Smart Technologies B.V.*, represented by Daan Hoek is intended to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled *"From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship"*.

In terms of contribution to the costs of the project, UV Smart Technologies B.V. intends to provide a cash contribution of EUR 250.000 over 8 years, and another type of contribution by supplying instruments worth EUR 75.000 in commercial value for research.

Signature:


Daan Hoek

Date:

24-feb-2022

 **UVsmart**
UV Smart Technologies B.V.
www.uvsmart.nl | info@uvsmart.nl
CoC 83929118



Patrijsweg 74
2289 EX, Rijswijk



+31 (0)85 060 98 00



info@uvsmart.nl

www.uvsmart.nl

**LETTER OF INTENT FROM PARTNERS/ORGANISATIONS
PARTICIPATING IN, AND/OR PROVIDING A (CASH/OTHER)
CONTRIBUTION TO THE PROPOSED PROJECT**

Call for proposals: Call for Flagship Programs – Convergence Alliance

Letter of intent

Van Dorp B.V., represented by Henk Willem van Dorp has an intention to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled "From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship".

Signature:  H.W. van Dorp

Date: 15 March 2022

Universitair Medisch Centrum Utrecht
Wilhelmina Kinderziekenhuis
Neonatologie
De heer Drs. E.J.H.S. Flipse
Postbus 85090
3508 AB UTRECHT

Laan van Nieuw Oost-Indië 334
2593 CE Den Haag
Postbus 93245
2509 AE Den Haag
Telefoon 070 349 51 11
Fax 070 349 51 00
www.zonmw.nl
info@zonmw.nl

Dossiernummer

80-87300-98-032

Ons kenmerk

2021/14285/ZONMW

Datum

21 april 2021

Onderwerp

Subsidiebesluit over dossiernummer 10130022010008

Contactpersoon

Simone Schellekens
Telefoon 070 349 54 66
safety@zonmw.nl

Geachte heer Flipse,

Op 2 februari 2021 heeft ZonMw uw subsidieaanvraag 'SAFE-TRANSFER NICU' ontvangen. In deze brief vindt u ons besluit en leest u wat u van ons kunt verwachten. Daarnaast leest u wat wij van u verwachten.

Besluit

ZonMw besluit tot honorering van uw subsidieaanvraag. U ontvangt een subsidie van maximaal € 134.200,- voor de duur van maximaal 12 maanden. Hiervan is € 3.500,- gereserveerd voor open access publicatie kosten. Dit bedrag is inclusief eventueel verschuldigde btw.

Onderbouwing besluit

De beoordelingscommissie Safety II en veiligheidsergonomie heeft positief geadviseerd over uw aanvraag. Daarbij is de aanvraag zelf, het referentenoordeel, het wederhoor, de presentatie tijdens het wederhoor en de discussie met de commissie meegewogen in het oordeel. Het advies van de commissie is naar oordeel van ZonMw op zorgvuldige wijze tot stand gekomen. ZonMw heeft dit advies overgenomen en ten grondslag gelegd aan haar beslissing.

Relevantie

De commissie heeft het volgende relevantieoordeel over uw subsidieaanvraag gegeven: **zeer relevant**.

Het relevantieoordeel is als volgt onderbouwd:

- Het onderzoek adresseert het veilig en tijdig overplaatsen van patiënten van de Neonatale Intensive Care Unit (NICU) naar andere ziekenhuizen. Daarbij wordt onderzocht welke aspecten essentieel zijn voor een veilige, maar ook tijdige overplaatsing en hoe deze aspecten geborgd kunnen worden in een proactieve, wendbare en lerende organisatie. De commissie is van mening dat uw onderzoek zal bijdragen aan een betere patiëntveiligheid en inzichten zal opleveren voor het Safety II onderzoeksdomein. Dit maakt uw onderzoek zeer relevant voor dit programma.

- De verwachte uitkomsten zijn zeer relevant voor het verbeteren van patiëntveiligheid op de NICU en geven inzichten in Safety II elementen. De uitkomsten zijn in de ogen van de commissie ook opschaalbaar en implementeerbaar voor andere IC-afdelingen.
- Het onderzoek heeft duidelijk aandacht voor de complexiteit van het zorgsysteem, getuige de uitgebreide beschrijving van de complexiteit van het NICU en de overdracht zelf. De commissie acht dit denkwerk vooraf aan het onderzoek, essentieel voor het realiseren van relevante uitkomsten.

Kwaliteit

De commissie heeft het volgende kwaliteitsoordeel over uw subsidieaanvraag gegeven: **goed**.

Het kwaliteitsoordeel is als volgt onderbouwd:

- De doelstelling is helder en concreet geformuleerd, het plan van aanpak en de gekozen methoden zijn goed. Dit voorstel is van hoge kwaliteit en is naar mening van de commissie zeer kansrijk.
- Alle relevante expertise is vertegenwoordigd in de projectgroep en relevante stakeholders zijn betrokken. Zowel kennis van de werkvloer als meta-kennis op het gebied van actieonderzoek is betrokken. De commissie vindt het sterk dat de patiëntjes actief worden vertegenwoordigd door de ouders.
- De commissie acht uw onderzoek zeer haalbaar. Hier heeft de toelichting in het wederhoor aan bijgedragen. Ook valt op dat de onderzoekers steeds realistisch en bedacht zijn op 'engagement' van werkvloer en de uitdagingen. Dit komt de haalbaarheid ten goede.
- De aanvraag is duidelijk over de invulling en fasen van het actieonderzoek. De cyclische aard van actie onderzoek is adequaat beschreven.

Financiën

Het subsidiebedrag is conform uw ingediende begroting. Van de verschillende posten dient bij de eindafrekening een specificatie te worden aangeleverd.

Specifieke voorwaarden

De commissie is van mening dat bij de monitoring van het project enkele onderwerpen specifieke aandacht behoeven. In het eerstvolgende voortgangsverslag dient u dan ook nader in te gaan op de volgende aanvullende voorwaarden:

- Zoals in het gesprek is benoemd vraagt de commissie u om studies te raadplegen die het borgen van veiligheid in een stroomversnelling duiden en hierover in het voortgangsverslag te rapporteren.

Gehonoreerde projecten Safety II en veiligheidsergonomie

Naast uw project zijn 10 andere projecten gehonoreerd binnen dit programma. Doel is om de belangrijkste bevindingen, de geleerde lessen, te benoemen en zo de werkzaamheid en werkbaarheid van Safety II verder in de praktijk te verankeren. Er zal een overkoepelende review worden gemaakt. ZonMw beraadt zich nog over de vormgeving van een bijeenkomst of interview om deze kennis op te halen. U wordt hier nader over geïnformeerd.

Start project

Het project dient uiterlijk **6 maanden** na de verzenddatum van deze brief te beginnen. Gaat het project later van start, dan vervalt de honorering van uw aanvraag. Hiervan kan alleen in zeer bijzondere gevallen worden afgeweken.

Regelgeving en algemene voorwaarden

Op dit besluit is de volgende regelgeving van toepassing:

- De Algemene Wet Bestuursrecht
- De Algemene subsidiebepalingen ZonMw. U kunt deze nalezen op: www.zonmw.nl/subsidievoorwaarden
- Procedure voor aanvragers. U kunt deze nalezen op: <https://www.zonmw.nl/nl/subsidies/hoe-werkt-subsidie-aanvragen>. Bij punt 17 t/m 21 leest u wat er gedurende het project van u wordt verwacht.

Betaling van de subsidie

De subsidie zal via voorschotbedragen aan u worden overgemaakt. De betalingen zijn afhankelijk van de ontvangst en goedkeuring van voortgangsverslagen en de eindverantwoording van uw project. U ontvangt het eerste voorschot als wij het ingevulde meldingsformulier hebben ontvangen.

Voortgangsverslag en projectleidersbijeenkomst

Om uw project te blijven monitoren zal eind december een voortgangsverslag worden opgevraagd. Om verdieping en discussie mogelijk te maken zullen wij in het verlengde van het voortgangsverslag een projectleidersbijeenkomst organiseren in januari/februari met (een delegatie van) de commissie. Hier kunnen mogelijk verbindingen tussen de verschillende projecten worden gelegd of geleerd worden van elkaar. ZonMw zal contact met u opnemen over de nadere invulling hiervan.

Eindverantwoording

Binnen 13 weken na afloop van uw project dient u een eindverantwoording in. Dit is een inhoudelijk eindverslag en de financiële eindverantwoording. De financiële verantwoording dient een specificatie van de verschillende posten te bevatten. Na ontvangst en goedkeuring van de eindverantwoording vindt de definitieve subsidievaststelling en afrekening plaats.

Tussentijdse wijzigingen

Laat het ons weten als er tussentijds iets in uw plan of uw organisatie verandert. Verandering in de opzet, planning, begroting of organisatie kunnen gevolgen hebben voor uw subsidie. Pas na goedkeuring door ZonMw zijn eventuele wijzigingen toegestaan.

Wat verwachten wij nu van u?

ZonMw kan u een voorschot voor de eerste periode van uw project verstrekken. Hiervoor dient u binnen 4 weken na de verzenddatum van deze brief:

- Bijgevoegd meldingsformulier ingevuld retour te sturen.
- Een Nederlandse publiekssamenvatting in ProjectNet in te vullen (maximaal 1000 tekens, inclusief spaties). ZonMw publiceert alle gehonoreerde projecten op haar website met een Nederlandse samenvatting in eenvoudige taal. Deze is bedoeld voor een breed geïnteresseerd publiek met verschillende achtergronden. Zie de schrijfwijzer op <http://www.zonmw.nl/nl/over-zonmw/logo-huisstijl>.

Vragen

Heeft u nog vragen? Neemt u dan gerust contact op met Simone Schellekens of Paula du Pont. Dat kan via e-mail: safety@zonmw.nl of via telefoonnummer: 070 349 54 66.

Houd u uw dossiernummer bij de hand zodat wij u snel kunnen helpen. Het oorspronkelijke nummer van uw aanvraag vervalt. Er geldt het nieuwe projectnummer: 10130022010008.

Bezwaar of klacht?

Bent u het niet eens bent met dit besluit? U kunt tot 6 weken na de verzenddatum van deze brief bezwaar maken. Liever hebben wij dat u eerst contact met ons opneemt. Wij beantwoorden graag uw vragen over deze brief. Blijft u het oneens? U kunt een bezwaarschrift sturen naar het bestuur van ZonMw, t.a.v. Commissie Bezwaarschriften ZonMw, Postbus 93 245, 2509 AE Den Haag.

Als u ontevreden bent over de wijze waarop ZonMw uw aanvraag heeft behandeld, kunt u ons dit laten weten of een klacht indienen. Meer informatie vindt u op de ZonMw website via www.zonmw.nl/signalerenklagenbezwaarmaken.

Ik feliciteer u met de honorering van uw subsidieaanvraag en wens u succes bij de uitvoering van uw project!

Met vriendelijke groet,
namens het bestuur,



Hannie Bonink
directeur programma's

**Bijlage(n)**

Meldingsformulier start project

Kopie

Prof. dr. M. Benders
Drs. K.A. de Bijl

Erasmus Universiteit Rotterdam
Erasmus School of Health
Policy and Management
De heer Prof. dr. W.B.F. Brouwer
Postbus 1738
3000 DR ROTTERDAM

Laan van Nieuw Oost-Indië 334
2593 CE Den Haag
Postbus 93245
2509 AE Den Haag
Telefoon 070 349 51 11
Fax 070 349 51 00
www.zonmw.nl
info@zonmw.nl

Dossiernummer
80-87300-98-037
Ons kenmerk
2021/14289/ZONMW

Datum
21 april 2021

Onderwerp
Subsidiebesluit over dossiernummer 10130022010009

Contactpersoon
Simone Schellekens
Telefoon 070 349 54 66
safety@zonmw.nl

Geachte heer Brouwer,

Op 2 februari 2021 heeft ZonMw uw subsidieaanvraag '*Met VERVE!: VERpleegkundigen in the lead voor kwaliteit- en patiëntVEiligheid*' ontvangen. In deze brief vindt u ons besluit en leest u wat u van ons kunt verwachten. Daarnaast leest u wat wij van u verwachten.

Besluit

ZonMw besluit tot honorering van uw subsidieaanvraag. U ontvangt een subsidie van maximaal € 140.985,- voor de duur van maximaal 12 maanden. Dit bedrag is inclusief eventueel verschuldigde btw.

Onderbouwing besluit

De beoordelingscommissie Safety II en veiligheidsergonomie heeft positief geadviseerd over uw aanvraag. Daarbij is de aanvraag zelf, het referentenoordeel, het wederhoor, de presentatie tijdens het wederhoor en de discussie met de commissie meegewogen in het oordeel. Het advies van de commissie is naar oordeel van ZonMw op zorgvuldige wijze tot stand gekomen. ZonMw heeft dit advies overgenomen en ten grondslag gelegd aan haar beslissing.

Relevantie

De commissie heeft het volgende relevantieoordeel over uw subsidieaanvraag gegeven: **zeer relevant**.

Het relevantieoordeel is als volgt onderbouwd:

- Het onderzoek adresseert een belangrijk en actueel thema, namelijk de rol van de verpleegkundigen in kwaliteits- en veiligheidsmanagement. De commissie is van mening dat dit onderzoek zeer relevante handvaten kan bieden voor het vergroten van de rol van de verpleegkundigen op patiëntveiligheid. Het onderzoek heeft aandacht voor de rol van gedrag en lerende cultuur in de Safety II benadering. Dit blijkt onder andere uit de koppeling van klinisch leiderschap aan de leertheorieën over 1e/2e/3e orde leren en de aandacht voor de cruciale factor die 'faciliteren' en 'ondersteunen' in de praktijk spelen. Hiermee draagt het onderzoek bij aan de gedachtevorming van de werkzaamheid van Safety II. Dit maakt uw onderzoek zeer relevant voor dit programma.

- Het onderzoek geeft aanknopingspunten voor het vormgeven en vergroten van identiteit en zeggenschap door verpleegkundigen. Dit is iets wat in de ogen van de commissie cruciaal is voor veilige en goede zorg. Tevens zijn de uitkomsten relevant voor ander ziekenhuizen en landelijke programma's.

Kwaliteit

De commissie heeft het volgende kwaliteitsoordeel over uw subsidieaanvraag gegeven: **voldoende**.

Het kwaliteitsoordeel is als volgt onderbouwd:

- De doelstelling en het plan van aanpak zijn voldoende helder en concreet geformuleerd. Het onderzoek kan aan kracht winnen door Safety II nog explicieter te maken voor de algehele zorg(verbeterings)praktijk. Het wederhoor heeft de commissie voldoende overtuigd dat dit onderzoek bijdraagt aan de gedachtevorming van Safety II. De commissie ziet dit graag terug in het voortgangsverslag, zie onderstaand.
- Projectgroep is van goede kwaliteit. Het team is een mooie vertegenwoordiging van relevante stakeholders.
- De commissie acht dit onderzoek ambitieus met een strakke tijdsplanning, waarbij er nog een aantal onduidelijkheden bestaan over welke instrumenten er gebruikt zullen worden. De toelichting in het wederhoor heeft de commissie niet geheel overtuigd van de haalbaarheid van dit onderzoek en blijft dit dus als risico beschouwen.

Financiën

Het subsidiebedrag is conform uw ingediende begroting. Van de verschillende posten dient bij de eindafrekening een specificatie te worden aangeleverd.

Specifieke voorwaarden

De commissie is van mening dat bij de monitoring van het project enkele onderwerpen specifieke aandacht behoeven. In het eerstvolgende voortgangsverslag dient u dan ook nader in te gaan op de volgende aanvullende voorwaarden:

- De commissie vraagt u in de voortgangsrapportage meer expliciet in te gaan op de kernelementen van Safety II om deze inzichtelijker te maken voor de verpleegkundigen.
- De commissie raadt u aan om ander management voldoende te betrekken. De commissie voorziet het risico dat de beslissingen van de verpleegkundigen elders teniet worden gedaan.

Gehonoreerde projecten Safety II en veiligheidsergonomie

Naast uw project zijn 10 andere projecten gehonoreerd binnen dit programma. Doel is om de belangrijkste bevindingen, de geleerde lessen, te benoemen en zo de werkzaamheid en werkbaarheid van Safety II verder in de praktijk te verankeren. Er zal een overkoepelende review worden gemaakt. ZonMw beraadt zich nog over de vormgeving van een bijeenkomst of interview om deze kennis op te halen. U wordt hier nader over geïnformeerd.

Start project

Het project dient uiterlijk **6 maanden** na de verzenddatum van deze brief te beginnen. Gaat het project later van start, dan vervalt de honorering van uw aanvraag. Hiervan kan alleen in zeer bijzondere gevallen worden afgeweken.

Regelgeving en algemene voorwaarden

Op dit besluit is de volgende regelgeving van toepassing:

- De Algemene Wet Bestuursrecht
- De Algemene subsidiebepalingen ZonMw. U kunt deze nalezen op: www.zonmw.nl/subsidievoorwaarden
- Procedure voor aanvragers. U kunt deze nalezen op: <https://www.zonmw.nl/nl/subsidies/hoe-werkt-subsidie-aanvragen>. Bij punt 17 t/m 21 leest u wat er gedurende het project van u wordt verwacht.

Datum
21 april 2021**Betaling van de subsidie**

De subsidie zal via voorschotbedragen aan u worden overgemaakt. De betalingen zijn afhankelijk van de ontvangst en goedkeuring van voortgangsverslagen en de eindverantwoording van uw project. U ontvangt het eerste voorschot als wij het ingevulde meldingsformulier hebben ontvangen.

Voortgangsverslag en projectleidersbijeenkomst

Om uw project te blijven monitoren zal eind december een voortgangsverslag worden opgevraagd. Om verdieping en discussie mogelijk te maken zullen wij in het verlengde van het voortgangsverslag een projectleidersbijeenkomst organiseren in januari/februari met (een delegatie van) de commissie. Hier kunnen mogelijk verbindingen tussen de verschillende projecten worden gelegd of geleerd worden van elkaar. ZonMw zal contact met u opnemen over de nadere invulling hiervan.

Eindverantwoording

Binnen 13 weken na afloop van uw project dient u een eindverantwoording in. Dit is een inhoudelijk eindverslag en de financiële eindverantwoording. De financiële verantwoording dient een specificatie van de verschillende posten te bevatten. Na ontvangst en goedkeuring van de eindverantwoording vindt de definitieve subsidievestiging en afrekening plaats.

Tussentijdse wijzigingen

Laat het ons weten als er tussentijds iets in uw plan of uw organisatie verandert. Verandering in de opzet, planning, begroting of organisatie kunnen gevolgen hebben voor uw subsidie. Pas na goedkeuring door ZonMw zijn eventuele wijzigingen toegestaan.

Wat verwachten wij nu van u?

ZonMw kan u een voorschot voor de eerste periode van uw project verstrekken. Hiervoor dient u binnen 4 weken na de verzenddatum van deze brief:

- Bijgevoegd meldingsformulier ingevuld retour te sturen.
- Een Nederlandse publiekssamenvatting in ProjectNet in te vullen (maximaal 1000 tekens, inclusief spaties). ZonMw publiceert alle gehonoreerde projecten op haar website met een Nederlandse samenvatting in eenvoudige taal. Deze is bedoeld voor een breed geïnteresseerd publiek met verschillende achtergronden. Zie de schrijfwijzer op <http://www.zonmw.nl/nl/over-zonmw/logo-huisstijl>.

Vragen

Heeft u nog vragen? Neemt u dan gerust contact op met Simone Schellekens of Paula du Pont. Dat kan via e-mail: safety@zonmw.nl of via telefoonnummer: 070 349 54 66.

Houd u uw dossiernummer bij de hand zodat wij u snel kunnen helpen. Het oorspronkelijke nummer van uw aanvraag vervalst. Er geldt het nieuwe projectnummer: 10130022010009.

Bezwaar of klacht?

Bent u het niet eens bent met dit besluit? U kunt tot 6 weken na de verzenddatum van deze brief bezwaar maken. Liever hebben wij dat u eerst contact met ons opneemt. Wij beantwoorden graag uw vragen over deze brief. Blijft u het oneens? U kunt een bezwaarschrift sturen naar het bestuur van ZonMw, t.a.v. Commissie Bezwaarschriften ZonMw, Postbus 93 245, 2509 AE Den Haag.

Datum
21 april 2021

Als u ontevreden bent over de wijze waarop ZonMw uw aanvraag heeft behandeld, kunt u ons dit laten weten of een klacht indienen. Meer informatie vindt u op de ZonMw website via www.zonmw.nl/signalerenklagenbezwaarmaken.

Ik feliciteer u met de honorering van uw subsidieaanvraag en wens u succes bij de uitvoering van uw project!

Met vriendelijke groet,
namens het bestuur,


Hanne Bonink
directeur programma's
**Bijlage(n)**

Meldingsformulier start project

Kopie

Dr. C. van Oostveen

Treant Zorggroep
T.a.v. mevrouw Nelissen
Raad van Bestuur
Postbus 2006
7900 BA HOOGEVEEN

Datum
25 maart 2021

Onderwerp
Offerte

Ons kenmerk
KA/mw

Uw kenmerk

Pagina
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Bijlage

Afdeling
HSMO

Bezoekadres
Burgemeester Oudlaan 50
Bayle Building

Postadres
Postbus 1738
3000 DR Rotterdam

T +31 10 408 8855
E ahaus@eshpm.eur.nl
W www.eur.nl/eshpm

Geachte mevrouw Nelissen,

Hartelijk dank voor uw verzoek om een offerte op te stellen voor een eerste project dat we samen met Treant Zorggroep mogen gaan uitvoeren. Dat verzoek was de uitkomst van een inspirerende bijeenkomst met van Treant-zijde Arjan Middel, Naomi de Boer, Annie Dorenbos en uzelf, en van ESHPM-zijde Martina Buljac (projectleider), Sylvia Elkhuisen, Sandra Sülz en ondergetekende op 7 januari jl.

De offerte heeft enige vertraging opgelopen door overleg over de leveringsvoorwaarden. Na overleg tussen de heer Salet van EUR/Legal en Middel van Treant met het verzoek om de Algemene Voorwaarden Contractonderzoek van EUR-leveringsvoorwaarden te accepteren, omdat deze het beste passen bij het doen van wetenschappelijk onderzoek en zo voorzien in relevante afspraken over wetenschappelijke integriteit en publicaties.

Inmiddels is gebleken dat van de zijde van Treant de voorwaarden van Erasmus mogen prevaleren onder de voorwaarde dat we de tekst uit de offerte m.b.t. publicaties, in afwijking van artikel 8 EUR, als volgt interpreteren:

- ESHPM en Treant zullen alle mogelijke publicaties met elkaar afstemmen.
- ESHPM zal de publicatie tijdig naar Treant sturen, zodat voldoende tijd bestaat om bezwaren kenbaar te maken en te motiveren (in elk geval 30 dagen).
- Tussen partijen wordt gestreefd om binnen 60 dagen een oplossing te bereiken, waarin de 30 dagen reactietijd van Treant zijn meegenomen.

We beogen een samenwerking over langere termijn en spraken over een projectplan voor fase 1 en 2 van de samenwerking. De resultaten van fase 1 worden door Treant Zorggroep gebruikt voor een nieuw meerjarenbeleidsplan. De resultaten van fase 2 richten zich op een nog te benoemen veranderproces.

Volgens het projectplan (zie bijlage) zijn de volgende activiteiten afgesproken:

Fase 1: Omgeving in kaart brengen met een DESTEP-analyse, het 7S-model en een SWOT-analyse

Fase 2: Veranderproces met interventies opzetten en volgen

In fase 1 werken we nadrukkelijk samen om de gevraagde deliverables (omschreven deelproducten) te kunnen leveren. Fase 2 heeft het karakter van de evaluatie van een veranderproces bijvoorbeeld rond een concreet

onderwerp in het operations management (inclusief HR-aspecten) van de ziekenhuizen of op het raakvlak van cure en care. Dit onderwerp moet nog worden vastgesteld, belangrijk is vervolgens dat er voor zo'n veranderproces gewerkt moet worden aan draagvlak binnen de organisatie. Na afloop van fase 2 is er een evaluatie van de samenwerking en worden eventueel nieuwe afspraken gemaakt over het vervolg.

Met betrekking tot de data stellen we het volgende voor. De opdrachtgever (Treant Zorggroep) is en blijft eigenaar van de data die beschikbaar wordt gesteld in fase 1, alsook van de data die in deze stappen wordt gegenereerd. De opdrachtnemer (Erasmus School of Health Policy & Management) wordt eigenaar van het tijdens het onderzoek in fase 2 verzamelde en ontwikkelde materiaal, met inbegrip van de in het kader van het project bewerkte databestanden, (ingevulde) vragenlijsten, gegenereerde rapportages en dergelijke. De Treant Zorggroep heeft het recht om de rapportage en de daartoe gerealiseerde onderzoeksresultaten te gebruiken, dat recht is niet exclusief.

We verwachten dat in de loop van de samenwerking (1) vakpublicaties voor in Nederland bekende media zoals bijvoorbeeld Medisch Contact, Zorgvisie, Qruux en (2) wetenschappelijke publicaties voor internationale peer reviewed journals tot stand worden gebracht. Bij vakpublicaties bepaalt de Treant Zorggroep de auteursvolgorde, bij de wetenschappelijke publicaties Erasmus School of Health Policy & Management. Bij het bepalen van de auteursvolgorde speelt verder natuurlijk een belangrijke rol de inbreng van de auteurs. Over publicaties zullen Treant Zorggroep en ESHPM tijdens het project afstemmen. In het geval van de wetenschappelijke publicatie anonimiseren we data conform het datamanagement plan.

ESHPM zal op drie momenten factureren, namelijk eerst na afloop van fase 1, vervolgens eind 2021 over het deel van fase 2 dat in 2021 is geleverd, en tenslotte na afronding van fase 2. Ieder half jaar wordt geëvalueerd en gezien of nieuwe onderwerpen kunnen worden opgepakt. Voor werkzaamheden die buiten de scope van fase 1 en 2 vallen zullen we een nieuwe maatwerkofferte opstellen.

Over fase 1 zullen we 36.250 euro exclusief BTW in rekening brengen. Over fase 2 15.000 euro, exclusief BTW. De werkelijke inzet voor fase 2 is geraamd op 30.000 euro; vanwege ons belang dat er in het kader van de samenwerking wordt gewerkt aan een of meerdere wetenschappelijke publicaties dragen we in fase 2 50 % bij uit eigen middelen bij (de zogenaamde in kind-bijdrage).

Het projectteam bestaat op dit moment uit Martina Buljac (projectleider), Sylvia Elkhuisen, Sandra Sülz, Annelie Oortwijn en ondergetekende. Afhankelijk van de keuze van een onderzoeksonderwerp voor fase 2 kan het projectteam worden gewijzigd om de collega's met de juiste specifieke kennis te kunnen inzetten.

We verzoeken u vriendelijk om, als u akkoord bent met deze offerte over fase 1 en 2 in onze samenwerking, een kopie voor akkoord getekend en van datum voorzien aan ons te retourneren.

In de afgelopen maanden is duidelijk geworden dat we uitkijken naar een hele mooie samenwerking, die naar wij hopen de organisatie-ontwikkeling van Treant Zorggroep ondersteunt maar ook leidt tot nieuwe wetenschappelijke kennis en inzichten. We verheugen ons op de uitdagingen die de samenwerking gaat bieden.

Pagina
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Met vriendelijke groet,

Ons kenmerk
KA/mw

Uw kenmerk

Kees Ahaus
Professor of Health Services Management & Organization (HSMO)
Sectieleider HSMO

Akkoord, de heer Brouwer, decaan:



Akkoord, mevrouw Nelissen/Treant:



Datum:

Received: 12/05/2018	FOA: AI18-029	Council: 05/2019
Competition ID: FORMS-E	FOA Title: HIV Drug Resistance: Genotype-Phenotype-Outcome Correlations (R01 Clinical Trial Not Allowed)	
1 R01 AI147330-01	Dual:	Accession Number: 4243767
IPF: 10001605	Organization: ERASMUS MEDICAL CENTER	
Former Number:	Department: Viroscience	
IRG/SRG: ZRG1 AARR-P (52)R	AIDS: Y	Expedited: N
<u>Subtotal Direct Costs</u> (excludes consortium F&A) Year 1: 498,424 Year 2: 499,567 Year 3: 499,778 Year 4: 498,747 Year 5: 492,537	Animals: N Humans: Y Clinical Trial: N Current HS Code: 30 HESC: N	New Investigator: Y Early Stage Investigator: N
<i>Senior/Key Personnel:</i>	<i>Organization:</i>	<i>Role Category:</i>
Charles Boucher M.D.	Erasmus University Medical Center Rotterdam	PD/PI
Jeroen van Kampen M.D.	Erasmus University Medical Center Rotterdam	Co-PD/PI
Rob Gruters Ph.D	Erasmus University Medical Center Rotterdam	Co-Investigator
David van de Vijver Ph.D	Erasmus University Medical Center Rotterdam	Co-Investigator
Daniel Kuritzkes M.D.	BRIGHAM AND WOMEN'S HOSPITAL	Other (Specify)-Project Lead
Ricardo Diaz M.D.	Federal University of Sao Paulo	Other (Specify)-Project Lead
MAURO SCHECHTER M.D.	Federal University of São Paulo	Co-Investigator
Benjamin Berkhout Ph.D	UNIVERSITY OF AMSTERDAM	Other (Specify)-Project Lead
Atze Das Ph.D	Academic Medical Center of the University of Amsterdam	Co-Investigator

APPLICATION FOR FEDERAL ASSISTANCE SF 424 (R&R)		3. DATE RECEIVED BY STATE	State Application Identifier
1. TYPE OF SUBMISSION*		4.a. Federal Identifier	
☐ Pre-application ☐ Application ☒ Changed/Corrected Application		b. Agency Routing Number	
2. DATE SUBMITTED	Application Identifier	c. Previous Grants.gov Tracking Number GRANT12754587	
5. APPLICANT INFORMATION		Organizational DUNS*: 4897197980000	
Legal Name*: ERASMUS MEDICAL CENTER Department: Viroscience Division: Street1*: Erasmus University Medical Center Rotterdam Street2*: Dr Molewaterplein 40 City*: Rotterdam County: State*: Province: Country*: NLD: NETHERLANDS ZIP / Postal Code*: 3015 GD			
Person to be contacted on matters involving this application Prefix: Mr. First Name*: Paul Middle Name: Last Name*: Huber Suffix: Position/Title: Project Manager Street1*: Dr Molewaterplein 40 Street2: City*: Rotterdam County: State*: Province: Country*: NLD: NETHERLANDS ZIP / Postal Code*: 3015 GD Phone Number*: +31107039791 Fax Number: Email: p.huber@erasmusmc.nl			
6. EMPLOYER IDENTIFICATION NUMBER (EIN) or (TIN)*		98-0574158	
7. TYPE OF APPLICANT*		W: Non-domestic (non-US) Entity	
Other (Specify):			

8. TYPE OF APPLICATION* ☒ New ☐ Resubmission ☐ Renewal ☐ Continuation ☐ Revision		If Revision, mark appropriate box(es). ☐ A. Increase Award ☐ B. Decrease Award ☐ C. Increase Duration ☐ D. Decrease Duration ☐ E. Other (specify) :
Is this application being submitted to other agencies? ☐ Yes ☒ No What other Agencies?		
9. NAME OF FEDERAL AGENCY* National Institutes of Health		10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER TITLE:
11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT* Novel mechanisms contributing to failure of dolutegravir-containing cART in clinical practice		
12. PROPOSED PROJECT Start Date* Ending Date* 07/01/2019 06/30/2024		13. CONGRESSIONAL DISTRICTS OF APPLICANT 00-000

Contact PD/PI: Boucher, Charles

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE

Page 2

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION Prefix: Prof. First Name*: Charles Middle Name: Last Name*: Boucher Suffix: M.D. Position/Title: Professor in Antiviral Therapy Organization Name*: Erasmus University Medical Center Rotterdam Department: Viroscience Division: Street1*: Wytemaweg 80 Street2: City*: Rotterdam County: State*: Province: Country*: NLD: NETHERLANDS ZIP / Postal Code*: 3015 CN Phone Number*: +31651203640 Fax Number: Email*: c.boucher@erasmusmc.nl	
15. ESTIMATED PROJECT FUNDING a. Total Federal Funds Requested* \$3,170,507.00 b. Total Non-Federal Funds* \$0.00 c. Total Federal & Non-Federal Funds* \$3,170,507.00 d. Estimated Program Income* \$0.00	16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?* a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON: DATE: b. NO ☒ PROGRAM IS NOT COVERED BY E.O. 12372; OR ☐ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW
17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances* and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001) ☒ I agree* * The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.	
18. SFLLL or OTHER EXPLANATORY DOCUMENTATION File Name:	
19. AUTHORIZED REPRESENTATIVE Prefix: Prof. First Name*: JPTM Hans Middle Name: Last Name*: van Leeuwen Suffix: Position/Title: Dean Organization Name*: ERASMUS MEDICAL CENTER	

From: Wensing, A.M.J. (Annemarie) <A.M.J.Wensing@umcutrecht.nl>

Sent: woensdag 19 januari 2022 15:22

To: Jeroen van Kampen <j.vankampen@erasmusmc.nl>; Nijhuis, M. (Monique) <M.Nijhuis@umcutrecht.nl>; Saadulla, M. (Mohammed) <M.Saadulla@umcutrecht.nl>; carole.devau@crp-sante.lu

Subject: Fwd: G.OPTICS: Proposal Approved - Annemarie Wensing (Netherlands), 2021-8515

Goood News!

Verstuurd vanaf mijn iPhone

From: admin@gileadmedaffairs.appiancloud.com <admin@gileadmedaffairs.appiancloud.com> on behalf of noreply@gilead.com <noreply@gilead.com>

Sent: Wednesday, January 19, 2022 5:30:28 AM

To: Annemarie Wensing <a.m.j.wensing@umcutrecht.nl>

Cc: Almudena Torres Cornejo <almudena.torrescornejo@gilead.com>; Eileen Roldan <Eileen.Roldan@gilead.com>; Brenda Ng <Brenda.Ng@gilead.com>; Melody Billena <Melody.Billena@gilead.com>; Rich Clark <Rich.Clark@gilead.com>; Dina Chinichian <Dina.Chinichian@gilead.com>

Subject: [EXTERNAL] G.OPTICS: Proposal Approved - Annemarie Wensing (Netherlands), 2021-8515

Reference Number: 2021-8515

Protocol Number: CO-US-380-6390

Research Title: A global registry on second generation and long-acting integrase inhibitor failures (ROSETTA)

Dear Annemarie Wensing,

I'm very pleased to inform you that Gilead Sciences has approved your proposal titled A global registry on second generation and long-acting integrase inhibitor failures (ROSETTA). This study CO-US-380-6390 was approved for funding in the amount of EUR,499814.7.

You will be contacted by your Gilead Clinical Operations Study Manager, Eileen Roldan , who will handle all operational aspects of the study. They will be your main contact; their contact information is below.

Study support is contingent on timely responses to communication, execution of contract and study start up. Please go into your G.OPTICS application and enter in your contracting contact information. Additional details to follow.

Eileen Roldan

Clinical Trial Manager

Gilead Sciences, Inc

eileen.roldan@gilead.com

Congratulations on the approval. We appreciate your interest in the Gilead HIV Treatment Medical Affairs Research Program. We look forward to a successful collaboration with you on this important study.

Sincerely,

Gilead Medical Affairs

Research Gilead Sciences, Inc.

Datum Februari 2020

Titel Samenwerkingsovereenkomst tussen Erasmus MC en Samenwerkende Rijnmond
Ziekenhuizen (SRZ)



Projectcode

Versie Samenwerkingsovereenkomst

Datum Februari 2020

Samenwerkingsovereenkomst tussen Erasmus MC en Samenwerkende Rijnmond Ziekenhuizen (SRZ)

SRZ-project Handen uit de Mouwen

PARTIJEN:

1. **Erasmus Universitair Medisch Centrum**, gevestigd te 's-Gravendijkwal 230, 3015 CE Rotterdam, meer in het bijzonder de afdeling Medische Microbiologie en Infectieziekten, hierna te noemen "MMIZ", ten deze rechtsgeldig vertegenwoordigd door Prof. Dr. A. Verbon, afdelingshoofd

en

2. **Stichting Samenwerkende Rijnmond Ziekenhuizen**, gevestigd te Capelle aan den IJssel, Rivium Quadrant 90 (gebouw E), 2909 LC Capelle aan den IJssel en, meer in het bijzonder het Netwerk Veiligheid en Kwaliteit, hierna te noemen "SRZ", ten deze rechtsgeldig vertegenwoordigd door dr. J.G. Boonstra, voorzitter SRZ.

Hierna afzonderlijk te benoemen als "Partij" en gezamenlijk te benoemen als "Partijen"

IN AANMERKING NEMENDE DAT:

- SRZ een samenwerkingsverband is tussen 9 ziekenhuizen in de regio Rijnmond; in dit project zijn tevens de volgende niet SRZ-ziekenhuizen aangesloten:
 - Albert Schweitzer Ziekenhuis
 - Admiraal de Ruijter Ziekenhuis
 - Beatrix Ziekenhuis
- SRZ belang hecht aan goede hygiëne binnen de ziekenhuizen;
- alle bij SRZ aangesloten ziekenhuizen in maart 2014 de intentie hebben uitgesproken om een actieve en stimulerende bijdrage te leveren aan de uitwerking van de afspraken, vastgelegd in het Masterplan 'Handen uit de mouwen' 3, gedateerd 14 december 2019 en die zijn vastgesteld in de AB vergadering van december 2019.
- MMIZ een toezegging van SRZ heeft ontvangen over de financiële vergoeding van de uit te voeren onderzoeksactiviteiten;
- De ingangsdatum van het onderzoek 1 januari 2020 is en loopt tot 31 december 2022.
- Het SRZ-project Handen uit de Mouwen wordt uitgevoerd onder verantwoordelijkheid van dr. W.W. Vrijland, voorzitter van het SRZ-netwerk Kwaliteit en Veiligheid ;
- Prof. dr. M.C. Vos aanspreekpunt is voor MMIZ;
- Prof. dr. M.C. Vos projectcoördinator is, waar het de coördinatie en uitvoering van de onderzoeksactiviteiten door het Erasmus MC betreft;
- drs. J.M.J.J. Vermeeren, stafadviseur kwaliteit en patiëntenzorg Erasmus MC en

programmamanager SRZ-project Handen uit de Mouwen, aanspreekpunt is voor SRZ;

- Partijen in verband met het bovenstaande afspraken wensen te maken over de voorwaarden waaronder de onderzoeksactiviteiten ten behoeve van het SRZ-project Handen uit de Mouwen in samenwerking met de SRZ-projectorganisatie zullen worden uitgevoerd;
- Partijen de voornoemde afspraken wensen vast te leggen en daarvoor deze overeenkomst hebben opgesteld;

Artikel 1: Aanleiding en Beoogde Resultaten	
1.1	Het doel van het project Handen uit de Mouwen is onder meer om te komen tot vermindering van potentieel onveilige situaties in het primaire proces (de patiëntenzorg) binnen alle bij SRZ aangesloten ziekenhuizen.

Artikel 2: Projectorganisatie en verantwoordelijkheden partijen	
2.1	MMIZ is opdrachtnemer in deze samenwerking en voert de onderzoeksactiviteiten uit ten behoeve van alle bij SRZ aangesloten ziekenhuizen.
2.2	SRZ is penvoerder van het project.
2.3	Onderzoeksactiviteiten in relatie tot dit project zijn: • Uitvoering en begeleiding van dataverzameling • Begeleiding van data analyses en rapportages
2.4	De onderzoeksresultaten zullen steeds schriftelijk worden vastgelegd.
2.5	Werkzaamheden worden niet door derden uitgevoerd, tenzij dit van te voren schriftelijk is vastgelegd en door beide partijen is goedgekeurd.
2.6	Werkzaamheden worden niet door derden uitgevoerd, tenzij dit van te voren schriftelijk is vastgelegd en door beide partijen is goedgekeurd.

Artikel 3: Geheimhouding en vertrouwelijke informatie							
3.1	Partijen zullen de aan haar door de andere partijen in deze overeenkomst ter beschikking gestelde kennis en gegevens uitdrukkelijk als vertrouwelijk behandelen. Dit geldt ook voor dié kennis en gegevens die niet worden verkregen door de uitvoering van het project en voor zover die kennis en gegevens uitdrukkelijk als 'vertrouwelijk' kunnen worden gewaarmerkt, (hierna te noemen 'Vertrouwelijke Informatie'). Partijen stemmen ermee in dat de kennis die ter beschikking gesteld wordt niet gebruikt wordt voor enig ander doel dan voor de uitvoering van het project.						
3.2	Deze verplichting geldt niet voor zover: <table border="1"> <tr> <td>a.</td><td>de vertrouwelijke Informatie reeds in het bezit is van de ontvangende partij(en) op het moment dat de vertrouwelijke informatie aan die partij(en) wordt verstrekt;</td></tr> <tr> <td>b.</td><td>de Vertrouwelijke Informatie van algemene bekendheid is of wordt, zonder dat dit het gevolg is van enig verwijtbaar handelen of nalaten t.a.v. de vertrouwelijke informatie door de ontvangende partij(en);</td></tr> <tr> <td>c.</td><td>vertrouwelijke informatie op rechtmatige wijze door de informatie ontvangende partij(en) word(t)(en) verkregen door een derde, dan wel uit eigen onderzoek van</td></tr> </table>	a.	de vertrouwelijke Informatie reeds in het bezit is van de ontvangende partij(en) op het moment dat de vertrouwelijke informatie aan die partij(en) wordt verstrekt;	b.	de Vertrouwelijke Informatie van algemene bekendheid is of wordt, zonder dat dit het gevolg is van enig verwijtbaar handelen of nalaten t.a.v. de vertrouwelijke informatie door de ontvangende partij(en);	c.	vertrouwelijke informatie op rechtmatige wijze door de informatie ontvangende partij(en) word(t)(en) verkregen door een derde, dan wel uit eigen onderzoek van
a.	de vertrouwelijke Informatie reeds in het bezit is van de ontvangende partij(en) op het moment dat de vertrouwelijke informatie aan die partij(en) wordt verstrekt;						
b.	de Vertrouwelijke Informatie van algemene bekendheid is of wordt, zonder dat dit het gevolg is van enig verwijtbaar handelen of nalaten t.a.v. de vertrouwelijke informatie door de ontvangende partij(en);						
c.	vertrouwelijke informatie op rechtmatige wijze door de informatie ontvangende partij(en) word(t)(en) verkregen door een derde, dan wel uit eigen onderzoek van						

	derden zonder dat daarbij op enigerlei wijze gebruik wordt gemaakt van de vertrouwelijke informatie.
3.3	Partijen zullen elkaar de in dit artikel bedoelde vertrouwelijke informatie alsmede eventuele andere kennis en gegevens ter beschikking stellen, indien en voor zover dat nodig is voor de uitvoering van het project.
3.4	Partijen verplichten zich tot geheimhouding voor de duur van vijf (5) jaar na beëindiging van het project, van elkaars als zodanig aangeduide vertrouwelijke informatie en achtergrondkennis waarvan zij bij de uitvoering van het project hebben kennis genomen en waarvan uitdrukkelijk is medegedeeld, of waarvan redelijkerwijs vermoed kan worden, dat geheimhouding in het belang van de wederpartij is.
3.5	Indien de verstrekende partij enige door haar mondeling verstrekte Informatie als vertrouwelijk wenst aan te merken, geeft zij dit bij het bekend maken aan de ontvangende partij(en) te kennen en dient zij deze Informatie binnen veertien dagen na de bekendmaking op schrift te stellen en met het opschrift 'vertrouwelijk' aan de ontvangende partij(en) te zenden. Zolang de bedoelde Informatie nog niet op schrift is gesteld, zal / zullen de ontvangende partij(en) niettemin geheimhouding met betrekking tot deze Informatie betrachten.
3.6	De verstrekende partij geeft bij het verstrekken van vertrouwelijke informatie aan, welke documenten, elektronische bestanden, tekeningen, modellen, gegevens, resultaten, ontwerpen, materialen, prototypen na afloop van het onderzoek dienen te worden vernietigd of aan haar te worden geretourneerd.
3.7	Vertrouwelijke informatie zoals in dit artikel omschreven, is die informatie die door de ene partij aan de andere partij(en) schriftelijk ter beschikking wordt gesteld en welke duidelijk is aangemerkt als zijnde vertrouwelijk. Mondeling of visueel verstrekte informatie is alleen dan vertrouwelijke informatie, indien deze binnen twee weken na verstrekking alsnog schriftelijk wordt samengevat door de verstrekende partij en door deze wordt aangemerkt als vertrouwelijk.
3.8	De deelnemers verplichten zich deze geheimhoudingsplicht ook op te leggen aan door hen ten behoeve van het project ingeschakelde derden, zoals uitzendkrachten, gedetacheerden en onderaannemers. Tevens geldt dat elke deelnemer er voor in staat dat haar personeel zich aan de geheimhoudingsplicht zal houden.

Artikel 4: Financiën	
4.1	De vergoedingen die betaald worden aan MMIZ door SRZ zijn conform de door het Algemeen Bestuur van SRZ vastgestelde begroting . De afgesproken maximale vergoeding per jaar, gedurende een periode van 2 jaar, is: € 45.530,00 (inclusief eventuele verschuldigde BTW)
4.2	Ter declaratie van uitgevoerde werkzaamheden stuurt MMIZ jaarlijks begin 2021 en begin 2022 een gespecificeerde factuur naar: Bureau SRZ, Rivium Quadrant 90, gebouw E 2909 LC Capelle aan den IJssel Als referentie zal MMIZ 'SRZ-project Handen uit de Mouwen' op de facturen plaatsen.

4.3	MMIZ zal een gedetailleerde administratie voeren van de in het kader van het project gemaakte kosten.
4.4	MMIZ draagt er zorg voor dat de resultaten van het onderzoek schriftelijk en in stoffelijke vorm worden vastgelegd.

Artikel 5: Intellectuele eigendomsrechten

5.1	Ziekenhuis specifieke data blijft te allen tijde eigendom van het ziekenhuis waar de data is verkregen, maar mag geaggregeerd en geanonimiseerd gebruikt worden voor het Onderzoek.
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Artikel 6: Publicatie en rapportage

6.1	Partijen zullen de Resultaten publiceren in onder meer wetenschappelijke tijdschriften en vakbladen.
6.2	Partijen mogen alleen publiceren over de Geaggregeerde Data.
6.3	Data waarvan de herkomst te herleiden is naar één van de deelnemende ziekenhuizen mag niet worden gebruikt in publicaties tenzij de data geanonimiseerd is en herleidbaarheid uitgesloten is.
6.4	Co-auteurschap zal worden vastgesteld op basis van de voor de beroepsgroep geldende richtlijnen voor auteurschap en volgens de Vancouver richtlijnen. Uitgangspunt is dat personen die een inhoudelijke bijdrage hebben geleverd aan het tot stand komen en uitvoeren van het Onderzoek, zullen worden uitgenodigd in de publicatie te participeren als co-auteur.
6.5	In alle publicaties voortkomend uit het Onderzoek, zal de medewerking van alle bij deze overeenkomst betrokken partijen worden vermeld. Voorstellen voor publicaties, presentaties, rapportages en andere publiciteit en de definitieve uitwerking daarvan dienen tenminste goedgekeurd te worden door de inhoudelijk projectleiders als vertegenwoordigers van de partijen.
6.6	Elke partij verplicht zich om uiterlijk drie weken voor de voorgenomen openbaarmaking c.q. inzending van een presentatie of manuscript gebaseerd op dit project een conceptversie voor commentaar voor te leggen aan de vertegenwoordigers van de andere partijen en aan alle vermelde auteurs. De co-auteurs verplichten zich om binnen een week te reageren op de inhoud van de presentatie of het manuscript. Bij gebleken onenigheid over te presenteren of te publiceren teksten dient voorafgaand aan de openbaarmaking overeenstemming bereikt te worden tussen de vertegenwoordigers van de partijen.
6.7	Partijen zullen de verslaglegging van het onderzoek naar de huidige maatstaven van onafhankelijke goed wetenschappelijke verslaglegging wederzijds niet frustreren, zodat er bij de publicaties aan alle voorwaarden die tijdschriften aan hun auteurs stellen, wordt voldaan.
6.8	Onderzoekersleiders, Onderzoekers en Projectadviseurs zullen in onderlinge samenwerking een schriftelijke eindrapportage van alle onderzoeksactiviteiten en bevindingen opstellen. De eindrapportage dient te voldoen aan de voorwaarden en tijdslijnen gesteld in de Subsidievoorwaarden.

Artikel 7: Aansprakelijkheid

- | | |
|-----|--|
| 7.1 | Alle deelnemers zetten zich in voor een goede uitvoering van het project en zullen zich houden aan de dienaangaande in deze overeenkomst en in het projectplan gemaakte afspraken. Indien één van de partijen toerekenbaar tekort komt bij de nakoming van zijn verplichtingen zijn de andere partijen gerechtigd, na schriftelijke ingebrekestelling, de overeenkomst met onmiddellijke ingang te beëindigen en hun schade te verhalen op de toerekenbaar tekortkomende partij. |
|-----|--|

Artikel 8: Duur – beëindiging

- | | |
|-----|--|
| 8.1 | Deze overeenkomst treedt in werking op 1 januari 2020 en wordt aangegaan voor de duur van twee jaar en wel tot en met 31 december 2022. Afspraken over Publicatie gelden tot 5 jaar na afloop van het project. |
|-----|--|

Artikel 9: Wijzigingen

- | | |
|-----|--|
| 9.1 | Wijzigingen van of aanvullingen op deze overeenkomst zijn alleen rechtsgeldig wanneer zij schriftelijk zijn vastgelegd en door bevoegde vertegenwoordigers van alle deelnemers zijn ondertekend. |
| 9.2 | Geen der Partijen zal wijzigingen in het Onderzoeksvoorstel aanbrengen zonder voorafgaand overleg en schriftelijke toestemming van de andere Partij, waarna dergelijke wijzigingen zullen worden beschouwd als onderdeel van het Onderzoeksvoorstel. |

Artikel 10: Overdraagbaarheid rechten en plichten

- | | |
|------|--|
| 10.1 | Het is geen van de deelnemers toegestaan zijn rechten en verplichtingen onder deze samenwerkingsovereenkomst over te dragen aan een derde zonder de uitdrukkelijke schriftelijke toestemming van de andere partij. |
|------|--|

Artikel 11: Geschillen

- | | |
|------|--|
| 11.1 | Bij geschillen tussen de deelnemers met betrekking tot deze overeenkomst, wordt in onderling overleg een neutrale partij uitgenodigd om een niet-bindend oordeel te geven. |
|------|--|

Artikel 12: Toepasselijk recht

- | | |
|------|--|
| 12.1 | Deze overeenkomst wordt beheerd door het Nederlands recht. |
| 12.2 | Voor alle geschillen die tussen Partijen omtrent de uitleg en/of uitvoering van deze overeenkomst mochten ontstaan, zullen Partijen trachten in goed overleg een oplossing te vinden. Geschillen die niet in goed overleg kunnen worden opgelost, zullen worden voorgelegd aan de bevoegde rechter in Rotterdam. |

Aldus overeengekomen en ondertekend,

Erasmus *Universitair* Medisch Centrum

Rotterdam, 17-12-2020

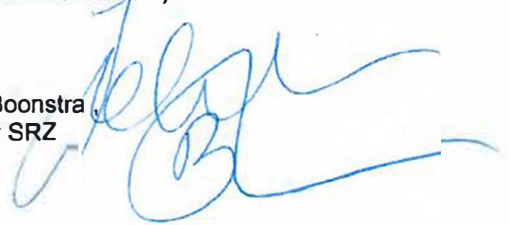


Professor dr. A. Verbon,
afdelingshoofd MMIZ

SRZ

Capelle aan den IJssel, 14/12/2020
..... 2020

dr. J.G. Boonstra
voorzitter SRZ



Erasmus MC
De heer Prof. dr. J.P.T.M. van Leeuwen
Postbus 2040
3000 CA ROTTERDAM

Laan van Nieuw Oost-Indië 334
2593 CE Den Haag
Postbus 93245
2509 AE Den Haag
Telefoon 070 349 51 11
Fax 070 349 51 00
www.zonmw.nl
info@zonmw.nl

Dossiernummer

50-56300-98-2105

Ons kenmerk

2021/28760/ZONMW

Datum

8 september 2021

Onderwerp

Subsidiebesluit over projectnummer 10430102110003

Contactpersoon

Veerle Bos
T: 070 515 0313
E: Covid19@zonmw.nl

Geachte heer Van Leeuwen,

Op 14 april 2021 heeft u bij ZonMw een subsidieaanvraag ingediend met de titel: *'Safe and Timely Cessation of Isolation of Patients with COVID-19, The SCIP-COVID-19 study'*. De commissie van het programma COVID-19 heeft tijdens de beoordelingsvergadering op 3 juni 2021 uw aanvraag beoordeeld. Zij was echter van mening dat er nadere aanpassingen nodig waren en heeft u verzocht een aangepaste aanvraag in te dienen. Op 20 juli 2021 heeft u een aangepast projectvoorstel ingediend. Deze aanvraag is vervolgens positief beoordeeld door de commissie.

ZonMw heeft een besluit genomen over uw subsidieaanvraag. In deze brief vindt u ons besluit en wat u van ons kunt verwachten. Daarnaast leest u wat wij van u verwachten.

Besluit

ZonMw besluit tot honorering van uw subsidieaanvraag. U ontvangt een subsidie van maximaal € 599.100,- voor de duur van maximaal 24 maanden. Dit bedrag is inclusief eventueel verschuldigde btw.

Onderbouwing besluit

De commissie realiseert zich dat deze aanvraag is ingediend binnen een hele bijzondere dynamiek en in een hele korte tijdspanne. Zij wil bij deze haar grote waardering uitspreken naar de projectgroep voor het realiseren van een aanvraag met betrokken partners binnen dit tijdsbestek.

De beoordelingscommissie van het programma COVID-19 (COVID-19 deelprogramma behandeling) heeft positief geadviseerd over uw aanvraag. Dit oordeel is gebaseerd op uw aanvraag, de beoordeling door de commissie, het interview tijdens de commissievergadering, de aangepaste aanvraag en het aanvullend wederhoor. Het advies van de commissie is naar het oordeel van ZonMw op zorgvuldige wijze tot stand gekomen. ZonMw heeft dit advies overgenomen en ten grondslag gelegd aan haar beslissing.

Relevantie

De commissie heeft het volgende relevantieoordeel over uw subsidieaanvraag gegeven: **zeer relevant**.

Het relevantieoordeel is gebaseerd op de volgende argumenten:

- Door het ontwikkelen van een algoritme waarmee kan worden bepaald wanneer gehospitaliseerde COVID-19-patiënten veilig uit isolatie kunnen worden gehaald voorziet deze studie in een belangrijke klinische behoefte.
- Het tijdig stopzetten van isolatie beperkt de fysieke en emotionele stress voor patiënten, de belasting voor beroepsbeoefenaren in de gezondheidszorg die voor geïsoleerde patiënten zorgen en de kosten voor de zorg van COVID-19 patiënten in het ziekenhuis.
- Door het vergelijken van verschillende testmethoden en het integreren van testresultaten zal het ontwikkelde algoritme om COVID-19-patiënten veilig uit isolatie te halen breed toepasbaar zijn.

Kwaliteit

De commissie heeft het volgende kwaliteitsoordeel over uw subsidieaanvraag gegeven: **goed**.

Het kwaliteitsoordeel is op de volgende argumenten gebaseerd:

- De adviespunten van de commissie zijn adequaat verwerkt.
- De commissie is positief over de multidisciplinaire aanpak in de studie. In het onderzoek wordt gebruik gemaakt van bestaande data om de laboratorium- en modelstudies te versterken.
- De studie maakt gebruik van data uit een groot multicenter cohort, waardoor verschillende isolatiemethodologieën die gebruikt zijn binnen dit cohort kunnen worden vergeleken.
- Het projectteam wordt door de commissie beoordeeld als zeer deskundig.
- Over de haalbaarheid van de aanvraag is de commissie positief. Er worden geen nieuwe patiënten geïncludeerd wat de werving en inclusie realistisch maakt.
- De commissie acht het aangevraagde budget realistisch voor het uitvoeren van de geplande activiteiten.
- De commissie heeft nog een aandachtspunt voor de aanvraag. Onder “aanvullende voorwaarden” staat beschreven welke voorwaarden gekoppeld zijn aan deze toekenning.

Aanvullende voorwaarden voor uw project

- De commissie vraagt u toe te lichten welke criteria worden gehanteerd om beëindiging van isolatie als ‘veilig’ te beoordelen.

Wij vragen u extra aandacht te hebben voor implementatie, opschaalbaarheid en borging van resultaten uit uw project. ZonMw wil de nieuwe kennis zo snel mogelijk beschikbaar maken voor het veld en de praktijk. U kunt, buiten reguliere rapportages om, door het programmasecretariaat gevraagd worden om informatie en updates over de voortgang van uw project (mondeling, schriftelijk of als een presentatie) aan te leveren. We gaan ervan uit dat u uw medewerking daaraan verleent.

Wettelijk kader en algemene voorwaarden

Op dit besluit is de volgende regelgeving van toepassing:

- De Algemene Wet Bestuursrecht
- Deze subsidieverlening is rechtstreeks gebaseerd op de Wet ZON
- De Algemene subsidiebepalingen ZonMw. U kunt deze nalezen op: www.zonmw.nl/subsidievoorwaarden
- Procedure voor aanvragers. U kunt deze nalezen op: <https://www.zonmw.nl/nl/subsidies/hoe-werkt-subsidie-aanvragen>. Bij punt 17 t/m 21 leest u wat er gedurende het project van u wordt verwacht.

Specifieke voorwaarden en verplichtingen

Bij dit besluit tot honorering gelden de subsidieverplichtingen die zijn opgenomen in bijgevoegde Subsidieverplichtingen COVID-19 programma; deze bijlage maakt integraal deel uit van dit besluit.

Start project

Het project dient zo spoedig mogelijk te beginnen.

Integriteit

Het project dient op een goede en integere manier uitgevoerd te worden. Hiervoor gelden onder andere de volgende principes: eerlijkheid, zorgvuldigheid, transparantie, onafhankelijkheid en verantwoordelijkheid. Deze principes zijn vastgelegd in de Nederlandse gedragscode wetenschappelijke integriteit. Indien bij een door ZonMw gefinancierd project een (mogelijke) integriteitsschending plaatsvindt, dient ZonMw hier zonder uitstel van op de hoogte te worden gesteld. Alle relevant documenten over de integriteitsschending dienen aan ZonMw te worden overhandigd.

Samenwerking/bijdragen van derden

U dient een kopie van de door alle partijen ondertekende versie van de consortium- en/of sponsorovereenkomst **uiterlijk 6 maanden na de startdatum** van het project aan ZonMw toe te zenden.

Op de ZonMw webpagina Subsidies en Samenwerking/bijdragen van derden vindt u meer informatie, voorbeelden van en de voorwaarden voor het opstellen van een consortium- en/of sponsorovereenkomst. Wij verzoeken u deze voorbeeldovereenkomsten en informatie te gebruiken bij het opstellen van de betreffende overeenkomst. Maak hierbij duidelijk wat de wijzigingen ten opzichte van de voorbeeldovereenkomst zijn.

Toelichting op hoogte subsidiebedrag

- Conform uw ingediende herziene begroting.
- Van het subsidiebedrag is € 5.000,- gereserveerd voor Open Access.
- Van verschillende posten dient bij de eindafrekening een specificatie te worden aangeleverd.

Betaling van de subsidie

De subsidie zal via voorschotbedragen aan u worden overgemaakt. De betalingen zijn afhankelijk van de ontvangst en goedkeuring van voortgangsverslagen en de eindverantwoording van uw project. Als er aan alle door ZonMw gestelde voorwaarden is voldaan, ontvangt u rond de startdatum van het project het eerste voorschot.

Eindverantwoording

Binnen 13 weken na afloop van uw project dient u een eindverantwoording in. Dit is een inhoudelijk eindverslag en de financiële eindverantwoording. De financiële verantwoording dient een specificatie van de verschillende posten te bevatten. Na ontvangst en goedkeuring van de eindverantwoording vindt de definitieve subsidievaststelling en afrekening plaats.

Tussentijdse wijzigingen

Laat het ons weten als er tussentijds iets in uw plan of uw organisatie verandert. Verandering in de opzet, planning, begroting of organisatie kunnen gevolgen hebben voor uw subsidie. Pas na goedkeuring door ZonMw zijn eventuele wijzigingen toegestaan.

Wat verwachten wij nu van u?

ZonMw kan u een voorschot voor de eerste periode van uw project verstrekken. Hiervoor dient u:

- **Binnen 4 weken** na de verzenddatum van deze brief een toelichting te sturen op de aanvullende voorwaarden.
- **Binnen 4 weken** na de verzenddatum van deze brief bijgevoegd meldingsformulier ingevuld retour te sturen.
- **Zo spoedig mogelijk** na de verzenddatum van deze brief een Nederlandse publiekssamenvatting in ProjectNet in te vullen (maximaal 1000 tekens, inclusief spaties). ZonMw publiceert alle gehonoreerde projecten op haar website met een Nederlandse samenvatting in eenvoudige taal. Deze is bedoeld voor een breed geïnteresseerd publiek met verschillende achtergronden. Zie de schrijfwijzer op <http://www.zonmw.nl/nl/over-zonmw/logo-huisstijl>.
- **Open Science:** ZonMw en NWO hebben het internationale statement ondertekend om in de strijd tegen COVID-19 onderzoeksresultaten en data zo snel mogelijk openbaar te maken. Wat dit voor de COVID-19-projecten betekent kunt u hier lezen <https://www.zonmw.nl/nl/actueel/nieuws/detail/item/nwo-en-zonmw-open-access-in-strijd-tegen-coronavirus/>

Vragen

Heeft u nog vragen? Neemt u dan gerust contact op met de medewerker die in het briefhoofd vermeld staat. Dat kan via e-mail: covid-19@zonmw.nl of via telefoonnummer: 070 515 0313. Het oorspronkelijke nummer van uw aanvraag vervalt. Er geldt een nieuw projectnummer zoals vermeld bovenaan deze brief. Houd uw nieuwe projectnummer bij de hand zodat wij u snel kunnen helpen.

Bezwaar of klacht?

Bent u het niet eens met dit besluit? Dan kunt u uw contactpersoon benaderen. Blijft u het oneens, dan kunt u overwegen om een bezwaarschrift in te dienen. U moet het bezwaarschrift opsturen binnen 6 weken na de verzenddatum van deze brief naar het bestuur van ZonMw, t.a.v. Commissie Bezwaarschriften ZonMw, Postbus 93 245, 2509 AE Den Haag.

Als u ontevreden bent over de wijze waarop ZonMw uw aanvraag heeft behandeld, kunt u ons dit laten weten of een klacht indienen. Meer informatie vindt u op de ZonMw website via www.zonmw.nl/signalerenklagenbezwaarmaken.

Ik feliciteer u met de honorering van uw subsidieaanvraag en wens u succes bij de uitvoering van uw project!

Met vriendelijke groet,
namens het bestuur van ZonMw,



Véronique Timmerhuis
Algemeen directeur



Bijlagen
Subsidieverplichtingen COVID-19 programma
Meldingsformulier start project 10430102110003

Kopie
Prof. dr. M.C. Vos, Erasmus MC
Dr. A.F. Voor in 't holt, Erasmus MC

INVESTIGATOR SPONSORED STUDY AGREEMENT
HUMAN SUBJECT RESEARCH

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THIS INVESTIGATOR SPONSORED STUDY AGREEMENT ("Agreement"), is effective as of the latest dated signature below, is by and between the Foundation for Liver and Gastrointestinal Research, registered under the name of Stichting voor Lever- en Maag – Darm Onderzoek ("SLO" or the "Institution"), having its principal place of business at Erasmus University Medical Center Rotterdam, Department of Gastroenterology & Hepatology, Wytemaweg 80, 3015 CN Rotterdam, the Netherlands ("Erasmus MC" or "Hospital") and BOSTON SCIENTIFIC INTERNATIONAL S.A. a French corporation, whose registered address is Parc d’Affaires Le Val Saint-Quentin, 2 rue René Caudron, 78960 Voisins-le-Bretonneux, France, registered with the Commercial Registry of Versailles under number B 420 668 402, and with intra-community VAT number FR 07 420668402 ("BSI").

RECITALS

- A. Erasmus MC's employee, Prof. Dr. Marco J. Bruno, the Principal Investigator, has developed a human subject study entitled Prevalence of MDRO carriage in patients undergoing an ERCP in three different countries (the "Study").
- B. Having reviewed the Protocol defining the Study, as well as the written request for support which included the nature and objective of the Study and evaluated its interest in supporting the performance of the Study, BSI desires to support the Study provided that the Institution, through the Principal Investigator, administers all aspects of the Study and that BSI only acts as a funding agency. Therefore, BSI does not play any role in the processing of personal data.
- C. The Institution has expressed its willingness to administer, through the Principal Investigator, all aspects of the Study under its own responsibility, according to the Protocol and in compliance with all applicable laws and regulations and scientific and ethical requirements.

The parties agree as follows:

ARTICLE I
Definitions

Section 1.01. Confidential Information. "BSI Confidential Information" means all information disclosed by or on behalf of BSI to Institution or Investigator, or any of their respective employees, agents or consultants, including, without limitation, information relating to a BSI Device, and all other information regarding BSI's past, present, or future research, technology, designs, products, know-how, ideas, investigational plans/protocols, processes, compositions of matter, product applications, operations, and trade secrets including without limitation the terms of this Agreement. "Institutions Confidential Information" means any proprietary information that is provided by Institution or Investigator that BSI agrees to accept.

Section 1.02. FDA. "FDA" means the United States Food and Drug Administration.

Section 1.03. Intellectual Property. "Intellectual Property" means inventions, whether or not patentable, copyrightable works, know-how, trade secrets, software, including source code and object code, compositions of matter, procedures, and experimental results created pursuant to the Agreement.

Section 1.04. Investigator. "Investigator" means the Principal Investigator and any sub-investigator performing the Study under the responsibility of Institution, either at the Erasmus MC or at a Participating Site.

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Section 1.05. Participating Site. A “Participating Site” is an institution, other than the Erasmus MC, with which Institution, as Study sponsor, contracts to perform the Study.

Section 1.06. Protocol. The Protocol means the Protocol for this Study which is incorporated herein by reference developed by Institution, attached as Exhibit B as may be amended from time to time after approval from the Ethics committee.

Section 1.07. Sponsor. “Sponsor” means Institution, the entity legally responsible for developing, implementing and managing the Study in accordance with the Protocol and all applicable laws and regulations.

Section 1.08. Study Effective Date. The MedTech Europe Code of Ethical Business Practice of 2 December 2015 requires the following: *“A Member Company may give consideration to a request for a Research Grant prior to ethics committee approval for the specific research project but shall not take any final decision regarding the Grant request unless and until the research receives formal ethics committee approval”*. Therefore the « Effective date » means the date when the Study receives formal ethics committee approval. In the exceptional case where no ethics approval is required (i.e. retrospective research, bench testing, etc.), the Effective date will be the date of the last signature appearing at the end of this Agreement below.

Section 1.9. Representatives. “Representatives” mean the directors, officers, employees, consultants and agents of the Institution.

ARTICLE II

Institution’s Responsibilities

Section 2.01. Study Implementation. Institution, not BSI, is the Sponsor of the Study and is responsible for all Investigators and Participating Sites and for complying with all applicable laws and regulations and scientific and ethical requirements. Institution is further responsible for registering the Study in accordance with all applicable laws, regulations and guidelines. BSI encourages Institution to register any clinical or non-clinical Study involving a BSI product on an external public registry for clinical trial transparency, e.g., www.ClinicalTrials.gov, as applicable, before enrollment of the first subject or the start of data collection.

The confidentiality of Subject records and Subjects’ personal health data (“PHD”) shall be maintained at all times by Institution. Institution warrants and represents that a research authorization (i.e. a formal explicit consent form) that is compliant with applicable laws and that allows for the use and disclosure of PHD in accordance with the Study, will be obtained for each Subject participating to the Study.

Institution will not use or disclose data subjects’ PHD in violation of any Laws. Institutions warrants to comply with the General Data Protection Regulation 2016/679/EU and related applicable laws in all respects, including implementation of all necessary administrative, physical and technical safeguards designed to protect PHD from loss and unauthorized use, access, alteration, destruction or disclosure and the assurance that any subcontractors or agents to whom Institution has provided such PHD, in accordance with the consent form, agree to abide by the restrictions and conditions imposed on Institution and potential participating sites hereto under the Laws regarding data protection and privacy. This includes taking reasonable steps to protect, including encrypting, the transmission and storage of PHD.

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Institution shall only collect and use PHD necessary for execution of the goals of the Study and only to the extent permitted by the consent form. Institution shall retain PHD collected and used for the Study in accordance with the Laws and as described in the consent form. Such consent form shall include any and all requirements in conformity with Laws regarding data protection and privacy. The consent form shall also include the explicit informed consent of each Subject to the potential transfer of personal health data out of the EU to other countries, such as the U.S, which do not provide the same level of protection of personal data. Institution shall conduct any data protection impact assessments and/or prior consultations with Government Authorities that may be required in respect of Processing carried out under the Agreement

Institution shall ensure that its personnel engaged in the Processing of Personal Data are informed of the confidential nature of the Personal Data, have received appropriate training on their responsibilities, and have executed written confidentiality agreements. Institution shall ensure that such confidentiality obligations survive the termination of the personnel engagement. Institution is also responsible for ensuring that when sharing personal data of its personnel or of participating sites, such sharing is done lawfully having duly informed concerned individual ad relying on a lawful ground for doing so.

In the event of a Security Incident relating to the Personal Data collected or received by Institution during the Study, Institution agrees to swiftly report the incident to competent authorities and take the mitigation measures needed.

When Study PHD is subject to disposal, it shall be destroyed in accordance with industry best practices and in compliance with Laws. In accordance with the Laws, Institution shall honor a Subject's request to access his/her available PHD relating to the Study. Institution shall take reasonable steps to maintain the accuracy and completeness of Study PHD. Institution agrees to maintain programs that are adequate to ensure protection of patient privacy to the level imposed by Laws and to provide means for patients to have their privacy questions and complaints addressed in a timely manner.

Section 2.02. Reporting.

BSI will not monitor the Study or receive any Study Personal Data, except as follows:

- (a) *Unanticipated Adverse Device Effects (UADEs)*: Institution agrees to report all *Adverse events, including but not limited to Unanticipated adverse device effects (UADEs)* to the relevant authorities per applicable laws and regulations and to BSI. An Investigator should report UADEs as soon as it is determined to meet the definition, even if complete information is not yet available.
- (b) *Complaint Handling*: Institution is solely responsible for reporting any Device complaints in accordance with the applicable regulations and notifying BSI.
- (c) *Interim reports*: To the extent applicable, Institution shall submit written status reports (e-mail reports acceptable), on a regular basis as described in Exhibit A, on the progress toward completion of milestones described in Exhibit A, in sufficient detail as to permit BSI to understand the progress of the Study. The report shall include notice to BSI of any anticipated or realized schedule delays or budgetary over-runs, together, if applicable, with a plan to remedy such delays or over-runs. In the event that Institution experiences unremedied or

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repeated delays in achieving the milestones described in Exhibit A, BSI reserves the right to either terminate this Agreement in accordance with Section 4.03(a) and/or to require that Institution submit a new application to obtain any additional funding required to complete the Study.

- (d) *Final Report*: Institution shall submit a comprehensive written final report for the Study to BSI within one hundred twenty (120) days after termination of the Study.

Section 2.03. Study Data. All documents, protocols, data, know-how, methods, operations, formulas, confidential information (excluding Confidential Information provided by BSI), and materials generated in the performance of this Agreement (“Study Data”) shall be the property of Institution. Institution agrees, however, to provide BSI with any and all Study Data upon request from BSI. Institution shall provide BSI with copies of the final form of any manuscripts, posters, slide presentations or abstracts relating to the Study. BSI shall have the right to use Study Data for regulatory submissions and internal research purposes without being entitled to modify any of the Study Data.

Section 2.04. Use of Confidential Information. Both parties may wish to disclose to the other party certain proprietary information that is confidential, in furtherance of the Study. For the avoidance of doubt: the disclosure of study Subject records and Subjects’ personal health data (“PHD”) is not permitted.

Both parties shall use reasonable efforts to identify information as Confidential at the time of the disclosure of such information; provided, however, that any failure to do so shall not constitute nor be deemed to constitute a waiver of confidentiality with respect to information that a reasonable person knowledgeable in the field would recognize as being confidential in nature. Notwithstanding the foregoing, BSI shall have a reasonable opportunity to accept or reject Institutional Confidential Information prior to being exposed to its proprietary contents. BSI Confidential Information and Institutional Confidential Information shall remain the property of the respective disclosing party, and shall not be used or disclosed by recipient except for the fulfillment of recipient’s obligations under, and in accordance with, this Agreement and in furtherance of the Study. BSI Confidential Information and Institutional Confidential Information will remain subject to the terms of this Agreement for a period of three (3) years from the conclusion or early termination of this Agreement.

- (a) Exclusions. The foregoing obligation of non-use and non-disclosure shall not apply to:

- 1) Information that at the time of disclosure is in the public domain;
- 2) Information that after disclosure is published or otherwise becomes part of the public domain through no fault of recipient;
- 3) Information that is disclosed to recipient after acceptance of this Agreement, by a third party who has an apparent legal right to make such disclosure; or
- 4) Information that is independently developed by recipient without use of disclosing party’s Confidential Information as evidenced by recipient’s contemporaneous written records;

INVESTIGATOR SPONSORED STUDY AGREEMENT
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- 5) Information that is published in accordance with the provisions of this Agreement;
- 6) BSI Confidential Information that is required by FDA regulations or the Protocol to be disclosed in subjects' informed consent;
- 7) Information that disclosing party provides written permission to disclose;
- 8) Information that is required to be disclosed by law or pursuant to a court order, provided that the recipient of the Confidential Information promptly notifies the other party and cooperates reasonably with the other party's efforts to contest or limit the scope of such order and takes reasonable and lawful actions to avoid and/or minimize the degree of such disclosure.

(b) **Obligations of Confidentiality.** In consideration of each and every disclosure of Confidential Information by BSI to Institution, Institution shall:

- 1) treat as Confidential and preserve the Confidentiality of all Confidential Information disclosed to it;
- 2) make no use of any Confidential Information disclosed to it except in connection with this Agreement without the prior written consent of BSI;
- 3) make no disclosures of any Confidential Information to any person not an officer or employee of Institution or to any Participating Site/Investigator without the prior written consent of BSI;
- 4) not attempt to reverse engineer, disassemble, or attempt to determine the physical construction or chemical composition of any devices or materials if provided under this Agreement without written authorization from BSI; and
- 5) limit disclosure of Confidential Information to Representatives who have a need to know for performance of the Agreement and who are themselves bound by a nondisclosure agreement having obligations of confidentiality substantially similar to those contained herein and Institution shall advise each Representative who receives any Confidential Information of the confidential nature of the information.

Section 2.05. Intellectual Property. Except for, and limited to, the use specified in the Protocol, BSI grants Investigator or Institution no express or implied intellectual property rights in the Device or in any methods of making or using the Device. Any Intellectual Property conceived or reduced to

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practice under this Agreement solely by Investigator or Institution employees shall be owned by Institution (“Institution Intellectual Property”); provided, however, no Institution Intellectual Property shall include Confidential Information provided by BSI. Institution agrees to disclose to BSI all Institution Intellectual Property directly related to the Study. Institution shall grant a non-exclusive, unlimited, perpetual, royalty-free license in and to any Institution Intellectual Property that constitutes an improvement to the Device. In the case of any other Institution Intellectual Property, Institution grants BSI a first option to acquire a license to Institution Intellectual Property. Institution and BSI will negotiate in good faith to determine the terms of a license agreement for each item BSI agrees to license. If BSI and Institution fail to execute a license agreement within six (6) months after disclosure of the Institution Intellectual Property to BSI, Institution shall be free to license the Institution Intellectual Property to any party upon such terms as Institution deems appropriate, without any further obligation to BSI. Nothing in the foregoing is intended to conflict with ownership of Intellectual Property, if any, involving BSI employees, which shall be determined in accordance with US or other local/international legal requirements governing joint inventorship.

To the extent permitted by the policy of a medical-professional journal, medical institution, educational institution, or medical association relating to any publication or presentation of Study Data or results hereunder, ownership of the copyright in any publication, abstract, or presentation approved under this section will belong to Investigator or Institution. To the extent that Investigator or Institution own such copyright and in the countries where permitted under local laws, Investigator and/or Institution hereby grants BSI an unlimited, perpetual and royalty free license to use, reproduce and distribute such publications, abstracts or presentations. Investigator agrees to sign and deliver to BSI any documents required to secure BSI’s rights under this Section 2.04.

Section 2.06. Publication. As used in this Agreement, “Publication” means any journal article, abstract, presentation, or other type of public disclosure that reports any Study Data. Institution shall have the right, and shall use reasonable efforts, to publish or otherwise publicly disclose Study data and results and shall have the final authority to determine the scope and content of any publications originating from the Study provided that

- a. before publishing, Institution shall submit copies of any proposed publication or presentation to BSI for review such review being strictly limited to verifying that no BSI confidential information appears in such publication and to confirm correct representation with respect to the technical operation of any BSI device that may be referenced in such publication, at least thirty (30) days in advance of submission for publication or presentation to a publisher or other third party. Institution shall in good faith consider any comments provided by BSI and shall delete any Confidential Information identified by BSI prior to submission. When reasonably requested by BSI, Institution will delay publication up to an additional sixty (60) days to allow BSI to protect its rights in patentable or copyrightable material. Institution shall properly acknowledge BSI’s support of the Study in publications, in particular in all oral and written presentations of the Study results, as required by Article 17 of the GMH Code.
- b. Standards; Disclosure of Support: For all Publications, Institution will comply with recognized ethical standards concerning publications and authorship in the Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals (<http://www.icmje.org/icmje-recommendations.pdf>)

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established by the International Committee of Medical Journal Editors. Institution shall properly acknowledge BSI's support of the Study in publications.

Section 2.07. Principal Investigator. Principal Investigator's relevant qualifications, including dates, location, extent, and type of experience, listed in his most recent curriculum vitae (CV), must be provided to BSI. Institution represents that Principal Investigator has never participated in an investigation or other research activity, which was terminated (disqualified) by FDA or other equivalent regulatory authority or BSI, except as explained in an attachment to this Agreement.

Section 2.08. Reassignment of Principal Investigator. If the services of the Principal Investigator become unavailable to Institution for any reason, Institution is entitled to designate another qualified member of its faculty who has never participated in an investigation or other research activity which was terminated (disqualified) by FDA or other equivalent regulatory authority or BSI, to sign on to the terms of this Agreement and to serve as the Principal Investigator of the Study ("Substitute Investigator"). Institution shall provide BSI with the name of the Substitute Investigator. If a Substitute Investigator has not been designated within sixty (60) days after the original Principal Investigator provides notice that he or she will cease his or her services under this Agreement, either party may terminate this Agreement upon written notice to the other party.

Section 2.09. Representations and Warranties. Institution represents and warrants and it is made an express condition of this Agreement, that:

- a. Institution will receive support from BSI as described in this Agreement solely for conducting the Study as outlined in the Protocol and in accordance with Exhibit A.
- b. Institution has not been offered or paid any remuneration, directly or indirectly, overtly or covertly, in cash or in kind to (i) refer any individual to another person (e.g., physician or hospital) for the furnishing or arranging for the furnishing of any BSI product or service; or (ii) purchase, lease, order, use, prescription or arrange for or recommend purchasing, leasing, ordering, using or prescribing of any BSI product or service.
- c. In the event that the Study will take place at multiple Participating Sites, Institution shall be solely responsible for the selection of such Participating Sites, and shall represent to such Participating Sites that it, not BSI, is the Sponsor of the Study and that Institution, not BSI, will be listed as sponsor in any Study or regulatory documentation prepared by such Participating Sites. Institution shall enter into a clinical study agreement with each Participating Site that contains terms consistent with the terms of this Agreement. In the event that BSI is inappropriately referred to as the sponsor of the Study to any third party, including Study subjects, Institution agrees to take reasonable and appropriate measures to remedy such event.
- d. Institution is solely responsible for any and all safety reporting and regulatory obligations associated with the conduct of the Study, including but not limited to, obtaining and maintaining an Investigational Device Exemption (IDE) application, if required. In the event that an IDE is required, Investigator or Institution shall provide BSI with a copy of the IDE and of FDA's approval thereof.
- e. Institution is solely responsible for designing and executing an appropriate plan for the oversight of patient safety (Safety Plan) during the Study.
- f. Institution and Investigator will comply with all applicable laws, rules and regulations in conducting the Study, including but not limited to European Standard ISO 14155 (2011), to

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the Declaration of Helsinki and Good Clinical Practice for conducting clinical studies (ICH GCP Guidelines), to the General Data Protection Regulation (EU) 2016/679 or any applicable European and Dutch health information privacy laws, and data security laws.

- g. Neither Institution nor any Investigator or Participating Site is debarred under Dutch law and that they have not and will not use in any capacity the services of any person debarred under such law with respect to services to be performed under this Agreement. Institution also represents and warrants that neither it nor any Investigator or Participating Site is excluded from any health care program.
- h. No Investigator is presently under any obligation which would conflict with the performance of the Study.
- i. In the event that BSI provides Institution with Devices as a part of its support for the Study, Institution (i) shall not use Devices provided hereunder except as specifically described in this Agreement or in the Protocol; (ii) shall not alter the Devices provided hereunder in any manner; (iii) shall not transfer any Devices provided hereunder to any other person or entity; and (iv) upon expiration or termination of this Agreement, shall discontinue use of any Devices provided hereunder and shall, upon direction of BSI, return or destroy (and certify destruction of) any remaining Devices.
- j. In the event that BSI does not provide Institution with Devices as part of its support for the Study, Institution shall purchase Device through normal commercial channels under a separate agreement with BSI through Institution's procurement office.
- k. Institution shall not (individually or jointly) submit any charge to any governmental agency or any other payor, including Study subjects, for any device, equipment, tests, or other Study materials that are provided by BSI at no charge, replaced at no charge, or which are reimbursed by BSI under this Agreement.
- l. Institution and Principal Investigator agree to comply with the internal rules of Institution governing research. Institution and Principal Investigator further agree to comply with International Conference on Harmonization Good Clinical Practice (ICH GCP) guidelines to the extent applicable to the Study. Further, Institution and Principal Investigator will also follow generally accepted research practices for this type of Study, including the internal rules of Institution governing research. As a condition of BSI's support, Institution must provide BSI with documentation of both the internal Institutional Review Board (IRB) approval - if applicable - of the final Protocol and annual renewals of that approval, if required for the Study. Institution will notify BSI promptly of any withdrawal or suspension of such IRB approval during the term of the Study.
- m. Institution shall maintain accurate and reasonably detailed books and records concerning the receipt, expenditure, or other disposition of all Study-related support, including but not limited to financial and materiel support received from BSI.

Section 2.10. Amendments to Protocol. If Institution modifies the Protocol in any way that significantly affects the administration or use of the Device, the objectives of the Study, or potential risks to Study subjects, Institution will promptly inform BSI. Continued support by BSI will be contingent on BSI's review and acceptance of these changes.

ARTICLE III

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BSI's Responsibilities

Section 3.01. Financial Support. BSI will provide financial Study support only to the Institution. BSI shall provide support for the Study as described in Exhibit A. Institution is responsible for appropriate dispersal of such funds to the Participating Sites or Investigators with which Institution has contracted. BSI's maximum commitment shall be subject to the limitations set forth in Exhibit A. BSI shall not be responsible for cost overruns or commitments made by Institution in excess of any financial limits set forth in this Agreement, unless Institution obtains the prior written consent of BSI. The parties acknowledge that any financial support provided under this Agreement constitutes Fair Market Value for Institution's performance of the Study and represents the real documented cost to Institution in performing the activities required to initiate and conduct the Study.

The parties further acknowledge that this Agreement, and any consideration paid under it, is not contingent upon Institution's purchase or use of BSI products, except such purchases that may be required for the performance of the Study. BSI makes no assurance that the financial support provided under this Agreement will be sufficient for Institution to accomplish the goals of the Study, and that the Institution will be expected to provide any additional funding necessary to complete the Study from sources other than BSI.

Section 3.02. Payments. Institution shall notify BSI upon achievement of a milestone in accordance with the provisions of Section 4 of Exhibit A.

Checks or wire transfers will be made payable to Institution upon receipt by BSI of confirmation of completion of each milestone described in Exhibit A and upon receipt of corresponding detailed invoices sent by Institution to BSI:

Payee Name (Institution)	Stichting Lever en Darm, Maag onderzoek (SLO) (Foundation for Live and Gastrointestinal Research)
Payee Address	p/a Erasmus MC, Gastrointestinal and Hepatology diseases.
Bank name and address	Rabobank Rotterdam, Postbus 10017, 3004 AA Rotterdam
IBAN	NL52RABO0148753663
SWIFT code	RABONL2U
Preferred mention to be shown	Project number: SLO 08.20.002
VAT-no	NL8039.12.705.B01
Chamber Of Commerce	ROTTERDAM no 41129062
Contact person for payments	Finance.CRB, Email: finance.crb@erasmusmc.nl

Any party may change the above-specified payment details by notice to the other party given in the manner prescribed in section 5.01.

Section 3.03. Disclosure of Payments to Health Care Professionals. In European countries where legally required, BSI will disclose information regarding this Agreement and related payments to Institution (e.g., name, purpose, amount paid, etc.) to the public and/or required government agencies through the relevant transparency disclosure platforms.

Thus, Boston Scientific Group plc may have the responsibility to register the amounts paid under this Agreement in the Transparency register Zorg (*Transparantieregister Zorg*). The Institution needs to fill in its Chamber of Commerce number on page 1 of this Agreement and the email address on which it will receive the verification via the Transparency register Zorg, as follows: finance.crb@erasmusmc.nl.

ARTICLE IV

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Term and Termination

Section 4.01. Term. This Agreement shall remain in effect until the agreed upon Study completion date stated in Exhibit A, unless earlier terminated as set forth herein. This Agreement may be extended or renewed only by mutual written agreement executed by duly authorized representatives of the parties.

Section 4.02. Termination by Institution. This Agreement may be terminated without cause by Institution upon sixty (60) days' prior written notice to BSI. If the Institution has to put an end to the Study, in the interest of patient safety and due to the requirement of a new regulatory obligation, it is the responsibility of the Institution to ensure that all such regulatory obligations are fulfilled.

In case of early termination of the Agreement by Institution, the parties will agree in writing if necessary on any non-covered patient rights for the patients enrolled in the Study.

Section 4.03. Termination by BSI. This Agreement may be terminated immediately by BSI if:

- (a) Institution fails to satisfy any Study milestone described in the Protocol and/or in Exhibit A and does not cure such failure within thirty (30) days;
- (b) BSI becomes aware that such termination is necessary to protect the health, safety and welfare of any Study subject;
- (c) Institution fails to comply with Sections 2.02, 2.08, 2.09 or 2.10 of this Agreement; or
- (d) BSI determines its continued support for the Study would be inappropriate, impractical or inadvisable to continue based on any applicable laws or regulations.

In case of early termination of the Agreement by BSI, the parties will agree in writing if necessary on any non-covered patient rights for the existing patients enrolled under in the Study.

Section 4.04. Termination for Insolvency. This Agreement may be immediately terminated by either party without prejudice to any other rights which the terminating party may have, should a party file a petition of any type as to its bankruptcy, be declared bankrupt, become insolvent, make an assignment for the benefit of creditors or go into liquidation or receivership.

Section 4.05. Effect of Termination. If the Study is terminated for any reason, Institution shall promptly return to BSI any unused funds, materials and/or equipment originally provided by BSI. Further, in no event shall BSI be liable for any payments described on Exhibit A that are scheduled after the date of termination. Notwithstanding the foregoing, BSI agrees to reimburse Institution for any prepaid or committed or necessary expenses directly related to the Study or any agreed upon termination and which the Institution is, upon reasonable effort, unable to recover.

Section 4.06. Survival. Termination of this Agreement by a party shall not affect the rights and obligations of the other party accrued prior to the effective date of the termination. In addition, the provisions of Sections 2.02 (Reporting), 2.03 (Study Data), 2.06 (Publications), 2.09 (Representations and Warranties), 4.06 (Survival), 5.03 (Publicity) and 5.07 (Choice of Law) shall survive the termination or expiration of this Agreement.

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ARTICLE V
Miscellaneous

Section 5.01. Notices. Any notice required or permitted to be given under this Agreement shall be in writing and shall be considered given when mailed by pre-paid registered or certified mail, return receipt requested, or delivered by hand, to the parties at the following addresses (or such other address as a party may specify by notice hereunder):

If to Institution:

Institution Name: Foundation for Liver and Gastrointestinal Research ("SLO")c/o Erasmus MC
Address: Wytemaweg 80
3015 CN Rotterdam
PO Box 2040
3000 CA Rotterdam The Netherlands
Attn: Prof. Dr. H.J. Metselaar, MD, PhD chair SLO

With a copy to:

Name: Erasmus MC University Hospital
Department of Gastroenterology & Hepatology
Address: Dr. Molewaterplein 40, Room NA-612
3015 GD Rotterdam. The Netherlands
Attn: Prof. Dr. M. J. Bruno, MD, PhD

If to BSI:

1. Invoice to be made out to and mailed to the Accounting Department at the below Invoice Address	2. Other Notices to be sent to below Mailing Address (if different):
To: Accounting Department Boston Scientific International S.A. Parc d’Affaires Le Val Saint-Quentin 2 rue René Caudron 78960 Voisins-le-Bretonneux France Tel. +33 (0) 1 39 30 97 00	Guidant Europe NV/SA A Boston Scientific Company Green Square Lambroekstraat 5D 1831 Diegem Belgium
Invoice to be made out in the name of: Soraya Kobouche E-mail: Soraya.Kabouche@bsci.com	Attn: [Grant Coordinator Name] E-mail: Soraya.Kabouche@bsci.com
Intra-community VAT No.: FR 07 420668402	Telephone No.: +32 (0) 2416 7011

Any party may change the above-specified recipient and/or mailing address by notice to the other party given in the manner herein prescribed. All notices shall be deemed given on the day when actually delivered as provided above (if delivered personally or by facsimile) or on the day shown on the return receipt (if delivered by mail or delivery service).

Section 5.02. Independent Contractor. Institution is not an employee, agent, joint venture, or partner of BSI.

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Section 5.03. Publicity. No party shall use the name of the other in connection with any product, promotion, or advertising without the prior written consent of the other which consent shall not be unreasonably withheld. Investigator and Sponsor hereby grant BSI the unrestricted right and permission to use and publish the Investigator and the Sponsor's names, along with the Study title, purpose and design, in BSI presentations to third party (i.e. external to BSI) audiences.

Section 5.04. Order of Precedence. The terms of this Agreement shall take precedence over other documentation in the interpretation and resolution of disputes concerning this Study.

Section 5.05. Assignment. Neither party shall assign this Agreement to another without the prior written consent of the other party; provided, however, that BSI may assign this Agreement to a successor in ownership of at least 51% of its business assets.

Section 5.06. Severability. In the event a court of competent jurisdiction holds any provision of this Agreement to be invalid, such holding shall have no effect on the remaining provisions of this Agreement, and they shall continue in full force and effect.

Section 5.07. Choice of Law. This Agreement will be governed and construed in accordance with the laws of the Netherlands.

Section 5.08. Entire Agreement. This Agreement constitutes the entire Agreement among the parties and supersedes any and all prior agreements, understandings or arrangements, whether oral or written. The parties may amend or modify this Agreement in such manner as may be agreed only upon a written instrument executed by authorized representative of all parties.

Section 5.09. Consents; Waivers. Any approval, authorization, waiver or consent required by this Agreement must be in writing, duly signed by an authorized representative of the granting party. The withholding of an approval, authorization, waiver or consent for regulatory, quality, or competitive reasons shall not be deemed unreasonable. The failure of a party to enforce at any time any of the provisions of this Agreement shall in no way be construed to be a waiver of any such provision, nor in any way to affect the validity of this Agreement or any part of it or the right of a party after any such failure to enforce each and every such provision. No waiver of any breach of this Agreement shall be held to be a waiver of any other or subsequent breach.

Section 5.10. Counterparts. This Agreement may be executed in one or more counterparts, each of which shall be deemed to be an original, but all of which shall be considered one and the same instrument.

Section 5.11. Binding Effect; No Third Party Beneficiaries. This Agreement shall be binding upon and inure to the benefit of, and shall be enforceable only by, the parties hereto, and their respective successors and permitted assigns. It is the explicit intention of the parties hereto that no person or entity other than the named parties or their successors is or shall be entitled to bring any action to enforce any provision of this Agreement, as a third party beneficiary or otherwise.

Section 5.12. Compliance. During the term of this Agreement, each party shall comply with all applicable laws, ordinances, rules and regulations in any way relating to this Agreement or the performance of such party's obligations hereunder (including obtaining and maintaining all licenses, permits and authorizations required by law). Thus, each party hereto expressly acknowledges that the wording of this Agreement complies in particular with OECD Anti-Bribery Convention of 1997, the U.S. Foreign Corrupt Practice Act, MedTech Europe Code of Ethical Business Practice and GMH Code, that prohibit any offer, payment, promise to pay, or authorization of any payment of money or

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anything of value, directly or indirectly, to public officials, health care professionals and health care organizations for the purposes of: (i) influencing a recipient to act, or to fail to act, contrary to or in violation of such recipient's official or fiduciary duties, *e.g.*, as an inducement or reward for the referral, purchase or ordering of items or services, (ii) inducing a recipient to use his or her influence with another public official or healthcare professional to influence that person to act, or to fail to act, contrary to or in violation of that person's official or fiduciary duties, and (iii) securing any unlawful or unethical business advantage. It is the intention of the parties hereto that this Agreement shall be performed in compliance with these and all other applicable laws and regulations. If any portion of this Agreement is found, by any court or agency with jurisdiction over the subject matter hereof, not to be in compliance with such laws, ordinances, rules, or regulations, that portion of the Agreement shall be deemed to be retroactively amended and reformed as necessary to comply with it, and the parties shall cooperate in taking whatever steps are necessary to ensure such compliance. Further, the parties understand and agree that this Agreement, and any consideration paid under it, is not in any way contingent upon Principal Investigator's or Institution's purchase, use, prescription and/or recommendation, or in any other way linked to any previous, current or potential future use, of Boston Scientific Corporation products or services.

Section 5.13. Indemnification.

- a. By BSI. The Study is not designed, sponsored or managed by BSI, therefore BSI provides no indemnification for Study conduct. However, BSI will indemnify and hold harmless, the Institution, its officers, agents, directors, trustees, subsidiaries, affiliates, employees and students, and their respective heirs, successors and assigns (collectively, "Institutional Indemnitees") from and against any and all liability, damage, judgment, loss or expense (including reasonable attorneys' fees and costs) ("Claims") that arises from: (i) BSI's use of the Study or of any Invention licensed to BSI under this Agreement; and (ii) any negligence or willful misconduct of BSI or any BSI Persons in the performance of, or its failure to perform, BSI's obligations under this Agreement. Provided, however, BSI shall have no obligation to indemnify, defend and hold harmless the Institution Indemnitees to the extent that any Claim is the result of negligence or willful misconduct on the part of the Institution's Indemnitees.
- b. By Institution. Institution shall indemnify and hold harmless BSI, its officers, directors, agents, employees, and affiliates ("BSI Indemnitees"), from and against any and all Claims that arises from: (i) any negligence or willful misconduct of the Institution Indemnitees in the performance of, or its failure to perform, Institution's obligations under this Agreement, or ii) Institution's failure to adhere to the terms of this Agreement. Provided, however, Institution shall have no obligation to indemnify, defend and hold harmless BSI Indemnitees to the extent that Claim is the result of negligence or willful misconduct on the part of the BSI Indemnitees.
- c. Each party's obligations to the other party under this Section are conditioned upon: (a) the indemnified party providing notice to the indemnifying party of any Claim promptly, but not later than 30 days after the party knows of such Claim; (b) the indemnified party permitting the indemnifying party to assume full responsibility for the defense of such Claim; (c) the indemnified party assisting the indemnifying party in the defense of such Claim; and (d) the

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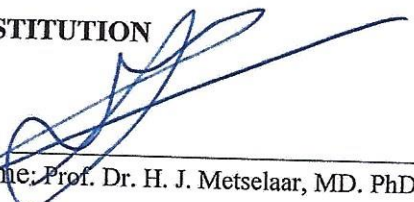
indemnifying party not compromising or settling any such Claim without the indemnified party's prior consent which shall not be unreasonably withheld, conditioned, or delayed, although the indemnified party's consent is not required if the compromise or settlement includes a full release of the indemnified parties without any admission of guilt or liability or current or future obligation, monetary or otherwise.

Section 5.14 Added Insurance. The parties represent and certify that each shall obtain and maintain insurance coverage, or a program of self-insurance, sufficient to meet its respective obligations under the Agreement. Upon request, the parties shall provide proof of such coverage to the other.

IN WITNESS WHEREOF, the parties hereto have caused this Agreement to be executed on their behalf by their duly authorized representatives.

INSTITUTION

BOSTON SCIENTIFIC INTERNATIONAL

By: 

Name: Prof. Dr. H. J. Metselaar, MD. PhD

Title : Chairman SLO

Date: 27/12/2020

By: _____

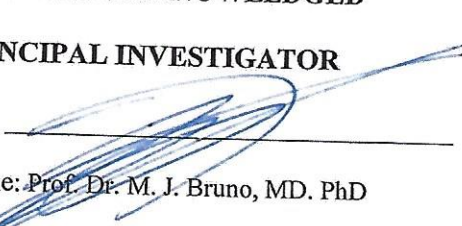
Name : Claudia Schulz

Title : Grant Committee Chair

Date: _____

READ AND ACKNOWLEDGED

PRINCIPAL INVESTIGATOR

By: 

Name: Prof. Dr. M. J. Bruno, MD. PhD

Title: Professor of Gastroenterology & Hepatology
Head of the department of Gastroenterology & Hepatology
Erasmus Medical Center

Date: 27-12-2020

By: 
NAME: DATTAGHI TOLAW
TITLE: U.P. OF ENDOSCOPY
ERASMUS
BOSTON SCIENTIFIC

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EXHIBIT A
SCOPE OF WORK, BUDGET AND PAYMENT SCHEDULE

1. Scope.

- 1.1. Interim reports, as described in Section 2.02(c) shall be provided to BSI on a quarterly basis.
- 1.2. The Agreement shall remain in effect until the last milestone as agreed in writing between the parties is completed.

2. Budget.

- 2.1 BSI shall pay Institution the amounts as described below. BSI shall pay Institution a maximum of \$350,000 USD. This amount includes any reasonable administrative indirect expenses and is inclusive of VAT. Payments will be made upon receipt of invoices.
- 2.2 Institution represents that the Institution-provided study budget upon which this Study support is based reflects an informed estimate of all funds required to complete the Study, including expenses relating to the publication of Study Results.

3. Payment delivery Schedule.

Payments will be made net thirty (30) days upon notification from Institution to BSI of the completion of the milestones outlined below and upon receipt of corresponding detailed invoices sent by Institution to BSI:

Milestone Date	Amount in Local Currency (inclusive of any applicable VAT)	Description of Milestone
Q4 2020	\$75,000 USD or \$61,182.04 Euro	Start-up: payable upon signing of this Agreement by all parties
Q1 2021	\$25,000 USD or \$20,394.01 Euro	At IRB/EC approval of 2 of the 4 sites
Q3 2021	\$25,000 USD or \$20,394.01 Euro	At IRB/EC approval of the other 2 of 4 sites
Q4 2021	\$50,000 USD or \$40,788.75 Euro	First 343 patients enrolled (can be from mix of the 4 sites)
Q2 2022	\$50,000 USD or \$40,788.75 Euro	Second 343 patients enrolled (can be from mix of the 4 sites)
Q4 2022	\$50,000 USD or \$40,788.75 Euro	Third 343 patients enrolled (can be from mix of the 4 sites)
Q2 2023	\$50,000 USD or \$40,788.75 Euro	Fourth 343 patients enrolled (can be from mix of the 4 sites)
Q4 2023	\$25,000 USD or \$20,394.01 Euro	Study analyzed; manuscript draft provided for review to BSC

In EU Member States, according to art. 44 of the Council Directive 2006/112/EC of 28 November 2006, the reverse charge mechanism may be applicable. In such case VAT should be self-assessed by the beneficiary of the service.

4. Certification of Milestone. Prior to payment, Institution shall send BSI documentation evidencing completion of the above milestones in writing to BSI to the Mailing Address or to the E-mail Address set forth

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in Section 5.01 (2. Other notices) of the Agreement. Payment for achievement of a milestone shall be made by BSI within thirty (30) days after documentation is received by BSI. Payment for achievement of a milestone shall be made by BSI within thirty (30) days after documentation is received by BSI.

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EXHIBIT B
PROTOCOL

**Prevalence of MDRO carriage in patients undergoing an ERCP in
three different countries**

With a comparison of MDRO presence in duodenum and rectum samples.

Judith A. Kwakman, M.D., MSc.^{1,2}

Prof. dr. Margreet C. Vos, M.D.²

Prof. dr. Marco J. Bruno, M.D.¹

1. Dept. of Gastroenterology and Hepatology, Erasmus MC University Medical Center, Rotterdam, The Netherlands.

2. Dept. of Medical Microbiology and Infectious Diseases, Erasmus MC University Medical Center, Rotterdam, The Netherlands.

Rotterdam, September, 2020

SUMMARY

Rationale: Over the last two decades it has become clear that one of the risks of endoscopic retrograde cholangiopancreatography (ERCP) is transmission of multi-drug resistant microorganisms (MDRO) through contaminated duodenoscopes. Since these instruments cannot be sterilized, they require an extensive, and unfortunately error prone, reprocessing technique. Despite several efforts to overcome this problem, new outbreaks are still being reported and new reprocessing techniques are not capable of producing a zero contamination rate (1). Recently, the FDA advised manufacturers and health care professionals to transition towards the use of disposable duodenoscopes (2). Using single use duodenoscopes would completely eliminate the risk of becoming infected or colonized by microorganisms derived from other patients. However, a complete transition towards disposable duodenoscopes requires a huge investment. Specifically in a country like the Netherlands, where MDRO prevalence is low, the question arises whether the burden of duodenoscope associated infections (DAI) outweigh the costs of these disposable duodenoscopes. In this study we will assess the prevalence of MDRO carriage in patients undergoing ERCPs. This will be carried out in three different countries since we expect the prevalence of MDRO carriage to vary between different areas. In the population of the Erasmus MC, we will look deeper into the possible transmission of microorganisms through duodenoscopes. In this population subjects will receive another rectum swab two weeks after

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ERCP and the duodenoscopes will be cultured before ERCP. Also, the microbiome of the duodenum will be assessed in this particular group and compared to the rectal samples.

Objective: To assess the prevalence of MDRO carriage among patients undergoing an ERCP.

Study design: Prospective observational multi-center study

Study population: Adult patients undergoing an ERCP (patients ≥ 18 years old)

Intervention (if applicable): Rectal swabs will be used to analyze MDRO carriage and questionnaires will be used to identify risk factors that can predict MDRO carriage. In the Dutch group, duodenum aspirate will be acquired during the ERCP procedure.

Main study parameters/endpoints: The main study endpoint is the percentage of patients carrying an MDRO when they undergo an ERCP.

Nature and extent of the burden and risks associated with participation, benefit and group relatedness: The burden and risks are expected to be very low, there are no expected risks associated with the collection of the samples. There are no benefits for the study participants.

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1) INTRODUCTION

a) MDRO outbreaks

Despite numerous efforts from manufacturers, government agencies and healthcare institutions, duodenoscope contamination is still a problem in current day ERCP practice. In previous studies by our department it was shown that 15% of ready-to-use duodenoscopes in Dutch ERCP centers was still contaminated with gut/oral flora(3). To this day, it remains unknown how often this duodenoscope contamination leads to duodenoscope-associated infections (DAI). We have to rely on outbreak reports to get an idea how often patients become infected through duodenoscopes. However, not much is known about incidental DAIs in non-outbreak settings. Most outbreaks of DAIs are attributable to MDROs (4). A few articles report DAIs based on susceptible microorganisms (5, 6), but presumably these DAIs are difficult to recognize since these microorganisms do not stand out.

b) MDRO prevalence in the Netherlands

A recent European overview on the prevalence of MDRO estimated that in 2015, 4982 cases of MDRO infections occurred in the Dutch population which accounted for 206 deaths and 35 disability-adjusted life-years (DALYs) per 100.000 population (7). This is lower than in the surrounding countries (Germany 75 DALYs, UK 90 DALYs, Belgium 125 DALYs) and even a lot lower compared to the overall numbers of the EU (190 DALYs). In 2015 the Netherlands had 16,9 million inhabitants. This means that in 2015, the chance of becoming infected by an MDRO was approximately 0.029%, in other words, 1 in every 3392 inhabitants became infected by an MDRO in that year. Although it is hard to define how much this risk is increased in hospitalized patients, especially in a tertiary center such as the Erasmus Medical Center, we assume the prevalence of MDRO to be higher in our population undergoing ERCP procedures.

In the MOVE study, another study in the Erasmus MC (results to be published), patients expected to be hospitalized for at least 48 hours were screened for MDRO carriage by collecting perianal swabs at the beginning and the end of their hospital stay. Of the 677 patients screened at admission, 33 (4.9%) carried Extended Spectrum Beta-Lactamase (ESBL) producing microorganisms. Eleven (2.9%) patients were negative at admission, but positive for ESBL at the end of their hospitalization. No carbapenem- or vancomycin-resistant microorganisms (CRE and VRE) were found at admission, only one (0.3%) patient carried a VRE at the end of their hospital stay.

c) Reusable duodenoscopes vs. disposable duodenoscopes

To eliminate the risk of becoming infected or colonized by microorganisms derived from other patients, it would be worth to consider using single-use, disposable duodenoscopes rather than reusable duodenoscopes. This might have profound economic consequences, partly because per-procedure costs of single-use, disposable duodenoscopes will be higher than that of reusable duodenoscopes. A recent cost analysis by Bang et al. (8) calculated that for large ERCP centers in the US the price of disposable duodenoscopes should be set at a maximum of \$612 to break-even with the costs of reusable duodenoscopes. For smaller centers this can be increased to \$1300 per duodenoscope. Disposable duodenoscopes are not on the market yet, so an official price is not yet available. However, the costs are expected to be somewhere around \$3000 per disposable duodenoscope. This would mean that a full transition towards disposable duodenoscopes would add significant costs to the healthcare system. The price of single use duodenoscopes could turn out to be lower, especially when more manufacturers will start selling them and try to compete with one another. Two manufacturers, Ambu and Boston Scientific, are expecting to bring out a disposable duodenoscope in 2020. Ambu already sells disposable bronchoscopes at a cost of \$300 per procedure (9). Another option to mitigate the additional costs associated with applying single-use duodenoscopes would be to use these duodenoscopes only in selected patients. We expect that in countries like the Netherlands, where MDRO prevalence is relatively low, a full transition towards disposable duodenoscopes will not be a cost-effective strategy and will encounter resistance among endoscopists. A scenario where disposable duodenoscopes will only be used in high risk patients, or patients known to be carrier of an MDRO could be a compromise to contain the spread of MDRO via contaminated non-disposable reusable duodenoscopes while controlling health care costs. No previous studies have measured the prevalence of MDRO carriage among ERCP patients.

d) Microbiome of the duodenum

If the plan to use disposable duodenoscopes on MDRO carriers would be implemented, all patients need to be screened on MDRO carriage. Most often this is tested through samples of the rectum and/or nose. Since the duodenum is far from both areas, one can wonder whether the outcomes of these tests reliably represent the

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presence of any MDROs in the duodenum. The microbiome of the duodenum has been investigated in a few studies, one of them comparing the microbiota in the duodenum to the microbiota in the rectum (10). This study revealed large differences in diversity of the microbiota in duodenum and rectum. With larger amounts of bacteria found in the rectum, but with less diversity compared to the microorganisms found in duodenum samples. If we would find no correlation between MDRO presence in the rectum samples and the duodenum samples, the strategy to use disposable duodenoscopes on MDRO carrying patients would become difficult to perform.

2) OBJECTIVES

a) Primary objective

- i)** What is the prevalence of MDRO carriage in patients undergoing ERCPs in three different countries?

b) Secondary objectives

- i)** What risk factors can be identified that predict carriage of MDRO in patients undergoing ERCP?
- ii)** What is the prevalence of colonization with MDRO after ERCP in patients that were free of MDRO carriage before ERCP?
- iii)** What is the prevalence of MDRO contamination of duodenoscopes in a Dutch tertiary care setting?
- iv)** Can we identify any transmission from contaminated duodenoscopes into patients?
- v)** Does the presence of MDRO in rectal swabs match the presence of these MDRO in duodenum samples?

3) STUDY DESIGN

a) Study design

This prospective observational study can be divided in two parts:

- The first part, which will answer the primary objective, will be conducted in three hospitals in three different countries (India, Italy and the Netherlands). Patients included in this study will be asked to provide a rectal swab sample and to fill out a questionnaire regarding potential risk factors for MDRO carriage.
- The second part will only be performed in the Erasmus MC (Rotterdam, the Netherlands). Here we will ask patients additionally to provide a duodenum fluid sample which will be acquired during the ERCP and a second rectal swab two weeks after the ERCP procedure. Also, the duodenoscopes will be sampled prior to ERCP procedure to measure the prevalence of MDRO contamination and to identify potential transmissions (see figure 1).

The microbiology department will be asked to store gut and oral microorganisms grown in clinically acquired patient cultures of patients who underwent ERCP. Patients will be cultured on clinical indication, which will not be influenced by this study. The resemblance between duodenoscope and patient strains will be reviewed retrospectively at the end of the study by molecular typing and whole genome sequencing.

Strains grown from the duodenoscope and patient cultures will be compared to each other to search for patterns and possible matches that indicate a transmission.

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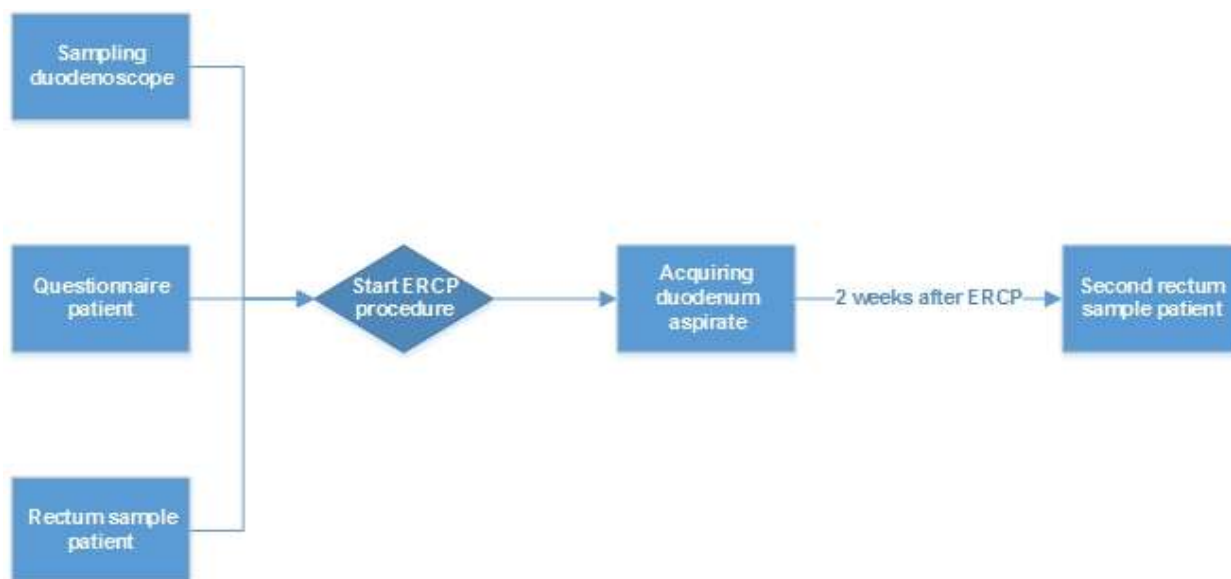


Figure 1. Flow scheme sampling of patients in the Erasmus medical center.

b) Sample size

The sample size was calculated according to the following formula and recommendations used by Naing et al (2006), specifically designed for prevalence studies(11):

$$n = \frac{Z^2 P(1 - P)}{d^2}$$

We used a confidence level of 95%, which means Z was set at 1.96. P is the expected proportion, for which we used the 4.9% as found in the MOVE study. As recommended by Naing et al, the precision (d) was set at $\frac{1}{2}P$ because P was smaller than 0.1. According to this formula, our sample size should include at least 286 patients. If we add 20% to correct for missing patients and double included patients, we learn that we will need to include at least 343 patients. If the percentage of the MOVE study is translatable to this population, this would mean we would find 17 MDRO carriers among these 343 patients. In the two foreign centers, we will also aim to include 343 patients per center. We expect that the number of MDRO carriers will be higher in these centers, however, we do not know how much higher. Studies on the prevalence of MDRO carriage among hospital patients are rare. The prevalence of colonization with ESBL-producing *Enterobacteriaceae* in Italian long-term care residents was measured to be 57.3% in 2015(12). In Dutch long-term care residents, the prevalence of ESBL-E was 14.5%(13). Although we expect the population undergoing ERCPs to have a lower prevalence of colonization compared to these long-term care residents, these numbers provide some insight in the large differences in prevalence of MDRO colonization between the Netherlands and Italy.

4) METHODS

a) Patient samples

The ESwabTM consists of a nylon flocked swab and a container which includes 1ml Amies medium. The use of such swabs is proven to be effective in detecting colonization of the digestive tract of both susceptible and drug-resistant microorganisms (14-16). A rectal swab has a higher detection rate compared to perianal swabs. One study (16) found a detection rate of 90.4% by rectal swabs for ESBL-positive patients, the other 10% were confirmed by cultures from other sites, mostly from urine samples. However, no other swab sample sites seem to have a higher detection rate. Rectal swabs will be collected prior to and two weeks after ERCP. The pre-ERCP sample will be collected at the moment the patient is prepared for the ERCP procedure. A trained nurse or student will acquire this sample before the duodenoscope is inserted. The sample two weeks after the procedure will be collected by the patients at home after

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instruction, the samples will be sent to the laboratory by mail. In case a patient is not able to collect the samples himself, the patient will be only included in the first part of the study.

b) Culturing patient samples

The medium of each swab will first be mixed with broths, either Enterococcosel broth with Amoxi 8mg/l and one with Vancomycine broth 50mg/l and both are incubated overnight at 35°C. The Enterococcosel broth will be inoculated on a VSA plate and incubated overnight at 35°C. The Vancomycine broth will be inoculated on a Chromid Carba Smart plate and an ESBL plate, both will be incubated over two nights at 35°C. Identification of microorganisms will be performed by the Maldi-biotyper. PCR and Vitek will be used to obtain the antibiotic resistance patterns. The standard laboratory operation procedure will be used to prove resistance patterns and –mechanism, including PCR on known resistance genes such as MeA, VanA, VanB, IMP, IMI, OXA23, OXA, 48 VIM1-2.

At the start of this research, patients are asked for their permission to store these samples for 15 years. In case these stored samples could be of use for other research purposes (to answer another question as they were initially collected for), patients will again be asked for their permission to use these samples.

c) Duodenoscope samples

Two separate samples will be collected from each duodenoscope:

- Swab of the distal tip (including forceps elevator, if present)
- Flush-brush-flush of the suction channel and biopsy channel

Sampling will be performed prior to the ERCP procedure according to a uniform sampling protocol, based on the Duodenoscope Surveillance Sampling and Culturing Protocols of the CDC (17). In contrast with the CDC protocol, but in line with the Dutch Guideline for monitoring of the microbiological safety of heat-labile flexible high-risk gastrointestinal endoscopes(18), a sample of the distal tip will be collected by using a flocked swab with a separate transport medium. 1ml of DE broth will be added to the swab medium after collection. This approach is chosen to enable differentiation between colonization of the channel and the distal tip. The flush-brush-flush method will use two flushes of 20ml sterile water which will be flushed through the suction channel and the working channel. The brush head (Pentax CS5522A) will be cut and collected in the flush sample as well. After both flushes are collected, 40 ml of neutralizer (Dey-Engley broth) will be added to the sample.

d) Culturing duodenoscope samples and storage

Culturing will be performed according to a protocol based on the Dutch guideline (18). The swab samples will be placed on blood agar plates. The flush samples will be filtered using a 450 µm filter and inoculated on R2A agar. Samples are incubated at 35°C and examined for growth for 72 hours (three days). The culture result at 72h will be used for the primary endpoint. Culture results are presented in Colony Forming Units (CFU)/20ml per microorganism per sample site. Cultures with >20 CFU/20ml or containing gut or oral flora will be included as a positive test result. Cultured gut and oral flora will be stored as separate isolates at -80°C for future molecular typing.

e) Microbiome of the duodenum

During the ERCP procedure, 20 ml of duodenum aspirate will be collected by the endoscopist. This sample will be used to assess the microbiome of the duodenum. In this microbiome we will search specifically for genes that encode for antibiotic resistance. These results will be compared to the results of the rectal swabs in order to find any resemblance in presence of MDROs at these two sample sites. First, in a pilot run, 20 duodenum samples will be used to test the isolation procedure. If enough material can be retrieved through this method, the microbiome of all samples will be assessed. Depending on the results of the first pilot, we will use v3-v4 amplicon flow cells with 50.000 reads to assess the microbiome.

f) Patient data

The microbiology department will store all gut and oral flora microorganisms grown in patient cultures of included patients from the ERCP date up till six months post-ERCP. We will compare duodenoscope

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cultures with clinical isolates to detect infections caused by a treatment with a contaminated duodenoscope. Matching microorganisms in patient cultures and duodenoscope cultures on species level will be subjected to molecular typing.

The electronic patient files system HIX will be used to collect information on infections if we discover a link between an infection and a contaminated duodenoscope used in that patient. This specific information includes medical history, medication and treatment of the infection. Patient data will be collected using the data system Open Clinica and will contain non-identifiable data (no names, birth dates, etc.).

g) Molecular typing

If a match is found between a contaminated duodenoscope and an infection in a patient treated with that contaminated duodenoscope, molecular typing will be performed on these samples to determine whether they are indistinguishable. Depending on the type of microorganism found in the cultures, different types of PCR can be used to perform molecular typing. For the most common found gut microorganisms, amplification-fragment length polymorphism (AFLP) or multi-locus VNTR analysis (MLVA) can be used to distinguish microbial isolates from each other. If a match is found through molecular typing, whole genome sequencing (WGS) will be performed on those samples. If WGS reveals a strong resemblance between the samples, this case will be scored as a DAI. Not all species can be typed as we foresee now, but new techniques will be developed when needed.

h) Primary outcome

The primary outcome of this study is a prevalence of MDRO carriage among patients undergoing ERCPs. We expect that the prevalence will be different for the three inclusion centers.

i) Secondary outcomes

- The questionnaires will be used to identify risk factors predicting MDRO carriage in patients undergoing ERCPs.
- In the second part of the study, a prevalence will be calculated for the number of patients becoming colonized with MDRO after ERCP. We will look into the duodenoscope cultures to identify whether MDROs were present on the duodenoscopes prior to ERCP.
- The microbiome of the duodenum will be assessed to compare the presence of MDRO in rectum samples with their presence in the duodenum.
- The duodenoscope samples will also be used to calculate the risk of performing ERCPs with a contaminated duodenoscope.
- In the Dutch cohort we will also look for potential DAIs, if we find any, we can provide a risk calculation of developing a DAI per ERCP procedure.

j) Ethical approval

The patients will be informed about the study and asked for their consent before the ERCP procedure. Informed consent is needed to collect the rectal samples, duodenum sample and in case of a possible DAI, to look into the patient files. Patients can leave the study at any time for any reason if they wish to do so without any consequences.

5) Budget

Item	Number	Years	Price per unit	Total	
Execution of the study/Data management	1	1,75	€ 29.925,0	€ 29.925,0	
Statistical analyses	1	0,25	€ 1.137,5	€ 1.137,5	
Data managing for culturing center 1 NL	1	1	€ 17.500,0	€ 17.500,0	

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Data managing for culturing center 2 Italy	1	1	€ 17.500,0	€ 17.500,0	
Data managing for culturing center 3 India	1	1	€ 17.500,0	€ 17.500,0	
Legal costs	1,00		€ 2.500,0	€ 2.500,0	
METC	1,00		€ 5.000,0	€ 5.000,0	
Article processing charge for open access publication	1,00		€ 3.000,0	€ 3.000,0	
Congress costs	3,00		€ 1.000,0	€ 3.000,0	
Culture costs per duodenoscope - labor and materials. 1x swab forceps elevator 1x brush 1x flush	343,00		€ 81,8	€ 28.040,3	
Culture costs per patient - labor and materials (Italy and India). 1x rectal screening swab	686,00		€ 17,0	€ 11.662,0	
Culture costs per patient - labor and materials (Netherlands). 2x rectal screening swab with transportation	343,00		€ 36,0	€ 12.348,0	
Microbiome testing duodenum samples	343,00		€ 9.567,0 (per total)	€ 9.567,0	
Storage costs duodenoscope cultures	343,00	15	€ 1,0	€ 5.145,0	
Storage costs patient cultures	1372,00	15	€ 1,0	€ 20.580,0	
Strain determination duodenoscope cultures (15% MGO pos.)	51,45		€ 20,0	€ 1.029,0	
Strain determination patient cultures	275,00		€ 20,0	€ 5.500,0	
PCR molecular typing	275,00		€ 11,0	€ 3.025,0	
Whole genome sequencing on potential DAI cases	30,00		€ 200,0	€ 6.000,0	
Subtotal					€ 199.958,8
Unexpected expenses	10%				€ 19.995,9
Overhead	20%				€ 39.991,8
Total					€ 259.946,4

6) References

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Grant for Clinical Research

This agreement for grant for scientific research (the "Agreement") is entered into the day of last signature below, by and between **PENTAX Europe GmbH**, a company organized under the laws of **Germany** with registered office at Julius-Vosseler-Strasse 104, 22527 Hamburg (hereinafter referred as "Company"), represented by **Dr. Kai Wolter**, and Erasmus University Medical Center Rotterdam, an institution organized in accordance with public law of the Netherlands (article 1.13,2 WHW), with principal place of business at 's-Gravendijkwal 230, 3015 CE Rotterdam, The Netherlands, acting for and on behalf of its Departments of Gastroenterology & Hepatology and Medical Microbiology & Infectious Diseases, (the "Institution"), lawfully represented by the Head of the Departments of respective Departments, acting on request of Prof. Dr. Marco Bruno (hereinafter referred to as "Investigator"), with reference to the following background :

- A. Company is engaged in the research, development and commercialization of medical devices in the field of duodenoscopy (hereinafter referred to as the "Field") ;
- B. Institution has certain skills and abilities in regard to the field of hospital management and scientific research and has the facilities and staff required for the performance of scientific research and has interest in participating in and/or conducting scientific research activities for purpose of the development and/or improvement of diagnostic, therapeutic and interventional procedures, in order to further medical science and improve patient care, in a manner consistent with the status of a university hospital and an academic research and teaching institution;
- C. Company wishes to support Institution in its objectives by the provision of support, the particulars of which are set below;
- D. Institution is authorized to enter into agreements for grants for scientific research, including this agreement.

In consideration of the mutual obligations hereinafter set forth, the parties hereto (Company and Institution, hereinafter also each referred to as "Party" and jointly as "Parties") expressly agree as follows:

1. Subject Matter

Company agrees to contribute to the costs of the clinical research with protocol (Appendix A) and title: "Validation of the reprocessing efficacy of a novel design duodenoscope contaminated with artificial soil", performed by Institution (further referred to as the "Study"), more specifically under the supervision of the Investigator.

2. Support

2.1 Financial support

A total amount of **149000** Euro (exclusive of VAT) shall be paid by Company to Institution as financial support in order to perform the above mentioned research relevant for the Study.

This amount shall be due before the actual start of the Study, to be paid within 14 days as from the respective invoice of Institution.

The amount shall be transferred to the following bank account:

Name:	Stichting Lever- en Darm en Maag Onderzoek (SLO)
VAT identification number:	80391207B01
Address:	Postbus 2040, 3000 CA Rotterdam
Payment currency:	EUR
Payment Method:	Bank transfer
Bank account holder:	Stichting Lever- en Darm en Maag Onderzoek (SLO)
Bank account number:	0148753663
IBAN:	NL52RABO0148753663
Bank name:	Rabobank Rotterdam, Postbus 10017, 3004 AA Rotterdam
Bank identifier code (SWIFT):	RABONL2U
Communication:	structured message on invoice: Pentax-REDUCE Study

In case services provided under this Agreement should be subject to VAT, the Institution shall be entitled to charge VAT at the legal rate in addition to the fees stated in this Agreement, provided the VAT is stated separately on the invoice made out to Company.

2.2. Additional support

2.2.1 The Company shall provide Institution with the necessary Material including all know-how regarding such Material as to enable Institution to perform the Study. Said know-how includes detailed instructions regarding the use and storage of Material.

2.2.2 It is understood that all costs (including but not limited to transportation costs and taxes, if any) related to the delivery of the Material to Institution will be borne by the Company. The transportation and importation into The Netherlands of the Material will be the responsibility of the Company. All and any documents and permits required for or in connection with the delivery of the Material and use thereof by Institution (other than Medical & Ethical approval) shall be arranged for by the Company at its sole risk and expense. The Material will be delivered at the premises of Institution free of charge and timely in form and substance as set out in Annex A to this Agreement, in such time as not to affect the timely start and continuation of the Study.

The Company undertakes to support the Institution by lending to it:

- 1 Pentax ED34-i10T2 duodenoscope
- 1 Pentax ED34-i10T duodenoscope
- 1 Olympus TJF-Q180V duodenoscope
- 1 older duodenoscope for method validation (for example Pentax ED-3490TK)

hereinafter referred to as the "Material".

for the purpose of the performance of the Study which will remain in the premises of the Institution until the Study is finished (but in no way longer than until the end of term of this Agreement) free of charge.

The Institution agrees and undertakes that any equipment provided to it by the Company shall be used only and exclusively for the study purpose, and in reliance on that representation the Company will take care of the maintenance and any service related costs during the time the equipment remains in the hospital.

3. Obligations of Institution

3.1 The Institution and the Investigator will use their reasonable efforts to carry out the Study as defined in Annex A to this Agreement fully in accordance with all applicable rules, regulations and academic standards and will use all reasonable effort to provide the

facilities, staff and personnel required for such purposes.

3.2 The Institution and the Investigator undertake this Study in furtherance of their goal of seeking new knowledge and furthering the tasks and objectives of Erasmus MC as a research and teaching institution and university medical centre. However, neither Erasmus MC nor the Investigator guarantees that the work carried out under or pursuant to this Agreement will lead to any particular result. Furthermore it is understood that Erasmus MC does not guarantee the success of the research efforts performed under this Agreement.

3.3 The institution shall comply with the actions and shall strive to comply to the following:

- The STUDY will be initiated within one (1) month AFTER receipt of material and the recruitment and start of a new PhD-student, WHICHEVER OCCURS LAST.
- The Study is expected to take a total of 13 months as shown in table below (Table 1)

	Time
Activity A: Study preparation	
Pilot phase A: Soil composition	3 months
Pilot phase B: Soiling method	1 month
Pilot phase C: New duodenoscopes	1 month
Activity B: Conduction of the study	5 months
Activity C Analysis and publication	3 months

Table 1

- The Study shall be completed within 15 months of execution of the Agreement.

3.4 The Institution will inform the Company of the progress of the Study upon written request. The Institution will provide the Company with a final written report (hereinafter the "Report") of the Study in English, not later than 6 months after the end of the Study. The final report shall be send to the Company by e-mail in pdf.

4. IP-Rights, Publication and Confidentiality

IP-Rights

4.1 All rights related to Results and Inventions under this Agreement will vest in Erasmus MC.

For the purpose of this Agreement "Results" shall mean the specific findings during the Study in the Field, including the Study data; "Invention" shall mean a patentable improvement or invention in the Field made by Institution as a direct result of the performance of the Study, including and limited to Inventions that directly relate to the Material provided by the Company.

In case of Results within the Field, same shall be notified to the Company and the Company shall be free to use the said Results for its own internal research purposes, only.

In case of Inventions within the Field from the side of Erasmus MC, Company will have a right of first refusal to negotiate in good faith the terms and conditions which will be applicable for the use of such Inventions. This right of first refusal will be valid or a period of 6 months as from the day on which the respective Inventions were made known to the Company for the first time, by or on behalf of Institution. If no agreement has been reached or if the Company has not reacted to the information regarding the Invention provided by or on behalf of Institution, Institution will be free to enter into

negotiations and/or agreement with any third party.

The rights related to Results and Inventions outside the Field will vest in Erasmus MC, for use at its discretion.

It is understood by both Parties, that Institution retains at all times the right to use Results and Inventions for its scientific and educational purposes.

4.2 In addition to rights granted in art. 4.1 here above,

The Institution hereby grants to the Company a non-exclusive license to the Company and any of its Affiliates to use the Study protocol (Annex A) and the Report or any part thereof including the material incorporated in the Report (photographs/videos/digital images and/or testimonials) (in the following collectively referred to as the "Licensed Documents") provided by him as follows:

Company is granted a non-exclusive license to use the Licensed Documents irrevocably and unlimited, especially worldwide and indefinite.

The Investigator and the Institution further agree that the name of the author, his title and the name of the Institution may be revealed in descriptive text or commentary in connection with the Report.

Affiliate of the Company for the purpose of this Agreement means any entity that directly or indirectly controls, is controlled by or is under common control with the Company, control for the purpose of this definition meaning direct or indirect ownership or control or more than 50% of the voting rights of the other entity.

Publication

4.3 Company recognizes the desire of Institution to publish results of academic research in scientific journals. Institution undertakes and agrees not to engage in any dissemination of data from and Results of the Study, without first obtaining the written consent of Company, which consent shall not be delayed and shall be given within a period of 30 (thirty) days from the receipt of Institution's written request for consent. In case Company has not responded in writing to such request within the above mentioned term, Institution will be free to proceed with the publication without any further delay.

Without prejudice to the paragraph here above, Company shall have the right to withhold such consent only in case the publication:

- a. Contains information that may be subject of a patent application, in respect of which no patent application has yet been filed;
- b. Contains confidential proprietary information originating from Company.

It is understood and agreed that in case of (a.) such withholding of consent will not lead to a delay of publication for more than three months as from the date on which the text of the publication has been sent to Company for the first time.

In case of (b.) Company will inform Institution within the thirty (30) day period stating that the proposed publication contains confidential proprietary information of Company. In such case Company will indicate the proprietary information and Institution will then adapt the proposed publication in such a way that it will not publish the proprietary information of Company.

In oral presentations or written publications concerning the Study, Institution will acknowledge Company's contribution to the Study, unless otherwise requested.

It is expressly understood between the Parties, that Company shall not publish any information regarding the Study or disseminate any results of the Study without Institution's prior written consent before Institution has been able to publish such information as a Science Citation Index (SCI) Publication:

- (a) SCI Publication shall mean that the publication has been fully referenced in the SCI with volume and page numbers;
- (b) The publication of a mere abstract shall not be considered a SCI Publication.

It is understood, that Company will not use the name of the Erasmus MC and/or the Investigator in any advertising or other form of publicity, without the express prior written permission of same institutions/persons.

Confidentiality

4.4 Without prejudice to article 4.3, Parties agree not to transfer, transmit or in any other way disclose and/or use for its own benefit any information that will be provided by or on behalf of the other Party, which is marked "Confidential" and or any information which is reasonably deemed to be confidential to any third party without the written consent of the other party for a period of five (5) years after termination of this Agreement.

The disclosure of confidential information for purpose of this Agreement can occur in writing as well as oral, during meetings or telephone conversations with representatives of Company. All oral disclosures by either Party to the other Party, which are then summarized in writing within fifteen (15) working days after the disclosure, are also covered by the provisions on confidentiality under this Agreement. Each Party shall protect said confidential information with the same degree of care as it applies to protect its own confidential information.

The obligations of confidence referred to here above shall not include or extend to any information which: (i) is or becomes generally available to the public otherwise than by reason of breach of this Agreement by one of the Parties; (ii) is known to Parties and is at its free disposal prior to its receipt from the other Party; (iii) is subsequently disclosed to one of the Parties from sources other than the disclosure by the other Party pursuant to this Agreement.

4.5 For the purpose of clarification the Parties explicitly state that no further fee or other monies shall be paid by Company for the rights granted under this § 4, but they shall be included in the financial support, mentioned above under § 2 (1).

5. Statement of purpose

This grant is not being given in exchange for any explicit or implicit agreement to purchase, prescribe, recommend, influence or provide favorable formulary status for any of Company's products. The amount of the grant is based upon the level of support to a scientific research that Company has chosen in its sole discretion to provide to Institution.

6. Independence of Institution

This Agreement does not in any way constitute an employment contract or agency agreement. Vis à vis Company, the Institution is an independent contractual partner and does not through this Agreement acquire any rights incompatible with this status.

7. Term

The term of this Agreement shall commence as of [25 February 2018 / date of last signature below] and shall continue in full force and effect until completion of the research and publication of the results according to Table 1.

Performance may be terminated by Institution at any time, if circumstances beyond its control preclude continuation of the Study. Upon termination of the Agreement, Institution will be reimbursed by Company for all costs and non-cancellable commitments incurred in the preparation and performance of the Study, such reimbursement not to exceed the total maximum costs as per Section 2 of this Agreement. For the avoidance of doubt; it is understood by the Parties, that the full costs of staff which was hired and/or deployed by Institution for the performance of the Study, are non-cancellable commitments of Institution.

8. Entire Agreement

This Agreement including the Appendix A contains the entire agreement of the Parties with respect to the subject matter hereof and may not be modified except in writing signed by all Parties. No party to this Agreement may assign or transfer its rights or obligations hereunder without the prior written consent of the other Party hereto.

9. Amendments

No amendment or modification of this Agreement shall take effect unless it is in writing, executed and delivered by representatives of each of the parties duly authorized to that effect.

10. Limitation of Liability and Choice of law

It is expressly understood, that nothing in this Agreement shall be construed as a warranty of the quality of the Results or fitness of the Results for any particular purpose. The performance of scientific research by Institution is experimental in nature. Institution's involvement in the Study is limited to "best endeavours" only. Institution shall conduct the Study in accordance with the principles governing good academic research practice. As a consequence hereof Institution and/or its Investigator cannot and shall not be held liable for any claims or damages by Company or any third party, in connection with or as a result of the Study, except in case of gross negligence or willful misconduct on the side of Institution. Company undertakes to indemnify and hold harmless Erasmus MC and its Principal Investigator at all times against all of such damages or claims, except in case of gross negligence or willful misconduct on the side of Erasmus MC.

This Agreement, its validity and interpretation, including the legal relation between the parties to it, shall be governed by and interpreted under the laws of The Netherlands. Any and all disputes between the Parties, which cannot be settled amicably, shall be subject to the exclusive jurisdiction of the Court in Rotterdam, the Netherlands.

IN WITNESS WHEREOF, the Parties have caused this Agreement to be executed in [number of copies, spelled] (x) original copies.

On behalf of COMPANY

On behalf of Institution

Dr. Kai W o l t e r

Prof. dr. Marco J. Bruno

General Manager QARA EMEA
Compliance Officer

Head of the Department of
Gastroenterology & Hepatology

Prof. dr. Annelies Verbon

Head of the Department of Medical
Microbiology & Infectious Diseases

Prof. dr. Margreet C. Vos

APPENDIX A: PROTOCOL

(Most recent version of the protocol added by reference)

Erasmus MC
Raad van Bestuur
Mr Prof. dr. J.P.T.M. van Leeuwen
Postbus 2040
3000 CA ROTTERDAM

Laan van Nieuw Oost-Indië 334
2593 CE Den Haag
Postbus 93245
2509 AE Den Haag
Phone 070 349 51 11
Fax 070 349 51 00
www.zonmw.nl
info@zonmw.nl

Filenumber
50-52900-98-062
Our reference
2019/30830/ZONMW

Subject
approval of your grant application, project number 549009005

Date
December 12, 2019

Dear Mr Van Leeuwen,

Contact
Linda van Gaalen
Telefoon 070 349 51 57
lgaalen@zonmw.nl

On November 12, 2019 you submitted a grant application to ZonMw entitled: *A Smart Surveillance Strategy for Carbapenem-resistant Pseudomonas aeruginosa (SAMPAN)*. This project aims to develop a globally-applicable smart surveillance strategy to guide action against the spread of CR-PA, with a focus on the human-water interface.

We are pleased to confirm positive assessment by the Peer Review Panel (PRP) of your application, of which the JPIAMR Call Steering Group has informed you.

This positive assessment means that ZonMw will grant the funding of this project, including a contribution from NWO-WOTRO for LMIC activities. Project number **549009005** has been assigned to this project. This letter explains which ZonMw terms and conditions apply to your grant, and what you are required to do before your project can commence.

JPIAMR Assessment

The assessment of your application was carried out by the PRP and was assessed in terms of quality and relevance to the joint call requirements. JPIAMR received 34 full applications in total, of which 12 proposals have been recommended for funding. Five grant applications with Dutch participants have been approved. ZonMw adopts the assessment and decision of JPIAMR and refers you to the JPIAMR email informing you of the outcome of the review procedure letter, from the call secretariat, for further information and details.

Funding

Amount of the grant

ZonMw's funding for your project will be for a total amount of € 298.986,60 for a maximum period of 36 months. This amount includes any payable VAT.

With this allocation, we agree with the hourly rates that have been specified. We hereby would like to emphasise that this approval of hourly rates only applies to the grant associated to this specific project and that no rights can be derived from this for future ZonMw grants.

However, we would like to ask you to submit a clear substantiation of the rates that are applied by the RIVM in the final financial statement.

We would also like to point out that you are obliged to keep track of time registration. Additionally, we ask you for a clear specification of the "material costs, equipment and consumables" in the final financial statement.

Grant conditions

As you know, the funding is subject to certain conditions. These grant conditions can be downloaded from the ZonMw website:
www.zonmw.nl/subsidievoorwaarden

Besides using the ZonMw logo, you are requested to include the JPIAMR and NWO logo as well when presenting your project. These can be downloaded from <http://www.zonmw.nl/nl/over-zonmw/logo-huisstijl/>; <https://www.jpiaamr.eu/document-library/support-material/>; <https://www.nwohuisstijl.nl/home-page/logo/het-logo/>. The same applies to publications: you are asked to mention ZonMw as well as JPIAMR and NWO.

Moreover, the Dutch law is applicable to this grant.

I would like to emphasise that ZonMw can only pay an advance when all requirements for conducting research are met. Please start any procedures in time, such as a positive determination from a recognised medical research ethics committee (Medisch Ethische Toetsings Commissie), the Dutch Central Committee on Research Involving Human Subjects (Centrale Commissie Mensgebonden Onderzoek), a project license of the Central Committee Animal Experiments (Centrale Commissie Dierproeven), or a license granted in accordance with the Population Screening Act (Wet Bevolkingsonderzoek). If you are unsure whether your project requires such, please check with the relevant authorities.

Integrity

Article 2, paragraph 3 of the General Terms and Conditions Governing Grants of ZonMw implies that the nationally and internationally accepted standards regarding scientific conduct - as laid down in the Dutch Code of Conduct for Scientific Practice (VSNU, last revised October 31, 2014), or similar codes for non-university institutions - are respected. ZonMw should be informed immediately in case of (potential) violation of the aforementioned standards in a project funded by ZonMw. All relevant documents must be submitted to ZonMw upon request.

ZonMw determines that the annex "Agreement funding scientific research 2008" and the addendum, according to Article 7 of the agreement, is not fully applicable to the funding. They will be applied as similar as possible, whilst not conflicting with the ZonMw General Terms and conditions. The General Terms and Conditions are always leading. For example, ZonMw always settles financial accounts based on actual costs.

What should you do now?

Important: written confirmation within four weeks

ZonMw can provide you with an advance payment for the first project year. This is only possible, however, if you have agreed to the grant conditions and once your project has actually commenced. We would thus request you to submit the following information in writing to ZonMw before **January 14, 2020**. You can use the enclosed information form for this purpose:

- Your agreement to the conditions applicable for allocation of the funding;

- The starting date of your project;
- The bank and reference data for the grant payments;
- With respect to approval by METC or CCD:
 - If no declaration(s) is (are) required, please send written confirmation of this.
 - If the declaration(s) is (are) required for the commencement of the project, send the declaration to ZonMw before the start of the project.
 - If the declaration(s) is (are) not required until later in the project, indicate when the declaration(s) is (are) necessary. This may be no later than one year after commencement of the project. ZonMw will then make an advance payment for the first year. ZonMw can make further advance payments only if a copy of the declaration(s) has been received.

You can use the enclosed information form (Report form for commencement of project) for this purpose.

I would like to point out that the project must commence **six months at the latest** after the date of this letter. If the project commences after this date, then our approval of your application will expire. Exceptions to this principle are possible only under very exceptional circumstances.

Consortium Agreement

A final draft of the consortium agreement (approved by all project partners, but not yet signed) needs to be submitted **within six months after your project proposal has been awarded a grant**. ZonMw will assess and has to approve the arrangements made for compliance with the applicable European law on state aid and ZonMw's general grant terms and conditions.

In this consortium agreement, you need to specify at least all items as described under legal aspects on the [ZonMw website](#) and the [ZonMw general terms and conditions](#). Here you can also find a skeleton agreement which you can use.

Sponsorship of a third party is allowed under the condition of a sound sponsorship agreement in which the role, involvement, any license agreements, foreground and background IP arrangements and financial commitments of the co-financing partner(s) are described.

ZonMw reserves the right to reverse the grant decision if the consortium agreement and/or sponsorship agreement (proof of co-financing) are not established, or not acceptable under the European law on state aid and/or ZonMw's general grant terms and conditions.

If you have any questions on this matter, contact your intellectual property (IP), contract and/or legal advisor of your organisation's valorisation department or technology transfer office (TTO). ZonMw advises you to involve your IP, contract or legal specialist in your application at the earliest possible stage.

Public Summary

ZonMw publishes all approved projects on its website with an accessible summary. This summary is intended for a broad, interested readership whose language skills are roughly equivalent to that of pupils who have completed pre-university secondary education (VWO). For further details, please refer to the style guide at <http://www.zonmw.nl/nl/over-zonmw/logo-huisstijl>.

We kindly request that you submit this summary as soon as possible, but **at least within four weeks after the date of this letter**. For this purpose, you can use the text field in ProjectNet with the heading Public Summary, which limits you to 1,000 characters including spaces.

Progress Report

ZonMw would like to remain informed about the progress of your project. Approximately halfway the duration of your project the programme secretary will send you a request to submit a progress report. You are requested to fill in the ZonMw midterm format, which, if relevant, you can base on the midterm report JPIAMR requires you to submit. In addition, you are obliged to report any interim changes to ZonMw in a timely manner. The changes are permitted only with the approval of ZonMw.

Utilisation of knowledge

Results from the research project can be implemented, play a part in policy making, become the next step in a scientific career or constitute the basis of a new project.

In order to gain insight in the use of the results from the study, we request information on the implementation and spread through several questions in the progress report(s) and final report.

For a period of four years following the conclusion of the study you are obliged to submit all publications and results of the project into ProjectNet.

During this period you are obliged to inform ZonMw of the use of these results.

Datamanagement

To allow future reuse of data from your project, ZonMw requires every grant recipient to draw up a data management plan. You can find information about the procedures and tools on [FAIR data and data management > Data management in your research project](#). Starting from this link, follow the steps. ZonMw does not review the plans. ZonMw monitors the outcome of your data management during and at the end of your project with a number of key items for FAIR data. Please use this [list for providing the information on key items](#).

Please send the data management plan and the key items (as far as you know them) as PDF files within 3 months to jpiamr@zonmw.nl. During your project you can make changes and updates.

In case you will not collect data, please inform ZonMw.

If this letter gives rise to any questions, please contact the individual listed in the letterhead. Please mention your project number in all correspondence with ZonMw. Now that your application has been approved, the original number expires and you are now assigned the following project number: **549009005**.

Once again I congratulate you on the approval of your grant application. I wish you success in carrying out your project!

Yours faithfully,
on behalf of the board,


✶ Hanne Bonink
programme director

Enclosure(s)

Report form for commencement of project (intended for project leader / representative)

Copy

Dr. J.A. Severin, Erasmus MC, Medische Microbiologie en Infectieziekten

Beste Dineke, Marian,

'Hierbij mijn akkoord voor de begroting "budget 0812" dd. 08-12-2021 11:41'.

Groet,
Mark

Beste Mark,

Graag je akkoord op de begroting van het project Onderzoek stationstoiletten zoals te vinden op de gezamenlijke schijf onder SDE ('Sustainable Design Engineering'), map 021-165.
Het project begint per 1 januari as en zal 5 maanden duren.

Als je nog vragen hebt hoor ik het graag.

Vriendelijke groeten,
Dineke

Dineke de Visser
Contract Manager

TU Delft - Faculty of Industrial Design Engineering

Overeenkomst Speciale Leerstoel Installatietechniek WOI-TU Delft (versie 0.7)

Partijen:

Stichting ter Bevordering van Wetenschappelijk Onderwijs en Onderzoek op het Gebied van de Installatietechniek, gevestigd te Korenmolenlaan 4, 3447 GG Woerden, ten dezen vertegenwoordigd door de heer A.L.A. van Gelder, voorzitter, hierna te noemen: WOI

Technische Universiteit Delft, gevestigd te Delft, vertegenwoordigd door de Rector Magnificus / Voorzitter, de heer prof. dr. ir. T.H.J.J. van der Hagen, hierna te noemen: TU Delft

Hierna gezamenlijk te noemen: Partijen

Overwegingen

1. Partijen hebben een Convenant Installatietechnisch Onderzoek WOI-TU Delft afgesloten, waarin de instelling van een speciale leerstoel een kernpunt is;
2. TU Delft stelt een door WOI gefinancierde speciale leerstoel in, die de kern zal vormen van een onderzoeksprogramma, dat geïnspireerd is op de ontwikkelingen in en behoeften van de praktijk. Bij dit programma zijn bedrijven uit de sector betrokken, zowel ontwerp, advies en uitvoering, en andere kennisinstellingen en onderwijsinstellingen, met name het HBO.

Artikel 2. Leerstoel

2.1 Speciale leerstoel

TU Delft zal een tijdelijke (voor 5 jaar) speciale leerstoel installatietechniek bij de Faculteit Bouwkunde van TU Delft instellen en hierop een hoogleraar benoemen. Het profiel voor de leerstoel zal door TU Delft met input van WOI worden opgesteld. De omvang van de benoeming bedraagt ten minste 0,3 fte en de benoemingsduur ten minste 5 jaar.

2.4 Uitvoering en kosten

WOI zal ten behoeve van en met medewerking van bedrijven in de installatietechniek en van andere intermediaire organisaties en brancheorganisaties in de installatietechniek de kosten financieren die gemoeid zijn met het aanstellen van een hoogleraar op bedoelde speciale leerstoel voor de bedoelde looptijd en omvang van de aanstelling, met een aantal bijkomende kosten voor een bedrag van maximaal 291.000,00 EUR. WOI betaalt hiervoor aan TU Delft, te beginnen in de maand van de benoeming van de bedoelde hoogleraar en voor een aaneengesloten periode van ten hoogste 60 (zestig) maanden een bedrag van EUR 4.850,00.

Mocht de benoeming tussentijds worden beëindigd en de leerstoel vacant raken, dan is WOI voor de periode waarin geen hoogleraar is benoemd geen betaling aan de TU Delft verschuldigd.

Annex: III Letter of support from department head of the main lead



Direct dial +31 10 70 33510
Fax number +31 10 70 33875
Room number Na-916
E-mail a.vanrossum@erasmusmc.nl
Our reference 22
Date

Call for proposals: Call for Flagship Programs – Convergence Alliance

Letter of commitment

This letter is to confirm that Department of Medical Microbiology and Infectious Diseases (MMIZ) at Erasmus Medical Center Rotterdam, represented by Prof. Dr. Annemarie van Rossum is committed to participate in the Flagship Convergence project coordinated by the Prof. Dr. Margreet Vos, of the MMIZ Erasmus MC entitled "From macro to micro-level; a microbial safe environment for hospital patient care; the MI-Safe-Hospital flagship".

The total project budget is 2.933.307,00 euros and MMIZ will provide in terms of contribution to the costs/other of the project, 850.765,00 euros a cash contribution, and/or in kind contribution of 0 euros.

Sincerely,

Prof. dr. A.M.C. van Rossum

Department of Medical Microbiology & Infectious Diseases

Erasmus MC Rotterdam

Postal address
Postbus 2040
3000 CA Rotterdam

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam

Parking
Wytemaweg 12
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Westzeedijk 361
3015 AA Rotterdam

Head of Department
Head a.i.:
Prof. dr. A.M.C. van Rossum

Diagnostical Unit
Head: A. Luijendijk
Med. Coordinator:
Dr. M. van Westreenen

Unit Infection Prevention
Head: A. Luijendijk
Med. Coordinator: Dr. J.A. Severin

Unit Research & Development
Head.: Dr. J.J. van Hellemond

Unit Clinical Infectiology
Med. Coordinator:
Dr. N.J. Verkaik
Med. Coordinator:
Dr. T.E.M.S. de Vries-Sluijs



Annex IV: References

IV References

- ¹Pirzadian J., S. P. Harteveld, S. N. Ramdutt, W. J. B. van Wamel, C. H. W. Klaassen, M. C. Vos, and J. A. Severin. 2020. *Novel use of culturomics to identify the microbiota in hospital sink drains with and without persistent VIM-positive Pseudomonas aeruginosa*. Sci Rep. 10:17052.
- ²Rauwers A. J., A. F. Voor in 't holt, J. G. Buijs, W. de Groot, B. E. Hansen, M. J. Bruno, M. C. Vos. 2018. *High prevalence rate of digestive tract bacteria in duodenoscopes: a nationwide study*. Gut. 67:1637-1645.
- ³Verfaillie C. J., M. J. Bruno, A. Voor in 't holt, J. G. Buijs, J. W. Poley, A. J. Loeve, J. A. Severin, L. F. Abel, B. J. Smit, I. de Goeij, and M. C. Vos. 2015. *Withdrawal of a novel-design duodenoscope ends outbreak of a VIM-2-producing Pseudomonas aeruginosa*. Endoscopy. 47:493-502.
- ⁴Voor in 't holt A. F., J. A. Severin, E. M. Lesaffre, and M. C. Vos. 2014. *A systematic review and meta-analyses show that carbapenem use and medical devices are the leading risk factors for carbapenem-resistant Pseudomonas aeruginosa*. Antimicrob Agents Chemother. 58:2626-37.
- ⁵World Health Organization. Health care-associated infections fact sheet. Available from: https://www.who.int/gpsc/country_work/gpsc_ccisc_fact_sheet_en.pdf
- ⁶European Centre for Disease Prevention and Control. 2018. Incidence and attributable mortality of healthcare-associated infections in intensive care units in Europe, 2008-2012. Stockholm: ECDC.
- ⁷European Centre for Disease Prevention and Control. Healthcare-associated infections- a threat to patient safety in Europe.
- ⁸Lancet. 2022 Feb 12;399(10325):629-655. Christopher JL Murray, et al., *Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis*. Lancet. 2022 Feb 12;399(10325):629-655. doi: 10.1016/S0140-6736(21)02724-0. Epub 2022 Jan 19.
- ⁹Cassini et al., 2019. *Attributable deaths and disability-adjusted life-years caused by infections with antibiotic-resistant bacteria in the EU and the European Economic Area in 2015: a population-level modelling analysis*. Lancet Infect Dis. 19(1): 56-66.
- ¹⁰Ngandu, N.K., Mmotsa, T.M., Dassaye, R. et al. *Hospital acquired COVID-19 infections amongst patients before the rollout of COVID-19 vaccinations, a scoping review*. BMC Infect Dis **22**, 140 (2022).
- ¹¹Abed Alah, M., et al., *Compliance and barriers to the use of infection prevention and control measures among health care workers during COVID-19 pandemic in Qatar: A national survey*. J Nurs Manag, 2021. **29**(8): p. 2401-2411.
- ¹²Stangerup, M., et al., *Hand hygiene compliance of healthcare workers before and during the COVID-19 pandemic: A long-term follow-up study*. Am J Infect Control, 2021. **49**(9): p. 1118-1122.
- ¹³Huang, F., et al., *COVID-19 outbreak and healthcare worker behavioural change toward hand hygiene practices*. J Hosp Infect, 2021. **111**: p. 27-34.
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¹⁵Diab-El Schahawi M., W. Zingg, M. Vos, H. Humphreys, L. Lopez-Cerero, A. Fueszl, J. R. Zahar, E. Presterl, ESCMID Study Group on Nosocomial Infections “The decontamination research working group”. 2021. *Ultraviolet disinfection robots to improve hospital cleaning: Real promise or just a gimmick?* Antimicrob Resist Infect Control. 12;10(1):33.

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¹⁷Loth M. 2021. PhD thesis entitled ‘*Design for Sanitation: How does design influence train toilet hygiene?*’

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¹⁹Buljac-Samardzic M., K. D. Doekhie, J. D. H. van Wijngaarden. 2020. *Interventions to improve team effectiveness within health care: a systematic review of the past decade*. Hum Resour Health. 18(1):2.

²⁰Spoon D., T. Rietbergen, A. Huis, M. Heinen, M. van Dijk, L. van Bodegom-Vos, E. Ista. 2020. *Implementation strategies used to implement nursing guidelines in daily practice: A systematic review*. Int J Nurs Stud. 111:103748.